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Explainable Artificial Intelligence (XAI) for Decision-Making in Graph-Based Recommendation System

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Abstract

The growing reliance on algorithmic decision-making has intensified concerns regarding transparency, fairness, and accountability in artificial intelligence systems. Recommender systems, which strongly influence user choices in domains such as e-commerce, entertainment, and social media, are particularly affected by these issues. This thesis proposes a series of frameworks that advance Explainable Artificial Intelligence (XAI) for recommender systems by integrating Graph Neural Networks (GNNs) with game-theoretic attribution methods to enhance both algorithmic transparency and fairness.

This research begins with a detailed exploration of traditional and deep learning-based recommendation approaches, establishing the theoretical foundations of explainability and fairness in machine learning. Building on this, the first contribution introduces a GNN-based recommender system that captures the complex structure of user-item interactions through graph representations, significantly improving the predictive performance over classical methods.

The second contribution presents an adaptive explainability framework using Shapley values, derived from cooperative game theory, to quantify and interpret the contribution of each feature in real time. This approach not only provides transparent, human-understandable recommendations but also enables adaptive fairness adjustments during the model inference.

The final contribution extends this work through hyperbolic GNN architectures combined with Shapley-value-based explanations to model the hierarchical relationships within recommendation data. Experimental results on benchmark datasets, including MovieLens and Amazon Product Reviews, demonstrate substantial improvements in precision, recall, interpretability, and fairness compared with existing baselines.

Overall, this thesis contributes a unified and principled framework that combines geometric deep learning, game-theoretic explainability, and fairness-aware optimisation to promote trustworthy and transparent recommender systems capable of delivering accurate, fair, and interpretable outcomes.

Keywords: *Explainable Artificial Intelligence (XAI), Recommender Systems, Graph Neural Networks (GNNs), Algorithmic Transparency, Algorithmic Fairness, Game-Theoretic Attribution, Shapley Values, Hyperbolic Embeddings, Interpretability, Adaptive Learning, Trustworthy AI, Hierarchical Recommendation*

Résumé

La dépendance croissante aux systèmes de décision automatisés soulève des préoccupations majeures concernant la transparence, l'équité et la responsabilité des algorithmes d'intelligence artificielle. Les systèmes de recommandation, qui influencent fortement les choix des utilisateurs dans des domaines tels que le commerce électronique, le divertissement et les réseaux sociaux, sont particulièrement concernés par ces enjeux. Cette thèse propose une série de cadres méthodologiques visant à faire progresser l'Intelligence Artificielle Explicable (XAI) appliquée aux systèmes de recommandation, en intégrant les Réseaux de Neurones Graphiques (GNNs) et l'attribution fondée sur la théorie des jeux, afin d'améliorer à la fois la transparence algorithmique et l'équité.

La recherche débute par une étude approfondie des approches classiques et des modèles d'apprentissage profond utilisés dans les systèmes de recommandation, établissant ainsi les fondements théoriques de l'explicabilité et de la justice algorithmique. La première contribution introduit un système de recommandation fondé sur les GNN, capable de modéliser la complexité des relations utilisateur-objet à travers des structures de graphes, tout en améliorant la précision prédictive par rapport aux méthodes traditionnelles.

La deuxième contribution présente un cadre d'explicabilité adaptatif reposant sur les valeurs de Shapley, issues de la théorie coopérative des jeux, permettant de quantifier et d'interpréter la contribution de chaque caractéristique en temps réel. Cette approche fournit non seulement des recommandations transparentes et interprétables par l'humain, mais permet également un ajustement adaptatif de l'équité au cours de l'inférence du modèle.

Enfin, la dernière contribution étend ce travail grâce à des architectures hyperboliques de GNN combinées à des explications basées sur les valeurs de Shapley, afin de mieux modéliser les relations hiérarchiques dans les données de recommandation. Les expériences menées sur les jeux de données de référence, tels que MovieLens et Amazon Product Reviews, démontrent des améliorations significatives en matière de précision, de rappel, d'interprétabilité et d'équité par rapport aux approches existantes.

Dans l'ensemble, cette thèse propose un cadre unifié et rigoureux combinant l'apprentissage profond géométrique, l'explicabilité fondée sur la théorie des jeux et l'optimisation équitable, afin de promouvoir des systèmes de recommandation trans-

parents, fiables et équitables, capables de fournir des résultats à la fois précis, interprétables et responsables.

Mots clés: *Intelligence Artificielle Explicable (XAI), Systèmes de Recommandation, Réseaux de Neurones Graphiques (GNN), Transparence Algorithmique, Équité Algorithmique, Attribution Théorique des Jeux, Valeurs de Shapley, Représentations Hyperboliques, Interprétabilité, Apprentissage Adaptatif, Intelligence Artificielle Fiable, Recommandation Hiérarchique*

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Acronyms

Artificial Intelligence and Learning

AI	A rtificial I ntelligence
ML	M achine L earning
XAI	E xplainable A rtificial I ntelligence

Recommender Systems

CF	C ollaborative F iltering
UBCF	U ser- B ased C ollaborative F iltering
IBCF	I tem- B ased C ollaborative F iltering
Top-N	T op- N recommendation

Graph-Based Models

GNN	G raph N eural N etwork
GNNs	G raph N eural N etworks
GCN	G raph C onvolutional N etwork
LightGCN	L ightweight G raph C onvolutional N etwork
GAT	G raph A ttention N etwork
GraphSAGE	G raph S ample and A ggregate
HGNN	H yperbolic G raph N eural N etwork
HGNN-SV	H yperbolic G raph N eural N etwork with S hapley V alue explanations

Explainability Methods

GNNShap	G NN-based Shap ley-value explanation method
MAGE	M odel- A gnostic G raph E xplanations
LIME	L ocal I nterpretable M odel-agnostic E xplanations

Matrix and Neural Baselines

MF	M atrix F actorisation
SVD	S ingular V alue D ecomposition
SVD++	S ingular V alue D ecomposition with implicit feedback
NCF	N eural C ollaborative F iltering

Evaluation Metrics and Optimisation

Precision@N	Precision at cutoff N
Recall@N	Recall at cutoff N
NDCG	Normalised Discounted Cumulative Gain
ℓ_2	ℓ_2 regularisation

List of publications

This thesis consists of an introductory review, followed by XX research articles. In the introductory part, these papers are cited according to their roman numerals.

- I. R. Nesmaoui et al., “Dynamic Recommender Systems with Real-time Shapley Value-based Contribution Adjustment,” *International Journal of Intelligent Engineering and Systems*, vol. 18, pp. 241–257, Sep. 2025. DOI: 10.22266/ijies2025.1031.16
- II. R. Nesmaoui et al., “Graph Neural Networks with Shapley-Value Explanations for Hierarchical Recommendation Systems,” *International Journal of Advanced Computer Science and Applications*, vol. 16, pp. 806–817, Sep. 2025. DOI: 10.14569/IJACSA.2025.0160977
- III. R. Nesmaoui et al., “A Collaborative Filtering Movies Recommendation System based on Graph Neural Network,” *Procedia Computer Science*, The 14th International Conference on Ambient Systems, Networks and Technologies Networks (ANT) and The 6th International Conference on Emerging Data and Industry 4.0 (EDI40), vol. 220, pp. 456–461, Jan. 2023, ISSN: 1877-0509. DOI: 10.1016/j.procs.2023.03.058. Accessed: Aug. 2, 2025. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1877050923005938>
- IV. R. Nesmaoui et al., “A Hybrid Machine Learning Method for Movies Recommendation,” en, in *Proceedings of the 5th International Conference on Big Data and Internet of Things*, M. Lazaar et al., Eds., Cham: Springer International Publishing, 2022, pp. 517–528, ISBN: 978-3-031-07969-6. DOI: 10.1007/978-3-031-07969-6_39

Introduction

Recommender systems are a fundamental component of many modern digital platforms, including e-commerce websites, video streaming services, and online content platforms [5]. Their main purpose is to assist users in discovering relevant items within large collections by analyzing their past interactions and preferences. As the amount of available data continues to increase, recommender systems have become essential for reducing information overload and improving the user experience [6]. Therefore, they have attracted significant attention from both academic and industrial researchers alike.

Traditional recommender systems are mainly based on collaborative filtering and content-based approaches. Collaborative filtering relies on similarities between users or items, whereas content-based methods use the item attributes and user profiles. Although these approaches are effective in many scenarios, they suffer from well-known limitations, such as data sparsity, cold-start problems, and limited modelling capacity [7]. To address these issues, more advanced techniques based on matrix factorization and neural networks have been proposed, allowing richer latent representations and improved recommendation accuracy.

Recently, graph-based recommender systems have emerged as powerful alternatives to traditional recommender systems. In these systems, users and items are represented as nodes in a graph, and their interactions are modelled as edges [8]. This representation allows the model to explicitly capture relational and structural information that is difficult to represent using traditional matrix-based methods. Graph Neural Networks (GNNs) have become a popular choice for this task because they can learn node representations by aggregating information from neighboring nodes through message-passing mechanisms [9][10]. GNN-based recommender systems have demonstrated strong performance, particularly in sparse data settings and complex interaction environments.

Despite their effectiveness, existing graph-based recommender systems have several limitations that restrict their practical applications. The first and most critical limitation is the lack of explanation. Most GNN-based models operate as black boxes [11], producing recommendations without providing clear reasons for their decisions. This makes it difficult for users to understand why specific items are recommended and for system designers to analyze, debug, and improve the models. As recommender

systems increasingly influence user behavior and decision-making, this lack of transparency raises concerns about trust, accountability, and fairness.

The second major limitation is the inability of many existing models to adapt effectively to changes in user preferences [12]. User interests are dynamic and evolve over time; however, most recommender systems rely on static training procedures and historical interaction data to recommend content. Consequently, outdated or less relevant interactions may continue to influence recommendations, reducing their relevance in dynamic settings. Although periodic retraining can partially mitigate this issue, it is often inefficient and does not fully address the need for continuous adaptation in the model.

The third limitation concerns the modelling of the hierarchical structures. Many real-world recommendation scenarios involve hierarchical information, such as product categories, content taxonomies, and genre structures. Most existing recommender systems, including many GNN-based approaches, rely on Euclidean embedding spaces that are not well suited for representing hierarchical and tree-like relationships. This mismatch can lead to distorted representations and reduced recommendation quality in hierarchical settings [2].

These limitations reveal a clear research gap in current recommender system methods research. Although graph-based models offer strong predictive performance, they generally lack built-in explainability, have limited adaptability to evolving user behavior, and struggle to represent hierarchical structures in a principled way. Existing explainability methods are often applied after a recommendation is generated, providing post hoc explanations that do not influence the model’s behavior. Similarly, adaptive mechanisms and hierarchical modelling techniques are rarely integrated into a unified framework.

Therefore, the research problem addressed in this thesis is the design of recommender systems that are simultaneously accurate, explainable, adaptive, and capable of modelling hierarchical relationships. More specifically, the problem is how to integrate Explainable Artificial Intelligence (XAI) mechanisms directly into graph-based recommender systems to improve transparency and adaptability without sacrificing performance.

To address this problem, this study proposes a set of novel XAI-based graph recommender frameworks that differ from existing approaches in several key ways. First,

explainability is not treated as a separate or post hoc component but is integrated directly into the recommendation process. The proposed methods use Shapley values, a well-established concept from cooperative game theory, to quantify the contribution of individual user–item interactions to recommendation outcomes in a principled and interpretable manner. This provides faithful explanations that are directly linked to the model’s internal decision-making process.

Second, the proposed approach introduces adaptive recommendation mechanisms in which Shapley-based contribution scores are used during inference to dynamically adjust the influence of interactions. This allows the recommender system to respond more effectively to changes in user behavior by strengthening relevant interactions and reducing the impact of less informative ones. Unlike traditional static models, this adaptive mechanism improves robustness in dynamic environments while maintaining interpretability.

Third, this thesis addresses hierarchical recommendation scenarios by combining graph neural networks with hyperbolic embedding spaces. Hyperbolic geometry provides a natural representation of hierarchical and tree-structured data, allowing the preservation of deeper and more meaningful hierarchical relationships. By integrating hyperbolic graph neural networks with Shapley-value-based explainability, the proposed models offer a unique combination of geometric expressiveness and transparent decision-making that is not present in existing approaches.

The strengths of the proposed methods lie in their unified designs. They simultaneously address the explainability, adaptability, and hierarchical modelling within a single graph-based recommendation framework. Compared with existing techniques, the proposed approaches provide clearer explanations, improved responsiveness to user preference changes, and more faithful representations of structured data while maintaining competitive recommendation performance.

This thesis is organized as follows. Chapter 1 presents the theoretical background related to recommender systems, graph neural networks, explainable artificial intelligence, and geometric representation learning. Chapter 2 reviews the related work and positions the contributions of this thesis with respect to the existing approaches.

Chapter 3 defines the experimental protocol and evaluation framework used throughout this thesis, including datasets, data splitting strategies, baseline methods, evaluation metrics, and training procedures. This chapter serves as a common reference for all

the experimental studies.

Chapter 4 introduces a graph neural network–based collaborative filtering approach for movie recommendation and analyzes its performance in comparison with classical collaborative filtering and matrix factorization baselines. Chapter 5 extends this framework by proposing adaptive recommender systems based on real-time Shapley-value contribution adjustment. Chapter 6 extends the proposed approach to hierarchical recommendation by integrating Shapley value explanations within hyperbolic graph neural networks.

Finally, Chapter 7 concludes the thesis by summarizing the main contributions and outlining directions for future research.

1 Background and Theoretical Foundations

Introduction

This chapter presents the theoretical background and fundamental concepts underpinning the research conducted in this thesis. Its purpose is to provide the necessary foundation for understanding the proposed methods and contributions developed in the subsequent chapters. This chapter introduces key concepts related to recommender systems, graph-based modelling, graph neural networks, and explainable artificial intelligence, which together form the core of the research framework.

This chapter begins with an overview of recommender systems, outlining their objectives, characteristics, and typical problem formulations. It then reviews traditional recommendation techniques and discusses their main limitations, highlighting the motivations for using more advanced models. A graph-based representation of recommender systems is subsequently introduced, followed by a detailed presentation of graph neural networks and their application to recommendation tasks.

In addition, this chapter introduces the principles of explainable artificial intelligence, with particular emphasis on explanation methods relevant to recommender systems. The theoretical foundations of the Shapley values are presented as a principled approach to model explainability. Finally, the chapter discusses representation learning from a geometric perspective and introduces hyperbolic embedding spaces for modelling hierarchical data. Together, these concepts establish the theoretical basis required for the design, analysis, and evaluation of the explainable graph-based recommender systems proposed in this study.

1.1 Recommender Systems Overview

Recommender systems are computational tools designed to assist users in identifying items of interest in large and complex information spaces. Their primary objective was to predict user preferences and provide personalized suggestions that reduce information overload and improve decision-making. Recommender systems are widely used in many application domains, including e-commerce, multimedia streaming platforms, social networks, online news services, and digital libraries [13]. In these environments,

users are typically exposed to a vast number of available items, making manual exploration inefficient or impractical.

At the conceptual level, recommender systems analyze historical data describing the interactions between users and items. These interactions may take different forms, such as explicit feedback, where users provide ratings or reviews, or implicit feedback, where preferences are inferred from behaviors such as clicks, views, purchases, and browsing history [14]. Based on this information, the system learns patterns that allow it to estimate the likelihood of a user being interested in an unseen item being displayed. The output of a recommender system is typically a ranked list of items tailored to specific users [15], [16].

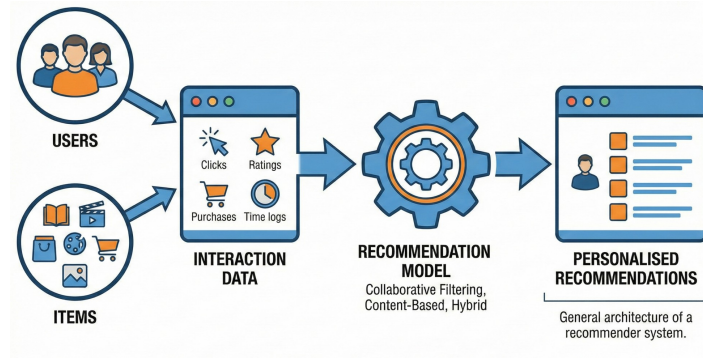


Figure 1: General architecture of a recommender system.

Recommender systems can be formally described as a prediction problem. Given a set of users, a set of items, and a collection of observed interactions between them, the goal is to estimate a relevance score for each user–item pair [17]. These scores were used to rank the items and generate recommendations. Depending on the application, the task may involve rating prediction, top- N recommendation, or item ranking. Although the underlying objectives may vary, most recommender systems share this formulation [18].

Several key characteristics distinguish recommender systems from traditional information retrieval systems. First, recommender systems are inherently personalized, meaning that different users receive different recommendations for the same item catalogue [19]. Second, they often operate in dynamic environments where user preferences, item availability, and contextual factors change over time. Third, recommender systems must balance multiple objectives, such as accuracy, diversity, novelty, and user satis-

faction. These characteristics introduce additional complexity and motivate the use of advanced modelling techniques [20].

From a data perspective, recommender systems typically rely on sparse and noisy interaction matrices [21]. In many real-world scenarios, users interact with only a small fraction of the available items, resulting in incomplete observations. Moreover, user feedback may be biased or inconsistent, further complicating the learning process [22]. These data characteristics strongly influence the design of recommendation algorithms and have driven the development of increasingly sophisticated models.

In recent years, the role of recommender systems has expanded beyond pure accuracy. As these systems increasingly influence user behavior and decision-making, issues such as transparency, trust, and fairness have gained importance. Users may wish to understand why certain items are recommended, whereas system designers require tools to analyze model behavior and detect potential biases [23]. This broader perspective has contributed to the growing interest in XAI and trustworthy recommendation systems [1].

Overall, recommender systems represent a complex and multidisciplinary research area that combines elements of machine learning, data mining, human–computer interaction, and information systems [16]. This section introduces the general concept and objectives of recommender systems, providing a foundation for the discussion of traditional and advanced recommendation techniques in the following sections.

1.2 Traditional Recommendation Techniques

Traditional recommendation techniques form the foundation of most early recommender systems and remain widely used in practice because of their simplicity and interpretability [24]. These techniques aim to predict user preferences by analyzing either the characteristics of items, the behavior of similar users, or a combination thereof. The most common traditional approaches include content-based filtering, collaborative filtering, and hybrid recommendation methods.

Content-based filtering recommends items to users by matching the item attributes with user preferences. In this approach, each item is described using a set of features, such as keywords, categories, or metadata, and a user profile is built based on

the features of items with which the user has previously interacted [4], [25]. Recommendations are generated by computing the similarity between the user profile and candidate items. Content-based methods have the advantage of being personalized and independent of other users' data [26]. They are also relatively transparent because recommendations can often be explained in terms of the shared item features. However, these methods tend to suffer from limited novelty and overspecialization because they mainly recommend items similar to those already consumed by the user.

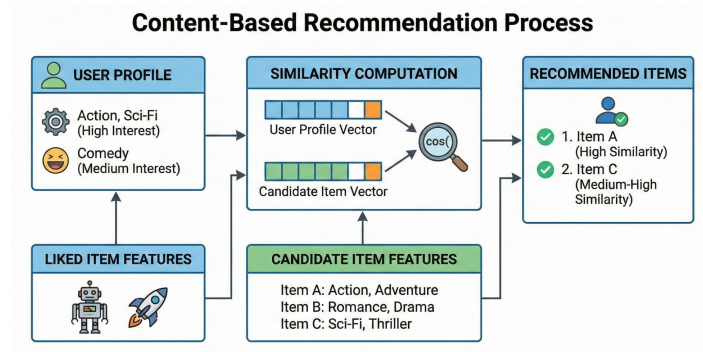


Figure 2: Principle of content-based recommendation.

Collaborative filtering is another widely adopted recommendation technique that relies on the collective behavior of users rather than explicit item features [27]. The central idea is that users with similar preferences in the past are likely to have similar future ones. Collaborative filtering can be divided into two main categories: user-based and item-based filtering. User-based collaborative filtering predicts a user's interest in an item by analyzing the preferences of other users with interaction patterns similar to those of the target user [28]. Item-based collaborative filtering, on the other hand, focuses on similarities between items and recommends items that are similar to those that the user has previously liked.

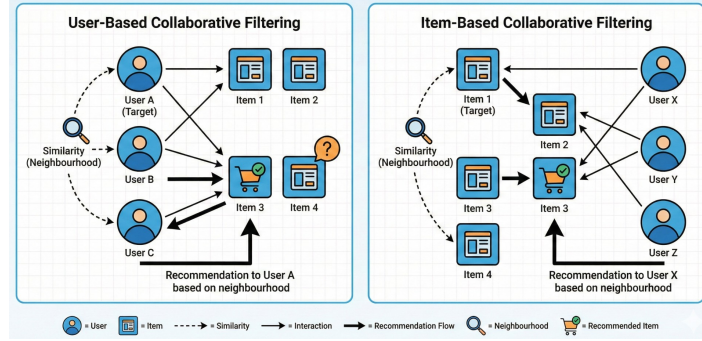


Figure 3: User-based and item-based collaborative filtering.

Collaborative filtering methods are attractive because they do not require detailed item descriptions and can capture complex user preference patterns [3]. However, they have several limitations. Data sparsity is a major issue because users typically interact with only a small fraction of the available items, making similarity estimation unreliable. Collaborative filtering methods also struggle with the cold-start problem, in which new users or items have insufficient interaction data to generate reliable recommendations. In addition, scalability is a concern as the number of users and items increases.

Hybrid recommendation techniques aim to combine the strengths of content-based and collaborative filtering approaches while mitigating their weaknesses [19]. Hybrid systems may integrate both types of information at different stages of the recommendation process, such as by combining similarity scores, incorporating content features into collaborative models, or switching between methods depending on data availability. By leveraging multiple information sources, hybrid approaches can improve recommendation accuracy and robustness, particularly in cold-start scenarios.

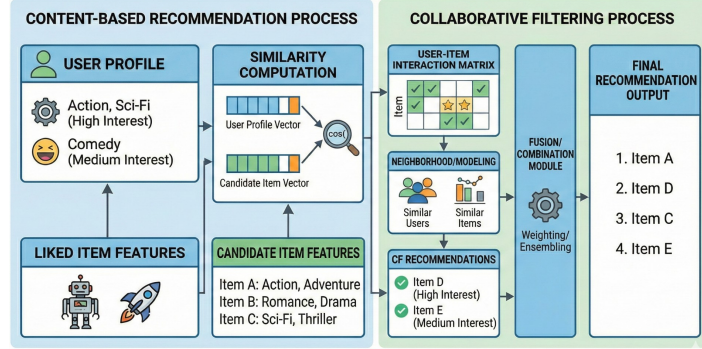


Figure 4: Hybrid recommendation framework combining content-based and collaborative filtering.

Despite their widespread use, traditional recommendation techniques have limited modelling capacity and struggle to capture complex relational structures and high-order interactions. Their reliance on handcrafted similarity measures and shallow representations restricts their ability to exploit large-scale, highly structured data fully. These limitations have motivated the development of more expressive models, including matrix factorization, neural-based approaches, and graph-based recommender systems, which are discussed in the following sections.

1.3 Limitations of Classical Recommender Systems

Despite their widespread adoption, classical recommender systems exhibit several inherent limitations that restrict their effectiveness in large-scale and complex environments such as e-commerce platforms. These limitations arise mainly from the nature of user-item interaction data, the modelling assumptions underlying traditional techniques, and the increasing complexity of real-world recommendation scenarios. Understanding these limitations is essential for motivating the transition to more advanced recommendation models [29].

One of the most significant challenges faced by classical recommender systems is data sparsity. In many real-world applications, users interact with only a small subset of available items, resulting in sparse user-item interaction matrices [30]. Sparse data make it difficult to accurately estimate the similarities between users or items, which directly affects the performance of similarity-based recommendation techniques [31]. As

sparsity increases, recommendations become less reliable, particularly for users with limited interaction histories.

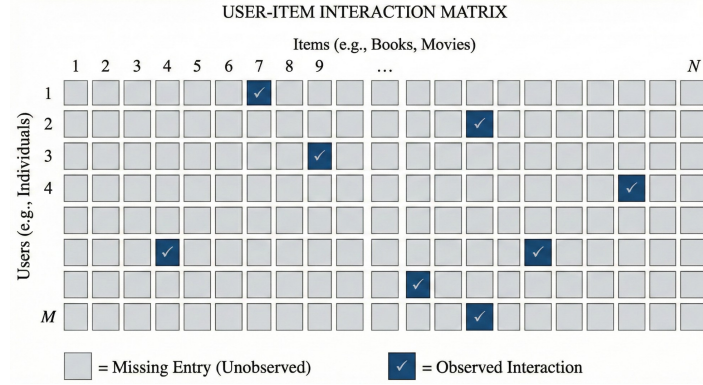


Figure 5: Illustration of data sparsity in a user-item interaction matrix.

The cold-start problem is closely related to the data sparsity. This issue arises when new users or items are introduced into the system without sufficient interaction data [32]. For new users, the system lacks information to infer preferences, whereas for new items, there are no interactions available to establish the similarities. Classical recommender systems, particularly collaborative filtering approaches, struggle in such situations because they rely heavily on historical data. Consequently, recommendations for new users or items are often inaccurate or unavailable [33], [34].

Another important limitation is its scalability. As the number of users and items increases, the computational cost of traditional recommendation techniques increases significantly. Similarity computations in collaborative filtering often require pairwise comparisons between users or items, which can be prohibitively expensive in large-scale systems [35]. Although various optimization strategies have been proposed, scalability remains a critical concern, particularly in applications with millions of users and items.

Classical recommender systems also suffer from limited modelling capability. Many traditional techniques rely on linear assumptions and shallow representations that cannot capture complex nonlinear user preferences and high-order interaction patterns. Consequently, these models may fail to exploit the rich contextual information or subtle relationships in large and heterogeneous datasets. This limitation restricts their ability to adapt to diverse user behaviors and evolving recommendation environments [36].

In addition, classical recommender systems provide limited support for modelling structured and relational data. User-item interactions are typically represented in a matrix form that does not naturally capture relationships beyond direct interactions. Information such as social connections, item similarities, or hierarchical structures must be incorporated through ad hoc extensions, often leading to fragmented and inefficient models.

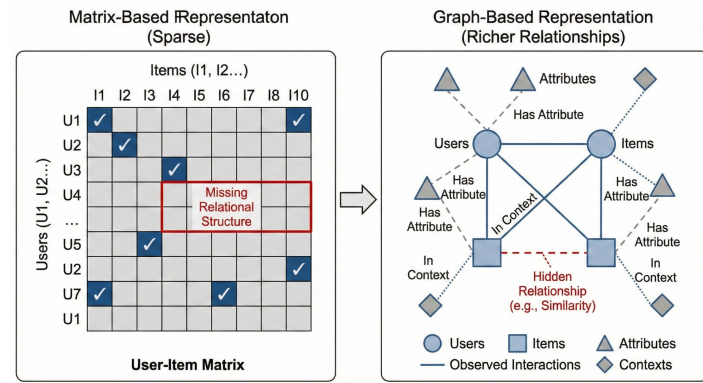


Figure 6: Comparison between matrix-based and graph-based representations in recommender systems.

Finally, traditional recommender systems offer limited transparency and interpretability. Although some content-based methods can provide simple explanations based on shared features, many collaborative filtering approaches generate recommendations without a clear justification. Users and system designers often cannot understand why specific items are recommended, which can reduce trust and hinder system analyses. This lack of interpretability becomes increasingly problematic as recommender systems influence important user choices [37].

Taken together, these limitations highlight the need for more expressive, scalable, and transparent recommendation models. In particular, there is a clear motivation to move beyond matrix-based representations toward graph-based approaches that can naturally model complex relationships and higher order interactions. These considerations provide the foundation for the introduction of graph-based recommender systems and graph neural networks in the following sections.

1.4 Graph-Based Recommender Systems

Graph-based recommender systems have emerged as an effective approach for modelling complex relationships in recommendation tasks. Unlike traditional matrix-based methods, which represent user-item interactions as a two-dimensional matrix, graph-based approaches model the recommendation problem as a graph structure [38]. In this representation, users and items are treated as nodes, and their interactions are represented by edges. This formulation allows recommender systems to naturally capture relational and structural information that is difficult to represent using conventional methods[39].

The most common graph representation used in recommender systems is the user-item interaction graph. This graph is typically modelled as a bipartite graph, where one set of nodes corresponds to users and the other corresponds to items [37]. An edge between a user and an item indicates an interaction, such as a rating, click, or purchase. Edge weights can be used to represent the interaction strength or confidence of the interaction. This graph-based representation preserves the original interaction structure and enables the use of graph learning techniques to propagate information across connected nodes [1].

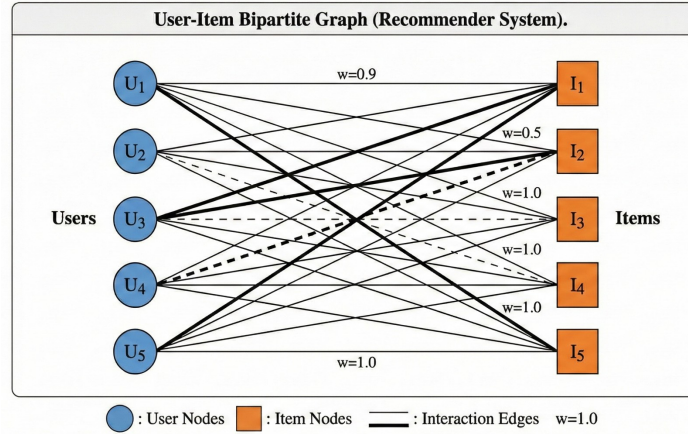


Figure 7: User-item interaction modelled as a bipartite graph.

One of the main advantages of graph-based recommender systems is their ability to model the higher-order connectivity. In a graph, a user may be connected not only to directly interacted items but also indirectly to other users and items through multiple hops. These higher-order connections can reveal latent preference patterns, such as

users with similar tastes or items with shared audiences, even when direct interactions are limited. By exploiting multi-hop neighborhood information, graph-based models can generate more informative and robust recommendations [40].

Graph-based representations also allow the integration of heterogeneous information into recommendation processes. In addition to users and items, graphs can include other types of nodes, such as item attributes, categories, tags, and social connections between users. This results in heterogeneous graphs that capture multiple types of relationships within a single structure. Such representations are particularly useful for incorporating contextual, semantic, and structural information into recommendation models without relying on ad hoc feature engineering [41].

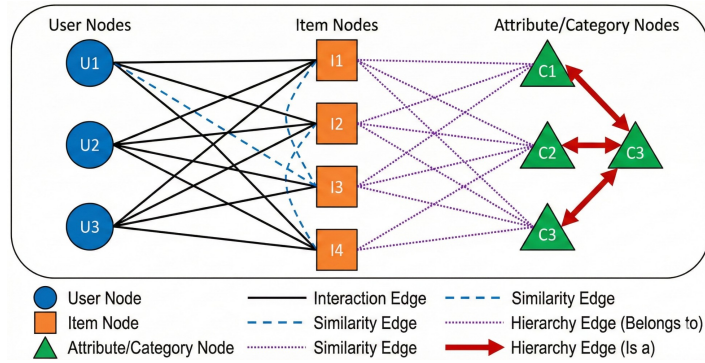


Figure 8: Example of a heterogeneous graph for recommender systems.

Another important characteristic of graph-based recommender systems is their flexibility in modelling different recommendation tasks. Graph representations can support rating prediction, top- N recommendation, and item ranking by learning node embeddings that capture user preferences and item characteristics [42]. These embeddings can be learned using various graph-based techniques, including random walks, graph embedding methods, and, more recently, graph neural networks. Compared with traditional similarity-based approaches, graph-based models offer greater expressiveness and improved scalability when combined with appropriate learning algorithms [43].

Despite their advantages, early graph-based recommender systems relied primarily on shallow-graph embedding techniques or handcrafted propagation rules. Although these methods improved upon classical approaches, they were often limited in their ability to fully exploit complex graph structures and non-linear interaction patterns. This

limitation motivated the adoption of deep learning techniques for graph data, leading to the development of recommender systems based on graph neural networks.

Therefore, graph-based recommender systems represent a critical transition between traditional recommendation techniques and modern deep-learning approaches. By explicitly modelling user-item interactions as graphs, they provide a natural and powerful framework for capturing relational information, handling sparsity, and integrating diverse data sources. This perspective sets the foundation for the introduction of graph neural networks, which extend graph-based recommendation models by enabling end-to-end representation learning through neighborhood aggregation and message passing, as discussed in the following sections.

1.5 Graph Neural Networks

Graph Neural Networks (GNNs) are a class of deep learning models that are specifically designed to operate on graph-structured data. Unlike traditional neural networks, which assume grid-like or sequential input structures [44], GNNs can process data represented as nodes connected by edges. This capability makes them particularly suitable for applications in which the relationships between entities play a central role, such as in social networks, molecular graphs and recommender systems [45].

At a high level, the objective of a GNN is to learn a vector representation or embedding for each node in the graph [46]. These embeddings are constructed by iteratively aggregating information from the neighborhood of the node. Through this process, each node representation captures both its features and the structural context provided by the connected nodes. By stacking multiple GNN layers, information can be propagated across increasingly larger neighborhoods, allowing the model to capture higher-order dependencies in the graph structure [47].

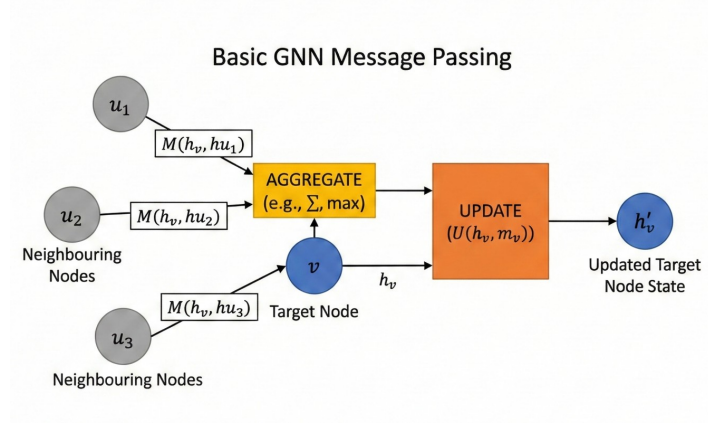


Figure 9: Message passing and neighbourhood aggregation in a graph neural network.

Most GNN architectures follow a common message-passing framework in which, during each layer of the network, a node receives messages from its neighbors, aggregates these messages using a permutation-invariant function, such as summation or averaging, and updates its representation through a learnable transformation. Formally, this process can be described as a sequence of aggregation and update operations applied to all nodes in the graph. This design ensures that GNNs are invariant to node ordering and can generalize across graphs of various sizes and structures [47].

Several GNN architectures have been proposed in the literature, each differing in the manner in which the messages are computed and aggregated. Graph Convolutional Networks (GCNs) extend the concept of convolution to graph structures by normalizing neighborhood information and applying shared linear transformations. GraphSAGE introduces neighborhood sampling and aggregation functions to improve the scalability of large graphs. Graph Attention Networks (GATs) incorporate attention mechanisms that allow the model to assign different importance weights to neighboring nodes, enabling more flexible information aggregation [18].

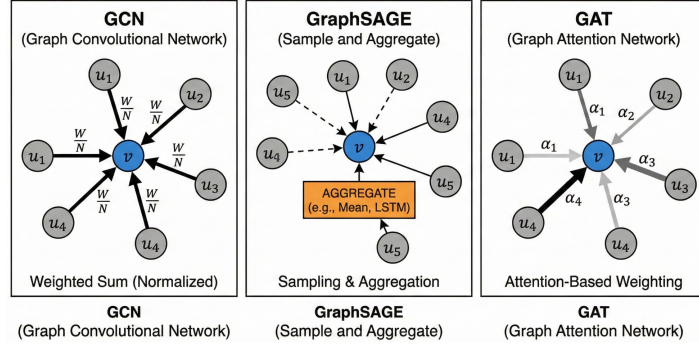


Figure 10: Illustration of common graph neural network architectures.

GNNs can be applied to different types of learning tasks, including node-, edge-, and graph-level prediction. In node-level tasks, the goal is to predict the labels or properties of individual nodes based on their representation. In edge-level tasks, the model predicts the relationships between pairs of nodes, such as the existence or strength of a link. Graph-level tasks involve learning the representation of an entire graph, which can then be used for classification or regression. In recommender systems, GNNs are most commonly used for node- and edge-level prediction, where users and items are represented as nodes, and recommendations are generated by estimating the relevance of user-item pairs [48].

One of the main strengths of GNNs is their ability to exploit both node features and graph structures within a unified framework. By jointly learning from attributes and connectivity patterns, GNNs can capture complex interaction dynamics that are difficult to model using traditional techniques. This makes them particularly effective in sparse and highly relational settings, where indirect connections provide valuable information [3], [18].

Despite their advantages, GNNs face certain challenges. As the number of layers increases, the node representations may become overly similar, a phenomenon known as oversmoothing. In addition, the computational complexity and memory requirements can increase rapidly for large graphs, making scalability a key concern. These challenges have motivated the development of simplified and specialized GNN architectures, particularly in recommender systems.

In summary, graph neural networks provide a powerful and flexible framework for learning representations from graph-structured data. Their ability to model higher-order

relationships and integrate structural information makes them well-suited for recommendation tasks. The following section focuses on the application of graph neural networks to recommender systems, with particular attention to architectures specifically designed for collaborative filtering.

1.6 Graph Neural Networks for Recommendation

The application of Graph Neural Networks (GNNs) to recommender systems has gained significant attention due to their ability to model complex relational structures and high-order user–item interactions. By representing users and items as nodes in a graph and learning their representations through neighbourhood aggregation, GNN-based recommender systems provide a natural extension of classical collaborative filtering approaches. In this setting, the recommendation task is typically formulated as a link prediction or ranking problem on a user–item interaction graph [3], [18].

In GNN-based recommendation models, each user and item node is associated with a latent embedding that is iteratively updated by aggregating information from connected nodes. For a user node, this aggregation captures information from interacted items, while for an item node, it captures information from users who have interacted with it [49]. Through multiple layers of message passing, the learned embeddings encode both direct and indirect interaction patterns, allowing the model to capture collaborative signals beyond immediate neighbours. This mechanism enables GNN-based models to address data sparsity by exploiting higher-order connectivity in the interaction graph [50].

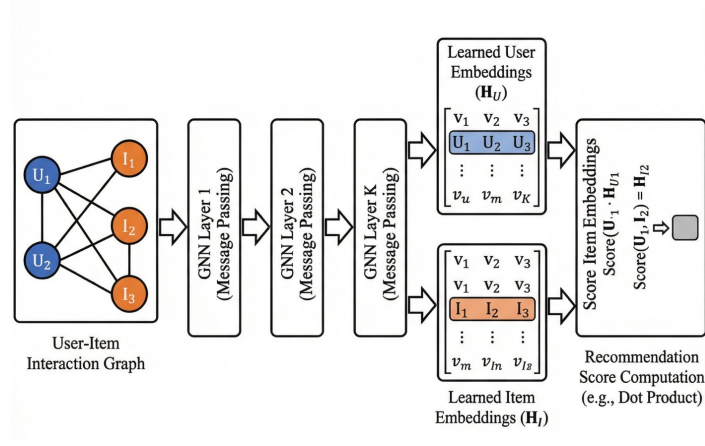


Figure 11: General workflow of a graph neural network-based recommender system.

One of the key advantages of using GNNs for recommendation is their ability to unify representation learning and collaborative filtering within a single end-to-end framework. Unlike traditional methods that rely on predefined similarity measures, GNN-based models learn similarity implicitly through message passing and parameter optimisation. This allows them to capture non-linear preference patterns and complex dependencies between users and items. Furthermore, GNNs can easily incorporate additional information, such as item attributes or social links, by extending the graph structure [51].

Several GNN-based architectures have been proposed specifically for recommender systems. Early approaches adapted general-purpose GNN models, such as Graph Convolutional Networks, to user-item graphs. These models demonstrated the effectiveness of graph convolution for recommendation but also revealed limitations related to over-parameterisation and unnecessary feature transformations. To address these issues, simplified architectures tailored to collaborative filtering have been introduced [52].

Light Graph Convolutional Networks (LightGCN) represent a notable example of such specialised architectures. LightGCN removes non-linear activation functions and feature transformation matrices, retaining only neighbourhood aggregation operations. By focusing on pure collaborative signals, LightGCN reduces model complexity and improves both efficiency and recommendation performance. The final user and item embeddings are obtained by combining representations from multiple propagation layers, allowing the model to capture information from different graph depths [3].

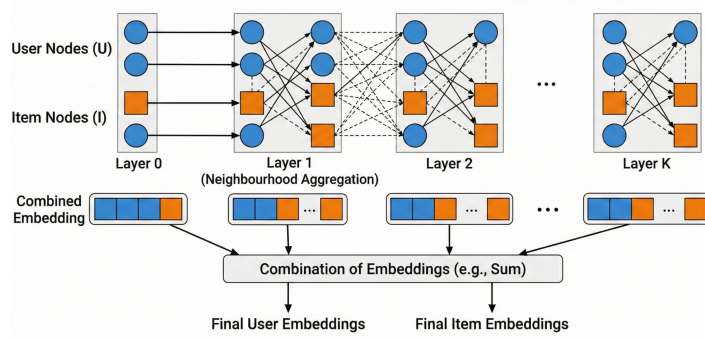


Figure 12: Propagation and embedding aggregation in Light Graph Convolutional Networks.

GNN-based recommender systems typically generate recommendations by computing a relevance score between user and item embedding. This score can be obtained using simple operations such as inner products or cosine similarity, and items are ranked accordingly for each user. Model training is often performed using pairwise or pointwise loss functions that encourage higher scores for observed than unobserved interactions. This training paradigm aligns well with the top- N recommendation objectives commonly used in practice [18], [51].

Despite their strong performance, GNN-based recommender systems introduce new challenges. Their increased model complexity can make training and inference computationally demanding, particularly for large-scale graphs. In addition, GNN-based models are generally less interpretable than traditional recommendation techniques because the learned embeddings and propagation processes are difficult to explain in human-understandable terms. This lack of transparency motivates the integration of explainable artificial intelligence techniques into GNN-based recommendation frameworks.

Overall, graph neural networks provide a powerful foundation for modern recommender systems by effectively modelling relational data and high-order interactions. However, their black-box nature and limited interpretability highlight the need for more explainable and adaptive extensions. These considerations motivate the exploration of explainable artificial intelligence approaches for GNN-based recommender systems, which is the focus of the following sections.

1.7 Explainable Artificial Intelligence (XAI)

Explainable Artificial Intelligence (XAI) refers to a set of methods and techniques designed to make the decisions of machine learning models understandable to humans. As machine learning systems are increasingly deployed in real-world applications, transparency has become a major concern. Complex models, particularly deep learning architectures, often operate as black boxes, producing accurate predictions without providing insights into the underlying reasoning. XAI aims to address this issue by allowing users, developers, and stakeholders to interpret, trust, and effectively manage intelligent systems that use AI [53], [54].

Explainability is especially important in decision-making systems that directly influence users, such as recommendation systems. Recommendations can affect user behavior, purchasing decisions and information exposure. When users cannot understand why specific items are recommended, their trust in the system may decrease [55]. From a system design perspective, explainability is essential for debugging models, identifying errors, detecting biases, and ensuring compliance with ethical and regulatory requirements of AI models. Consequently, explainability is increasingly viewed as a fundamental requirement rather than an optional feature [11], [55].

XAI methods can be broadly categorized according to the scope and nature of the explanations provided. One common distinction is between global and local explanations for this. Global explanations aim to describe the overall behavior of a model, providing insights into how predictions are made across datasets [56], [57]. In contrast, local explanations focus on individual predictions and seek to explain why a specific output is produced for a given input. In recommender systems, local explanations are often preferred because users are typically interested in understanding the reasoning behind specific recommendations. [58], [59]

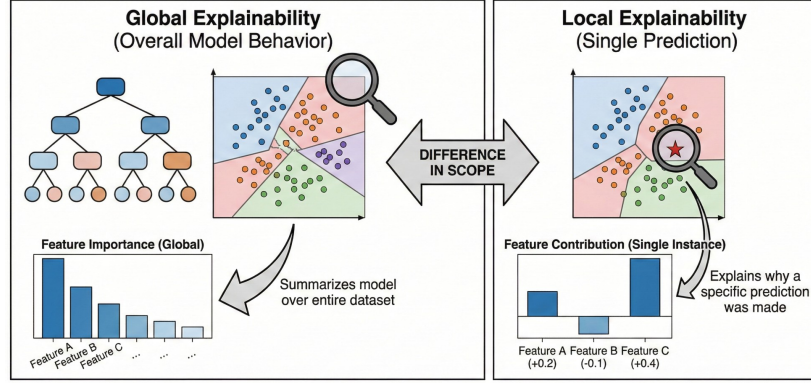


Figure 13: Difference between global and local explanations in explainable artificial intelligence.

Another important distinction in XAI is between model-agnostic and model-specific explanation methods. Model-agnostic approaches treat the underlying model as a black box and generate explanations by analyzing the relationship between the inputs and outputs [60]. These methods can be applied to a wide range of models without requiring access to internal parameters. In contrast, model-specific approaches exploit the internal structure of a particular model to generate explanations. Although model-specific methods may offer higher fidelity, they are often limited to certain architectures.

Explainability in recommender systems presents unique challenges compared to other machine learning tasks. Recommendations are often generated from implicit feedback and latent representations rather than explicit input features [1]. Consequently, it can be difficult to attribute recommendations to factors that are meaningful and interpretable. Moreover, recommender systems must balance the explanation quality, recommendation accuracy, and computational efficiency. Poorly designed explanations may oversimplify the model’s behavior or misrepresent the true decision process [2].

These challenges are further exacerbated in graph-based recommendation systems. Graph Neural Networks rely on iterative message passing and neighborhood aggregation, which makes it difficult to isolate the contribution of individual nodes and edges. Information flows across multiple hops in the graph, and the final prediction is influenced by complex interactions between users, items, and their neighbors. Consequently, traditional explanation techniques designed for tabular or image data are not directly applicable to graph-based models [37].

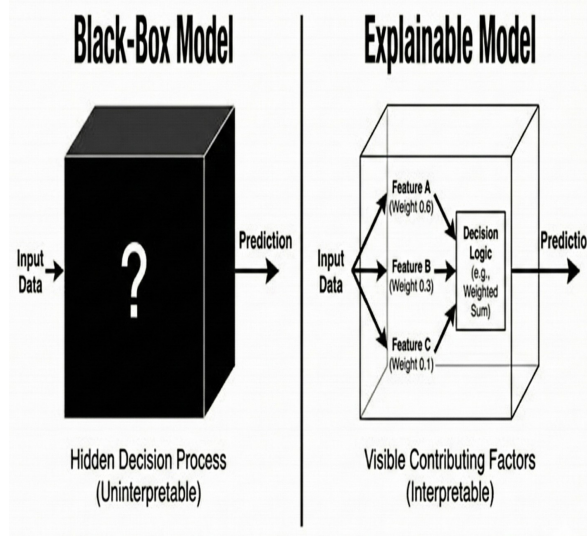


Figure 14: Conceptual comparison between black-box and explainable machine learning models.

To address these issues, recent research has focused on developing explanation methods that are specifically designed for recommender systems and graph-based models. These methods aim to identify the most influential users, items, or interactions contributing to a recommendation and present this information in an interpretable manner. Effective XAI techniques should provide explanations that are faithful to the model’s internal reasoning, consistent across similar predictions and understandable to end users.

In this thesis, explainability is not considered a post-hoc addition to an existing recommender system but an integral component of the recommendation process itself. By embedding explanation mechanisms directly into graph-based models, it is possible to improve transparency while influencing model behavior. This perspective motivates the use of theoretically grounded explanatory frameworks that can be integrated into graph neural networks.

In summary, Explainable Artificial Intelligence provides essential tools for improving the transparency, trust, and accountability of machine learning systems. Its application to recommender systems is particularly important because of the personalized and influential nature of the recommendations. However, existing XAI methods face significant challenges when applied to graph-based recommender systems, highlighting the need for specialized explanation techniques for GRSs. The following section introduces Shapley values as a principled approach to explainability that addresses many of these

challenges.

1.8 Shapley Values for Model Explainability

Shapley values originate from cooperative game theory and provide a principled framework for fairly distributing the total gain of a coalition among its participants [2]. In the context of machine learning, Shapley values have been adapted to quantify the contribution of individual features, data points, and components to model predictions. Their strong theoretical foundation and well-defined properties have made them one of the most widely adopted approaches for explainable artificial intelligence.

In a cooperative game, each player contributes to the overall outcome, and the Shapley value assigns a contribution score to each player based on their marginal impact on all possible coalitions. In machine learning, the “players” may correspond to input features, interactions, nodes, or edges, while the “game outcome” represents the model prediction. The Shapley value of a player is computed as the average marginal contribution of that player across all possible player subsets [61]. This formulation ensures that the contribution attribution is independent of the order in which the players are considered.

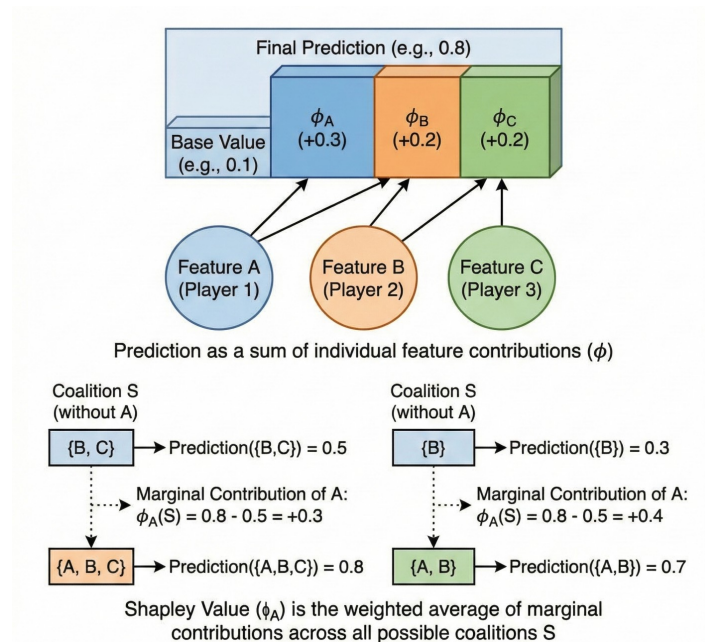


Figure 15: Intuitive illustration of Shapley value-based contribution attribution.

Shapley values satisfy several desirable theoretical properties that make them particularly attractive for model interpretation [62]. The efficiency property ensures that the sum of all Shapley values equals the total prediction. The symmetry property guarantees that players with identical contributions receive equal attribution. The dummy property assigns a zero contribution to players who do not influence the outcome. Finally, the additivity property allows the explanations to remain consistent when multiple models are combined. Together, these properties provide strong guarantees of fairness and consistency in attribution.

In recommender systems, Shapley values can be used to explain why a specific item is recommended to a user by quantifying the contribution of individual user–item interactions, features, or neighboring nodes. Compared with heuristic explanation methods, Shapley-based explanations are grounded in a formal mathematical framework, which increases their credibility and interpretability. This is particularly important in personalized recommendation scenarios, in which explanations must be faithful to the underlying model behavior [1], [37].

The application of Shapley values to complex machine learning models presents computational challenges. The exact computation of Shapley values requires evaluating the model over all possible subsets of players, which is computationally infeasible in large-scale systems. To address this issue, various approximation techniques have been proposed, including sampling-based methods and model-specific optimizations. These approximations trade off computational efficiency and explanation accuracy but can provide practical solutions for real-world applications [63].

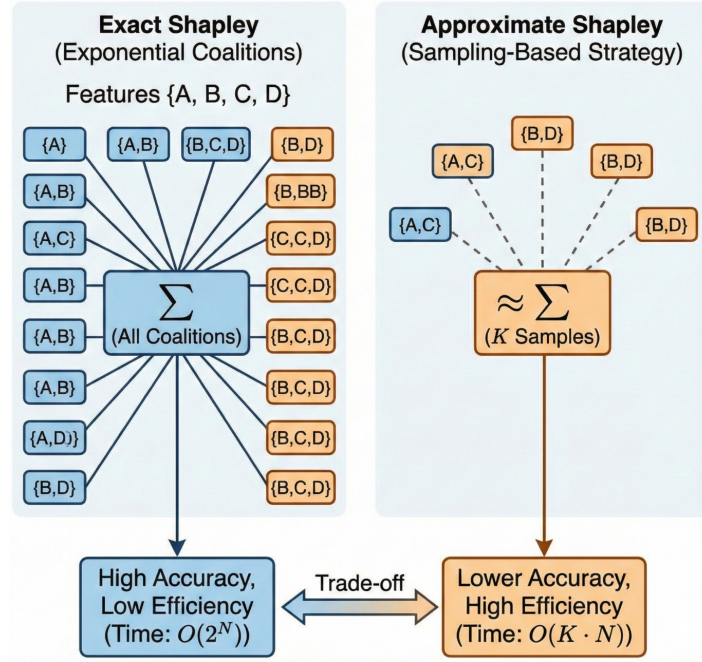


Figure 16: Comparison between exact and approximate computation of Shapley values.

The application of Shapley values to graph-based models introduces this additional complexity. In graph neural networks, predictions depend not only on individual nodes or features but also on the structure of the graph and the interactions between neighboring nodes [16]. As information propagates across multiple hops, isolating the contributions of a single node or edge is challenging. Nevertheless, Shapley values provide a flexible framework that can be adapted to graph settings by defining players as nodes, edges, or subgraphs, and carefully designing coalition sampling strategies.

Shapley-based explanation methods are particularly well-suited for graph-based recommender systems because they can capture both structural and relational contributions [64]. By attributing recommendation outcomes to specific interactions or graph components, these methods enable a deeper understanding of how collaborative signals influence predictions. Moreover, Shapley values can be integrated into the learning or inference process, allowing explanations to influence model behavior rather than being generated solely after predictions are made [2].

In summary, the Shapley values offer a theoretically grounded and flexible approach to model interpretability. Their strong axiomatic properties, applicability to complex models, and ability to provide fine-grained attribution render them well-suited for

explaining recommender systems and graph neural networks. However, their computational complexity and adaptation to graph-structured data require careful methodological design. These considerations motivate the development of specialized Shapley-based explanation frameworks for graph-based recommender systems, which are explored in subsequent sections.

1.9 Geometry in Representation Learning

Representation learning aims to map complex data objects into continuous vector spaces in which semantic relationships can be captured through geometric properties such as distance and direction. In recommender systems, representation learning is commonly used to embed users and items into a latent space in which proximity reflects the similarity of preferences. The choice of geometric space plays a crucial role in determining the effectiveness of the preservation and exploitation of these relationships by learning algorithms [65].

Most traditional recommender systems and neural models rely on Euclidean vector spaces for representation learning. Euclidean geometry is well-suited for modelling flat and homogeneous relationships, offering computational simplicity and intuitive distance measures. However, Euclidean spaces have a limited capacity to represent hierarchical and tree-like structures in data [66]. As the depth and complexity of hierarchical relationships increase, Euclidean embeddings tend to distort distances, making it difficult to simultaneously preserve both local and global structural properties [67].

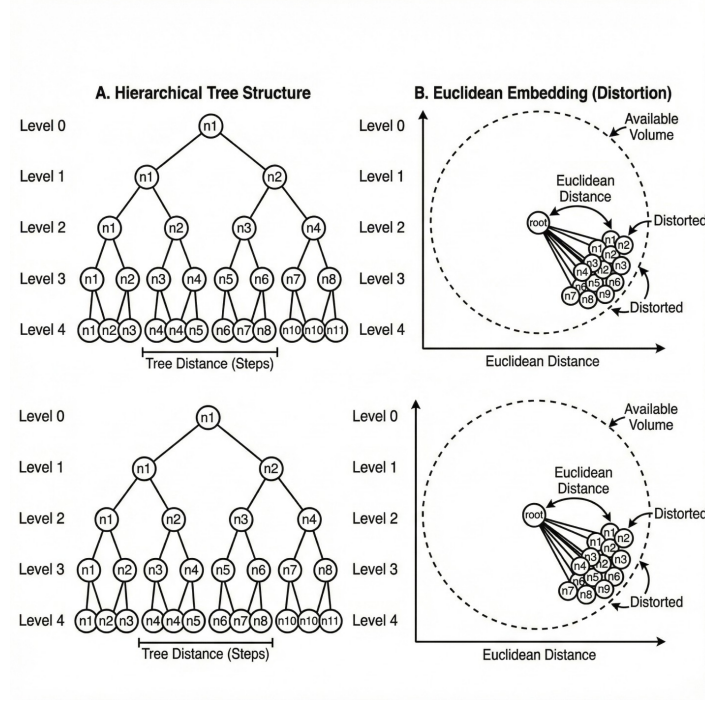


Figure 17: Distortion of hierarchical structures when embedded in Euclidean space.

Hierarchical relationships are common in many real-world recommendation systems. Examples include product category taxonomies, content genre hierarchies and organizational or semantic structures. In such settings, the number of nodes often increases exponentially with the hierarchy depth. Euclidean spaces grow polynomially with the radius, which limits their ability to embed hierarchical data without significant distortion. This mismatch motivates the exploration of alternative geometric spaces that better align with hierarchical structures [66].

Hyperbolic geometry has emerged as a promising solution for representing hierarchical and tree-like data. Unlike Euclidean space, hyperbolic space exhibits exponential volume growth, which naturally matches the hierarchical-graph structure. This property allows nodes at different hierarchy levels to be embedded at increasing distances from the origin while preserving relative relationships. Consequently, hyperbolic embeddings can represent hierarchies with lower distortion and greater efficiency [2].

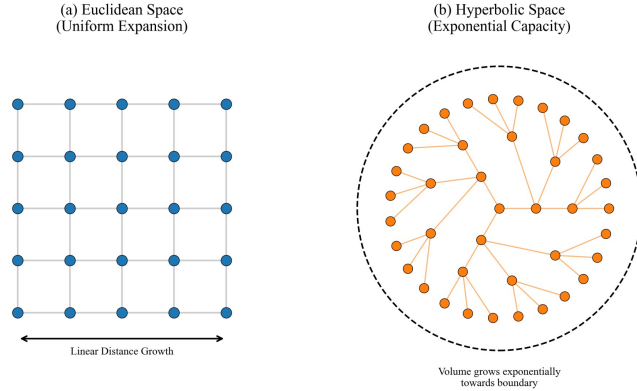


Figure 18: Comparison between Euclidean and hyperbolic spaces for hierarchical data representation.

Several models have been proposed to represent hyperbolic spaces in machine learning applications. Among these, the Poincaré ball model is one of the most widely used owing to its intuitive geometric interpretation and compatibility with gradient-based optimizations. In the Poincaré ball model, embeddings lie within a unit ball, and the distances increase rapidly as the points approach the boundary. This property enables the natural encoding of hierarchy depth, where more general concepts are placed closer to the center, and more specific concepts are located near the boundary.

Hyperbolic geometry has been successfully applied to graph embedding and neural networks in graph-based learning. Hyperbolic graph neural networks extend conventional GNNs by performing message-passing and aggregation operations directly in the hyperbolic space. This allows for better preservation of hierarchical relationships during the representation learning. For recommender systems that incorporate hierarchical information, such as category structures or multilevel item relationships, hyperbolic representations offer clear advantages over Euclidean representations.

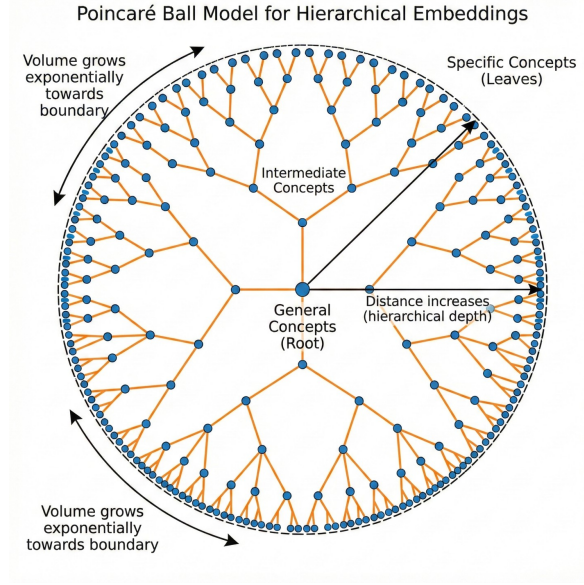


Figure 19: Illustration of hierarchical embeddings in the Poincaré ball model.

Despite their advantages, hyperbolic representations pose new challenges. Operations such as distance computation, aggregation, and optimization are more complex than those in Euclidean space and require specialized mathematical tools to perform. In addition, integrating hyperbolic representations with explainability methods is nontrivial because many existing XAI techniques are designed for Euclidean settings rather than hyperbolic ones. These challenges highlight the need for carefully designed models that balance geometric expressiveness, computational efficiency, and interpretability.

In summary, geometry plays a fundamental role in representation learning in recommender systems. Although Euclidean embeddings are suitable for flat relational structures, they are limited in their ability to represent such data. Hyperbolic geometry provides a more appropriate framework for such scenarios, enabling faithful and compact representations of hierarchical relationships. These properties motivate the use of hyperbolic graph neural networks in this study and provide the foundation for the hierarchical and explainable recommendation models introduced in later chapters.

Summary

This chapter presented the theoretical foundations and key concepts underlying the research conducted in this thesis. It began with an overview of recommender systems, outlining their objectives, characteristics, and typical problem formulations. Traditional recommendation techniques were then reviewed, and their main limitations were discussed, including data sparsity, cold-start issues, limited modelling capacity, and lack of transparency.

The chapter subsequently introduced graph-based recommender systems as a natural extension of classical approaches, highlighting their ability to model relational and high-order interaction structures. The fundamental principles of graph neural networks are presented, followed by a discussion of their application to recommendation tasks, with particular emphasis on collaborative filtering and specialized architectures such as Light Graph Convolutional Networks.

Explainable Artificial Intelligence has been introduced as a critical component for improving transparency, trust, and accountability in recommender systems. The theoretical foundations of Shapley values are presented as a principled approach to model explainability, along with a discussion of their advantages and computational challenges, particularly in graph-based settings. Finally, this chapter examines the role of geometry in representation learning, highlighting the limitations of Euclidean embeddings and motivating the use of hyperbolic geometry for modelling hierarchical structures.

Together, these concepts establish the theoretical basis for the explainable, adaptive, and geometry-aware recommender systems developed in this thesis. The following chapter reviews existing work in the literature and positions the contributions of this thesis with respect to current approaches in graph-based recommendation and explainable artificial intelligence.

2 Related Work and Research Positioning

Introduction

This chapter reviews existing research related to recommender systems, graph-based recommendation models, and explainable artificial intelligence. The objective of this chapter is to analyze prior work to identify the limitations of existing approaches and to clearly position the contributions of this thesis within the current state of the art.

This chapter first reviews traditional recommender systems and early graph-based approaches, followed by GNN-based recommendation models. It then examines existing explainable recommender systems and Shapley-value-based explanation methods, highlighting their strengths and limitations. Approaches for dynamic and adaptive recommendations are also discussed, as well as methods designed for hierarchical recommendation and hyperbolic representation learning.

Through this analysis, this chapter identifies several research gaps related to the lack of integrated explainability, limited adaptability to evolving user preferences, and insufficient handling of hierarchical structures in current recommender systems. These gaps motivate the unified explainable and geometry-aware graph-based frameworks proposed in this thesis.

2.1 Traditional Recommender Systems

Traditional recommender systems represent the earliest and most established approach to personalizing recommendations. Their primary objective is to estimate the relevance of items for users based on historical interaction data, which are typically represented as a user-item interaction matrix. Let $\mathcal{U} = \{u_1, \dots, u_{|\mathcal{U}|}\}$ denote the set of users and $\mathcal{I} = \{i_1, \dots, i_{|\mathcal{I}|}\}$ the set of items. User feedback is commonly modelled as a matrix $\mathbf{R} \in \mathbb{R}^{|\mathcal{U}| \times |\mathcal{I}|}$, where each entry r_{ui} represents the observed interaction between users u and item i . [68]

Collaborative filtering methods are the core of traditional recommendation systems. Neighborhood-based collaborative filtering predicts unknown interactions by exploiting the similarity patterns among users or items. In user-based collaborative filtering, the predicted preference of user u for item i is computed as a weighted aggregation of the

preferences expressed by users similar to u . Item-based collaborative filtering follows a similar principle by aggregating preferences over items similar to i . Although these methods are conceptually simple and intuitive, their performance strongly depends on reliable similarity estimation, which is challenging in sparse interaction settings.[69]

Matrix factorization techniques were introduced to overcome the limitations of neighborhood-based methods. These approaches approximate the interaction matrix $\mathbf{R}$ as the product of two low-rank matrices,

$$\mathbf{R} \approx \mathbf{P}\mathbf{Q}^\top,$$

where $\mathbf{P} \in \mathbb{R}^{|\mathcal{U}| \times d}$ and $\mathbf{Q} \in \mathbb{R}^{|\mathcal{I}| \times d}$ represent latent user and item embeddings in a d -dimensional space. The predicted interaction score between user u and item i is given by

$$\hat{r}_{ui} = \mathbf{p}_u^\top \mathbf{q}_i,$$

where $\mathbf{p}_u$ and $\mathbf{q}_i$ denote the corresponding latent vectors. Matrix factorization significantly improves scalability and accuracy by capturing latent preference patterns; however, it relies on linear interaction functions and shallow representations [69].

To increase the expressive power, neural-based recommender systems extend matrix factorization by replacing linear interaction functions with nonlinear neural architectures. Neural collaborative filtering models compute interaction scores using a function $f(\mathbf{p}_u, \mathbf{q}_i)$ parameterized by a neural network. While this formulation enables the modelling of complex preference patterns, it also increases model complexity and reduces interpretability. The learned representations are often difficult to analyze, and the contribution of individual interactions to a recommendation is unclear[3], [18].

Despite their effectiveness, traditional recommender systems have several limitations. Most approaches assume static user preferences and do not explicitly model temporal dynamics or evolving behaviors. Furthermore, the matrix-based formulation restricts the ability of these models to capture relational structures beyond direct user–item interactions, such as indirect dependencies, contextual relationships, and hierarchical information. From an explainability perspective, traditional methods either provide no explanations or rely on simple heuristics that do not accurately reflect the internal decision-making process of the model.[70]

These limitations highlight the need for recommendation frameworks that extend beyond matrix-based representations and linear interaction modelling. In particular, the

inability of traditional recommender systems to capture high-order interactions and relational structures has motivated the transition toward graph-based recommender systems, which model user–item interactions as graphs and are discussed in the following section.

2.2 Graph-Based Recommender Systems

Graph-based recommender systems were introduced to overcome the limitations of matrix-based recommendation models by explicitly modelling the relational structures between users and items. Instead of representing interactions as entries in a user–item matrix, these approaches formulate the recommendation problem as a graph, allowing indirect relationships and high-order connectivity patterns to be exploited.

Let $G = (V, E)$ denote a user–item interaction graph, where the node set $V = \mathcal{U} \cup \mathcal{I}$ consists of users and items, and the edge set $E \subseteq \mathcal{U} \times \mathcal{I}$ represents observed interactions. An edge $(u, i) \in E$ may be associated with a weight w_{ui} that indicates the interaction strength, such as a rating or confidence score. This bipartite graph formulation preserves the original interaction structure while enabling graph-based analysis and learning.

Early graph-based recommender systems relied on random walks and graph traversal techniques to estimate the relevance of items. In these approaches, recommendation scores are derived from the probability of reaching an item node starting from a user node through random walks on the graph. Methods such as personalized PageRank compute relevance scores by propagating preference signals through an interaction graph. Although effective in capturing indirect relationships, these methods require careful tuning and often lack scalability in large-sized graphs.[71]

Subsequently, graph embedding methods were proposed to learn the low-dimensional vector representations of nodes in interaction graphs. These approaches aim to map nodes to a latent space such that graph proximity is preserved. Formally, a graph embedding function $f : V \rightarrow \mathbb{R}^d$ is learned such that the distance between embeddings reflects the structural similarity. Recommendation scores can then be computed using the similarity measures between the user and item embeddings. Although graph embeddings improve scalability and capture higher-order relationships, they often rely on shallow models and fixed embedding objectives that are not directly optimized for the

recommendation tasks.[72]

Several hybrid approaches have combined graph modelling and matrix factorization. In these methods, graph-based regularization terms are added to the matrix factorization objective to enforce similarity constraints derived from the interaction graphs. A typical formulation can be expressed as

$$\min_{\mathbf{P}, \mathbf{Q}} \sum_{(u,i) \in E} (r_{ui} - \mathbf{p}_u^\top \mathbf{q}_i)^2 + \lambda \sum_{(v,w) \in E'} \|\mathbf{z}_v - \mathbf{z}_w\|^2,$$

where E' represents additional graph relations and $\mathbf{z}$ denotes the latent representations. Although such formulations integrate graph structures into latent factor models, they still rely on predefined similarity constraints and do not fully exploit nonlinear relational patterns.[1]

Despite their advantages, early graph-based recommender systems have several limitations. Many approaches treat graph structures as static and rely on handcrafted propagation rules or fixed embedding strategies. Moreover, the separation between representation learning and recommendation objectives can result in suboptimal performances. From an interpretability perspective, most graph-based methods provide limited insight into why specific items are recommended because relevance scores are often derived from complex propagation processes that are difficult to interpret [72].

These limitations have motivated the adoption of deep learning techniques for graph-structured data, leading to the development of recommender systems based on graph neural networks. By enabling end-to-end representation learning through iterative neighborhood aggregation, graph neural networks address many of the shortcomings of earlier graph-based approaches. However, as discussed in the following section, GNN-based recommender systems introduce new challenges related to model complexity and interpretability.

2.3 Graph Neural Network–Based Recommendation

Graph Neural Network (GNN)-based recommender systems extend graph-based recommendation models by enabling end-to-end representation learning on user–item interaction graphs. By iteratively aggregating information from neighboring nodes, GNNs can capture high-order collaborative signals that are difficult to model using traditional graph embedding or matrix factorization techniques. Consequently, GNN-based

approaches have become the dominant paradigm in modern recommender system studies [73].

Let $G = (V, E)$ denote a user-item interaction graph, where $V = \mathcal{U} \cup \mathcal{I}$. In a GNN-based recommender system, each node $v \in V$ is associated with a latent representation $\mathbf{h}_v^{(0)} \in \mathbb{R}^d$, which is iteratively updated through message passing. At layer k , the representation of node v is computed as

$$\mathbf{h}_v^{(k)} = \phi^{(k)} \left(\mathbf{h}_v^{(k-1)}, \bigoplus_{u \in \mathcal{N}(v)} \psi^{(k)}(\mathbf{h}_u^{(k-1)}) \right),$$

where $\mathcal{N}(v)$ denotes the neighborhood of node v , $\psi^{(k)}$ is a message function, $\bigoplus$ is a permutation-invariant aggregation operator, and $\phi^{(k)}$ is an update function. This general formulation captures a wide range of GNN architectures used for recommendations [74].

Early GNN-based recommender systems adapted graph convolutional networks to user-item graphs. These models propagate information across the interaction graph using normalized adjacency matrices, allowing user and item embeddings to incorporate signals from multi-hop neighbor nodes [18]. Although effective in capturing collaborative information, these approaches often include feature transformation and nonlinear activation functions that are not strictly necessary for recommendation tasks and may introduce additional noise [73].

To address these issues, simplified architectures tailored for collaborative filtering have been proposed in the literature. Light Graph Convolutional Networks (LightGCN) remove feature transformations and non-linearities, focusing solely on neighborhood aggregation [3]. In LightGCN, the embedding update at layer k is defined as

$$\mathbf{h}_v^{(k)} = \sum_{u \in \mathcal{N}(v)} \frac{1}{\sqrt{|\mathcal{N}(v)| |\mathcal{N}(u)|}} \mathbf{h}_u^{(k-1)}.$$

The final representation of a node is obtained by combining embeddings from multiple layers,

$$\mathbf{h}_v = \sum_{k=0}^K \alpha_k \mathbf{h}_v^{(k)},$$

where α_k denotes the contribution weight of layer k . Recommendation scores are typically computed using the inner product between user and item embeddings.

GNN-based recommender systems achieve strong empirical performance by exploiting high-order connectivity and collaboration signals. However, this performance comes at the cost of increased model complexity and reduced interpretability [74]. The iterative aggregation process makes it difficult to trace a recommendation back to specific user-item interactions or graph components. As information propagates across multiple hops, the contributions of individual neighbors become entangled, resulting in a black-box behavior [75].

Another limitation of existing GNN-based recommender systems is their static nature. Most models assume fixed interaction graphs and do not explicitly consider evolving user preferences or temporal dynamics during the inference process. Although temporal extensions exist, they often increase complexity without addressing underlying interpretability challenges [76].

From an explainability perspective, most existing GNN-based recommenders provide limited or no explanation of their prediction. Although post-hoc explanation methods can be applied, these approaches are often heuristic and do not faithfully reflect the internal reasoning of the model. Moreover, explainability is rarely integrated into training or inference processes, limiting its impact on model behavior [77], [78].

In summary, GNN-based recommender systems represent a significant advancement over traditional and early graph-based approaches by enabling expressive and scalable representation learning. Nevertheless, their black-box nature, limited adaptability, and lack of integrated explainability remain challenges to be addressed. These limitations motivate the development of explainable and adaptive GNN-based recommender systems, which form the core focus of this thesis and are discussed in the following sections.

2.4 Explainable Recommender Systems

Recommender systems aim to provide human-understandable justifications for recommendation outcomes, addressing concerns related to transparency, trust, and user acceptance. Early work in this area focused on generating intuitive explanations alongside recommendation results, often using heuristic or rule-based methods. Although such methods improve the user experience, they frequently lack a strong connection to the underlying recommendation model. [79], [80]

One of the earliest forms of explainable recommendation relies on content-based methods. In these approaches, recommendations are justified by highlighting the shared attributes between items previously consumed by a user and newly recommended items. Formally, given an item feature vector $\mathbf{x}_i$ and a user preference vector $\mathbf{p}_u$, the explanations are derived from the similarity function $s(\mathbf{p}_u, \mathbf{x}_i)$. Although these explanations are easy to interpret, they are limited to feature-based reasoning and are not applicable to collaborative filtering models that operate on latent representation.[81]

Neighborhood-based explanation methods provide another class of explainable recommender systems. These approaches explain a recommendation by referring to similar users or items that influence the prediction. For example, in user-based collaborative filtering, the predicted score for a user–item pair can be expressed as a weighted sum of neighbours’ preferences,

$$\hat{r}_{ui} = \sum_{v \in \mathcal{N}(u)} w_{uv} r_{vi},$$

where $\mathcal{N}(u)$ denotes the neighborhood of user u and w_{uv} represents the similarity weights. Explanations are then constructed by identifying the influential neighbors. Although this formulation provides a clear interpretive mechanism, it becomes unreliable in sparse settings and does not extend naturally to more complex models.[82]

Matrix factorization-based recommender systems pose additional challenges in terms of explainability. Because predictions are computed as inner products of latent vectors, $\hat{r}_{ui} = \mathbf{p}_u^\top \mathbf{q}_i$, the dimensions of these latent representations typically lack direct semantic meaning. Several approaches have attempted to interpret latent factors by associating them with observable features or constraining factor representations. However, these techniques often reduce model flexibility and provide only partial explanations that may not faithfully reflect the internal reasoning of the model.[83]

With the adoption of deep learning models, the explainability of recommender systems has become even more challenging. Neural collaborative filtering and graph neural network-based recommenders introduce multiple layers of nonlinear transformations and aggregation operations. Consequently, it becomes difficult to isolate the contribution of individual inputs or interactions to the recommendation outcome. Post-hoc explanation methods, such as gradient-based attribution or perturbation analysis, have been applied to neural recommenders; however, these approaches often suffer from instability and limited interpretability issues.[84]

Recent research has explored model-agnostic explanation frameworks that treat recommender systems as black boxes. These methods generate explanations by approximating the local behavior of a model around a specific prediction. Although such approaches can be applied to a wide range of models, they typically rely on simplifying assumptions and surrogate models that may not accurately capture the true decision-making process of the recommender system.[84]

A key limitation shared by most existing recommender systems is the separation between the recommendations and explanations. Explanations are often generated after a recommendation is produced and do not influence the model’s learning or inference processes. Consequently, the explanations may be unfaithful or inconsistent with the underlying model behavior. This limitation is particularly pronounced in graph- and GNN-based recommender systems, where complex relational dependencies are central.

In summary, existing recommender systems provide valuable insights into recommendation outcomes but are often limited by heuristic design, shallow interpretability, and lack of integration with complex models. These limitations highlight the need for explanation methods that are both theoretically grounded and tightly integrated into the recommendation process. Such methods should be capable of attributing recommendations to meaningful components while preserving model performance, motivating the exploration of the Shapley value-based explainability approaches discussed in the following section.

2.5 Shapley-Value-Based Explainability Methods

Shapley-value-based explanation methods have become one of the most influential approaches in explainable artificial intelligence because of their strong theoretical foundations. Derived from cooperative game theory, the Shapley value provides a principled method for attributing the outcome of a system to its individual components. In machine learning, they are widely used to quantify the contribution of input features, interactions, or structural elements to the prediction of a model. [85]

Formally, let $N = \{1, \dots, n\}$ denote a set of players and let $v : 2^N \rightarrow \mathbb{R}$ be a characteristic function that assigns a value to each coalition of players, such that: The Shapley

value of player i is defined as

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(n - |S| - 1)!}{n!} [v(S \cup \{i\}) - v(S)].$$

In machine learning applications, players may correspond to features, interactions, nodes, or edges, whereas the value function represents the model prediction or output score.

Shapley values have been adopted in several model-agnostic explanation frameworks, most notably SHAP, which provides efficient approximations of Shapley values for a wide range of models. These methods estimate feature contributions by sampling coalitions and evaluating marginal effects. Their popularity stems from the desirable axiomatic properties of Shapley values, including efficiency, symmetry, dummy, and additivity, which guarantee a fair and consistent attribution. [63]

In recommender systems, Shapley values have been applied to explain the predictions generated by collaborative filtering and matrix factorization models. In such settings, contributions can be assigned to input features, historical interactions, or neighboring users or items. By decomposing a predicted relevance score $\hat{r}_{ui}$ into additive components, Shapley-based explanations provide a more faithful interpretation than heuristic or rule-based methods. However, most existing applications are limited to relatively simple models and static interaction settings. [86]

Extending Shapley-value-based explainability to deep learning models introduces additional challenges. For neural recommender systems, the value function often depends on complex nonlinear transformations, rendering the exact computation of Shapley values infeasible. As a result, approximation methods based on Monte Carlo sampling or local surrogate models are commonly used in practice. Although these approximations improve scalability, they introduce variance and may reduce the stability of the explanation.[1]

The application of Shapley values to graph-based models and graph neural networks is an active area of research. In graph settings, the definition of players is less straightforward because the predictions depend on both the node features and graph structure. Existing approaches define players as nodes, edges, or subgraphs and evaluate their marginal contributions by perturbing the graph structure. Although these methods demonstrate the flexibility of Shapley values, they often incur high computational costs and rely on simplifying assumptions that limit their scalability.[58]

A key limitation of current Shapley-based explanation methods is their predominantly post-hoc nature. In most existing studies, the Shapley values are computed after model training and prediction, and the explanations do not influence the model’s behavior. Consequently, explainability remains decoupled from learning and inference, reducing its potential impact on robustness, adaptability, and its trustworthiness. This limitation is particularly evident in GNN-based recommender systems, where explanation methods rarely interact with message-passing processes.[87]

In summary, Shapley-value-based methods provide a theoretically sound and flexible framework for machine learning and recommendation systems. However, existing approaches are often limited by their computational complexity, approximation instability, and post hoc integration. A clear research gap remains in the development of Shapley-based explanation mechanisms that are tightly integrated into GNN-based recommender systems and actively influence recommendation outcomes. Addressing this gap is the central objective of the contributions presented in this thesis.

2.6 Dynamic and Adaptive Recommender Systems

Dynamic and adaptive recommender systems aim to model the temporal evolution of user preferences and item relevance. Unlike static recommender systems, which assume that user behavior is stationary over time, dynamic approaches explicitly account for changes in interaction patterns, preferences, and contextual factors. This line of research is motivated by the observation that user interests often evolve owing to trends, external influences, or changes in their personal circumstances.[88]

A common approach to modelling temporal dynamics is to extend the user–item interaction matrix with time-dependent information. Let $r_{ui}(t)$ denote the interaction between users u and item i at time t . Dynamic matrix factorisation models introduce time-varying latent representations, such that

$$r_{ui}(t) \approx \mathbf{p}_u(t)^\top \mathbf{q}_i(t),$$

where $\mathbf{p}_u(t)$ and $\mathbf{q}_i(t)$ represent the user and item embeddings at time t . These embeddings are typically regularized to enforce temporal smoothness and encourage gradual changes over time. While such models capture long-term preference drift, they often struggle to respond to rapid or short-term changes in the user’s behavior.[89]

Sequential recommender systems provide an alternative perspective by modelling user interactions as ordered sequences. In these approaches, the recommendation problem is formulated as predicting the next item in the user interaction sequence. Let $(i_1, i_2, \dots, i_T)$ denote a sequence of items with which a user interacts. The goal is to estimate the conditional probability [90]

$$P(i_{T+1} \mid i_1, i_2, \dots, i_T),$$

using models such as recurrent neural networks (RNNs) and attention-based architectures. Sequential models effectively capture short-term interests but often require large amounts of data and may overlook long-term preferences [91].

Recent studies have explored adaptive recommender systems that adjust model parameters and/or interaction weights during inference. These approaches aim to increase responsiveness to evolving user behaviors without requiring frequent retraining. However, adaptation mechanisms are often heuristic and may lack theoretical foundations. Moreover, the criteria used to adjust the model behavior are rarely interpretable, making it difficult to understand why certain interactions are emphasized or suppressed in the model.[92]

In graph-based recommender systems, temporal dynamics are commonly modelled by evolving interaction graphs. Let $G(t) = (V, E(t))$ denote a time-dependent interaction graph, where the edge set $E(t)$ changes as new interactions are established. Dynamic graph models update node representations based on recent changes in the neighborhood. Although these methods capture the temporal graph evolution, they often incur high computational costs and remain largely opaque in terms of interpretability. [2]

A fundamental limitation of existing dynamic and adaptive recommender systems is the separation between adaptation and explanation. Most approaches focus on improving prediction accuracy or responsiveness but do not provide clear explanations of the adaptive behavior. Consequently, it is difficult to determine which interactions or temporal patterns drive changes in the recommendations. This limitation is particularly problematic in GNN-based models, where adaptation mechanisms may further increase the model complexity and reduce the transparency.

In summary, dynamic and adaptive recommender systems represent an important step toward modelling evolving user preferences. However, existing approaches often rely on static or heuristic adaptation strategies and provide limited interpretability. There

remains a clear need for adaptive recommendation frameworks in which adaptation decisions are guided by principled and explainable mechanisms. This gap motivates the integration of Shapley-value-based contribution analysis into the recommendation process, enabling adaptive, responsive, and transparent behavior.

2.7 Hierarchical Recommendation and Hyperbolic Models

Hierarchical relationships are inherent in many real-world recommender system applications, such as product taxonomies, content categories, and semantic ontologies. In these settings, items are organized in multilevel structures, where higher-level nodes represent general concepts and lower-level nodes represent more specific entities [93]. Effectively modelling such hierarchies is important for capturing semantic relationships and improving recommendation quality, particularly in domains with structured item metadata.

Early approaches to hierarchical recommendation incorporated hierarchy information using regularization or constraint-based methods. In these models, hierarchical relations are used to enforce similarity between items belonging to the same category or propagate preferences across hierarchy levels. For example, given a hierarchy graph $H = (C, E_H)$ over categories, regularization terms are added to the recommendation objective to encourage consistency between parent and child representations. Although these approaches exploit hierarchical information, they remain limited by the linear modelling assumptions and shallow representations. [94]

Matrix factorization-based hierarchical recommenders extend standard latent factor models by introducing these constraints. Let $\mathbf{q}_i$ denote the latent embedding of item i and $\mathbf{q}_{c(i)}$ the embedding of its parent category **shot’2024**. A typical constraint enforces proximity between these representations,

$$\|\mathbf{q}_i - \mathbf{q}_{c(i)}\|^2 \leq \epsilon,$$

where ϵ controls the strength of the hierarchical dependency. Although such formulations improve hierarchical coherence, they rely on Euclidean geometry and struggle to efficiently represent the deep hierarchies.

Recent research has highlighted the limitations of Euclidean embedding spaces in modelling hierarchical data. In Euclidean space, the volume grows polynomially with the

radius, making it difficult to embed tree-like structures without distortion [95]. This limitation becomes more pronounced as the depth of the hierarchy increases. To address this issue, hyperbolic geometry has been proposed as a more suitable embedding space for hierarchical structures because of its exponential volume growth.

Hyperbolic space can be modelled using several equivalent representations, among which the Poincaré ball model is widely adopted in machine learning [96]. In this model, embeddings lie within a unit ball, and the distance between two points $\mathbf{x}$ and $\mathbf{y}$ is given by

$$d_{\mathbb{H}}(\mathbf{x}, \mathbf{y}) = \cosh^{-1} \left(1 + 2 \frac{\|\mathbf{x} - \mathbf{y}\|^2}{(1 - \|\mathbf{x}\|^2)(1 - \|\mathbf{y}\|^2)} \right).$$

This distance function allows nodes at deeper hierarchy levels to be placed closer to the boundary of the ball, whereas more general nodes remain near the center, naturally encoding the hierarchical relationships.

Hyperbolic embedding methods have been successfully applied to hierarchical graph representation learning. These approaches learn node embeddings that preserve hierarchical distances and ancestor–descendant relationships more faithfully than Euclidean embeddings. In recommender systems, hyperbolic representations are used to model category hierarchies, user preference generalization, and structured item relationships. However, many of these methods rely on static embedding and shallow optimization objectives. [97]

Recently, hyperbolic graph neural networks have been proposed to extend message passing and neighborhood aggregation to the hyperbolic space. In these models, node representations are updated using operations defined on hyperbolic manifolds, such as Möbius addition and exponential mapping. Let $\mathbf{h}_v^{(k)} \in \mathbb{H}^d$ denote the hyperbolic embedding of node v in layer k . Message passing is performed using geometry-aware aggregation functions that preserve manifold constraints. Although hyperbolic GNNs demonstrate strong performance on hierarchical data, their application in recommender systems remains relatively limited.[2]

Despite their advantages, existing hierarchical and hyperbolic recommender models have several limitations. Most approaches focus primarily on representation quality and recommendation accuracy, while providing limited interpretability. Explanations for recommendations in hyperbolic space are rarely addressed, and when they are, they are typically heuristic or post-hoc. Furthermore, hyperbolic models are often combined

with recommendation objectives in a loosely coupled manner without integrating explainability or adaptivity into the learning process.

In summary, hierarchical recommendation and hyperbolic models offer powerful tools for representing structured and hierarchical data in recommender systems. However, existing approaches largely overlook explainability and dynamic adaptation, particularly in graph neural network (GNN)-based frameworks. A clear research gap remains in the development of hyperbolic-graph-based recommender systems that provide principled and integrated explanations for their predictions. Addressing this gap is a key contribution of the present study.

2.8 Research Gap and Positioning

A review of the existing literature highlights significant progress in recommender systems, particularly through the adoption of graph-based models and graph neural networks. These approaches have substantially improved the ability of recommender systems to capture high-order interactions, relational structures, and collaborative signals. Simultaneously, research in explainable artificial intelligence has produced a wide range of techniques for interpreting machine learning models. However, despite these advances, several critical gaps remain at the intersection of graph-based recommendations, explainability, adaptivity, and hierarchical modelling.

First, although graph neural network-based recommender systems achieve strong predictive performance, they remain mostly opaque. Existing models primarily focus on accuracy optimization and treat interpretability as a secondary concern. When explanations are provided, they are typically generated using post-hoc methods that operate independently of the recommendation model. These explanations often rely on heuristic approximations and do not faithfully reflect the internal message-passing and aggregation mechanisms of graph neural networks. Consequently, current explainable approaches offer limited transparency and do not influence model behavior during training or inference.

Second, existing Shapley-value-based explainability methods, although theoretically grounded, are rarely integrated into recommender system pipelines. Most applications compute Shapley values after predictions are generated using approximation techniques that do not interact with the recommendation processes. Consequently, explainability

remains decoupled from learning and inference, reducing its potential to improve the robustness, adaptability, and trust in AI systems. There is a clear lack of frameworks in which Shapley-based contribution analysis is used not only to explain recommendations but also to guide the model behavior.

Third, current recommender systems exhibit limited adaptability to evolving user preferences and needs. Dynamic and temporal models attempt to address preference drift; however, they often rely on static update schedules, heuristic adaptation rules, or complex sequential architectures that increase the model opacity. The literature lacks principled adaptive mechanisms that explicitly quantify the contribution of individual interactions over time and use this information to adjust recommendations in a transparent and interpretable manner.

Fourth, hierarchical recommendation and hyperbolic representation learning have demonstrated strong potential for modelling structured data. However, existing approaches primarily focus on representation quality and recommendation accuracy, while largely overlooking explainability. Hyperbolic graph neural networks are rarely combined with explainable artificial intelligence techniques, and when explanations are provided, they are typically post-hoc and disconnected from the geometric structure of the model. This limits the practical interpretability of hierarchical recommendations.

Taken together, these gaps reveal the absence of a unified framework that simultaneously addresses explainability, adaptivity, and hierarchical modelling in graph-based recommendation systems. Existing research tends to address these aspects in isolation, resulting in fragmented solutions that fail to exploit their potential.

This thesis positions itself at the intersection of these research areas by proposing recommender systems in which Shapley-value-based explanations are tightly integrated into the recommendation process. Unlike existing approaches, explainability is treated as a core component rather than an auxiliary feature. Shapley values are used not only to generate faithful explanations but also to guide adaptive behavior by dynamically adjusting the influence of user-item interactions.

Furthermore, this study extends XRec to hierarchical settings by integrating Shapley-based explainability with hyperbolic graph neural networks. By jointly modelling hierarchical structures and interaction contributions within a principled geometric framework, the proposed approaches provide transparent and theoretically grounded explanations that align with the model’s internal reasoning process.

In summary, this study advances the state of the art by addressing the lack of integrated explainability, limited adaptivity, and insufficient hierarchical modelling in existing recommendation systems. By unifying graph neural networks, Shapley-value-based explainability, adaptive mechanisms, and hyperbolic geometry, this thesis proposes a novel and coherent framework for transparent, robust, and structure-aware recommendations.

Summary

This chapter reviewed existing research related to recommender systems, with a particular focus on graph-based recommendation models, GNN-based approaches, and explainable artificial intelligence techniques. Traditional recommender systems and early graph-based methods are discussed to highlight the evolution of recommendation models and the motivation for more expressive frameworks.

This chapter examined GNN-based recommender systems, emphasizing their ability to capture high-order interactions and relational structures while identifying their main limitations, particularly in terms of interpretability and adaptability. Existing recommender systems were reviewed, revealing that most explanation methods are heuristic, post hoc, or weakly connected to the underlying recommendation models. Shapley-value-based explainability methods were analyzed as a theoretically grounded alternative, but their limited integration into recommender system pipelines was identified as a key limitation of this approach.

Dynamic and adaptive recommender systems were also discussed, highlighting the lack of principled and interpretable adaptation mechanisms in existing approaches. In addition, this chapter reviewed hierarchical recommendation and hyperbolic models, noting that while hyperbolic geometry is well-suited for modelling hierarchical structures, explainability in such settings remains largely unexplored.

Based on this analysis, this chapter identifies clear research gaps related to the absence of unified frameworks that combine graph-based recommendations, explainability, adaptability, and hierarchical modelling. These gaps motivate the contributions of this study, which aims to integrate Shapley-value-based explainability directly into graph neural network-based recommender systems, extend adaptive recommendation mechanisms, and support hierarchical structures using the hyperbolic geometry.

The following section introduces the proposed XG-based recommendation framework and details the methodology used to integrate the explainability into the recommendation process.

3 Experimental Protocol and Evaluation Framework

Introduction

This chapter defines the experimental protocol and evaluation framework used in this thesis. Its purpose is to establish a unified and consistent basis for all experimental studies presented in subsequent chapters. By centralizing dataset descriptions, data splitting strategies, baseline methods, evaluation metrics, and training procedures, this chapter ensures that the experimental results are directly comparable and reproducible across different contributions.

The research presented in this thesis is organized into multiple paper-based chapters, each introducing a distinct methodological contribution. Although these contributions address different aspects of graph-based and explainable recommender systems, they rely on a common experimental setting for evaluation. Repeating the same experimental details in each chapter would lead to unnecessary redundancy and increase the risk of inconsistencies in the manuscript. Instead, this chapter serves as a reference point for all experimental assumptions and implementation choices.

The experimental protocol described here was designed to reflect realistic recommendation scenarios and support fair comparisons with established baseline methods. Particular attention is paid to preventing data leakage, ensuring a balanced evaluation across users, and adopting widely accepted metrics for top- N recommendation tasks. All hyperparameter selection and optimization procedures were defined consistently to isolate the effect of the proposed models from variations in the experimental setup.

The remainder of this paper is organized as follows. Section 3.1 describes the datasets used for the experimental evaluation. Section 3.2 presents the data-splitting strategies adopted for training, validation, and testing. Section 3.3 introduces the baseline recommendation methods used for the comparison. Section 3.4 defines the evaluation metrics employed to assess the recommendation quality. Section 3.5 details the training protocol and optimization procedures, and Section 3.6 summarizes the key implementation details relevant to reproducibility. Finally, Section 3.7 concludes the chapter and clarifies how this protocol is referenced in the subsequent experimental chapters.

All experimental results reported in Chapters 4, 5, and 6 follow the protocol defined in this chapter, unless explicitly stated otherwise.

3.1 Datasets

This study evaluated the proposed recommender system models on three widely used public benchmarks: MovieLens 100K, MovieLens 1M, and Amazon Product Reviews. These datasets differ in terms of scale, sparsity, and application domain, enabling a comprehensive evaluation of graph-based and explainable recommender systems under diverse conditions.

All three datasets consist of user–item interaction records and are suitable for collaborative filtering evaluation. Although they provide explicit ratings, the experiments in this thesis primarily adopt an implicit feedback formulation that reflects realistic recommendation scenarios and supports consistent top- N evaluation across datasets.

3.1.1 Dataset Structure

Each dataset is composed of interaction records represented as tuples of the form

$$(u, i, r_{ui}, t_{ui}),$$

where u denotes a user, i denotes an item, r_{ui} represents an explicit rating or review score, and t_{ui} corresponds to the timestamp of the interaction. The presence of an interaction indicates positive user feedback, whereas missing entries are treated as unobserved feedback.

The MovieLens datasets focus on the movie recommendation domain and exhibit relatively dense interaction patterns compared to large-scale e-commerce data. In contrast, the Amazon Product Reviews dataset spans multiple product categories and is characterized by extreme sparsity and large-scale user and item populations.

3.1.2 MovieLens Datasets

MovieLens 100K [98] and MovieLens 1M [98] are curated benchmarks released by the GroupLens research group. Both datasets contain explicit ratings collected from users

who voluntarily provided feedback on the movies. The MovieLens100K dataset is relatively small and well-balanced, making it suitable for controlled experimentation and method validation. MovieLens1M provides a large-scale setting that enables the evaluation of scalability and robustness [79].

Table 1: Structural characteristics of the MovieLens datasets.

Dataset	MovieLens 100K	MovieLens 1M
Number of users	943	6,040
Number of movies	1,682	3,706
Number of interactions	100,000	1,000,209
Feedback type	Explicit ratings	Explicit ratings
Rating scale	1–5	1–5
Temporal information	Yes	Yes
Sparsity level	Moderate	Moderate

The structured and relatively dense nature of the MovieLens datasets makes them well-suited for evaluating collaborative filtering models that rely on interaction patterns rather than auxiliary content features.

3.1.3 Amazon Product Reviews Dataset

The Amazon Product Reviews dataset provides a large-scale benchmark for recommendations in e-commerce environments. It contains user reviews and ratings across a wide range of product categories, such as electronics, books, and household products. Compared to MovieLens, this dataset exhibits significantly higher sparsity and greater heterogeneity in user behavior.

In this study, Amazon review data were used to construct user–item interaction graphs, in which observed reviews were treated as positive interactions. The scale and sparsity of this dataset make it particularly suitable for evaluating the robustness and scalability of GNN-based recommender systems. [21]

3.1.4 Implicit Feedback Conversion

Although all datasets provide explicit ratings, the experiments primarily adopted an implicit feedback formulation. An interaction is considered positive if a user has rated

Table 2: Structural characteristics of the Amazon Product Reviews dataset.

Characteristic	Amazon Reviews
Number of users	Large-scale (millions)
Number of items	Large-scale (millions)
Number of interactions	Tens of millions
Feedback type	Explicit ratings and reviews
Rating scale	1–5
Temporal information	Yes
Domain diversity	Multiple product categories
Sparsity level	High

or reviewed an item, whereas all unobserved user–item pairs are treated as unknown rather than negative feedback. This conversion enables consistent evaluation across datasets and aligns with the common practice in top- N recommendation tasks.

Any dataset-specific preprocessing or filtering steps, such as minimum interaction thresholds or category selection, are explicitly described in the corresponding experimental chapters when applicable.

Together, these datasets provide complementary evaluation settings that allow for the systematic analysis of graph-based, adaptive, and explainable recommender systems across different domains and scales.

3.2 Data Splitting Strategy

This section describes the data-splitting strategy adopted for all the experimental evaluations reported in this thesis. A unified splitting protocol was used across datasets and models to ensure a fair comparison and prevent information leakage. The strategy is designed to reflect realistic recommendation scenarios while remaining consistent with the experimental setups employed in the research papers on which this thesis is based.

3.2.1 User-level splitting.

All datasets were split at the user level to ensure that every user appearing in the test set was also present in the training data. This constraint is essential for collaborative filtering models that cannot generate recommendations for users without a prior interaction history. Users with insufficient interactions to support reliable splitting may be excluded following standard preprocessing procedures described in Section 3.1.

3.2.2 Temporal ordering.

Whenever timestamp information was available, the interactions were ordered chronologically for each user. This temporal ordering is used to simulate realistic recommendation settings in which future interactions must be predicted based on past behavior. In particular, the most recent interactions of each user are reserved for evaluation, whereas earlier interactions are used for training.

3.2.3 Train-validation-test split.

For each user, the interactions were divided into three disjoint sets: a training set, a validation set, and a test set. The training set contains the majority of historical interactions and is used for learning the model. The validation set was used for hyperparameter tuning and early stopping, whereas the test set was reserved exclusively for the final evaluation. This split was applied consistently across all datasets to ensure the comparability of the results.

3.2.4 Leave-one-out evaluation.

In line with the common practice in top- N recommendation, a leave-one-out evaluation protocol is employed for test evaluation. Specifically, for each user, a single interaction is held out as the test instance, and the model is evaluated based on its ability to rank this item among a set of unobserved items. When a validation set is used, a similar leave-one-out strategy is applied to the validation interactions.

3.2.5 Negative sampling for evaluation.

During evaluation, the held-out positive item for each user is ranked against a set of randomly sampled negative items with which the user has not interacted. This strategy reduces the computational cost while approximating the full ranking task. The same negative sampling procedure and random seeds were used across the models to ensure a fair comparison.

3.2.6 Dataset-specific considerations.

Although the general splitting strategy is shared across datasets, minor adjustments are applied when required by the dataset characteristics. For example, the scale and sparsity of the Amazon product review dataset may necessitate category-specific filtering or interaction thresholds to ensure a stable evaluation. Such adjustments are explicitly stated in the corresponding experimental chapter when applicable, and do not alter the core splitting principles described above.

This unified data-splitting strategy was applied consistently across all experimental chapters in this thesis. By combining temporal ordering, user-level constraints, and leave-one-out evaluation, the protocol supports a realistic assessment of recommendation performance while ensuring reproducibility and fair comparison across different models and datasets.

3.3 Baseline Methods

This section describes the baseline methods used for the experimental comparison in this study. The selected baselines cover classical collaborative filtering techniques, matrix factorization models, neural recommendation methods, graph-based approaches, and X-GNN models. Together, they provide a comprehensive reference set for evaluating the proposed methods.

3.3.1 User-Based Collaborative Filtering (UBCF)

User-based collaborative filtering predicts user preferences by aggregating the preferences of users with similar interests. For a target user u and item i , the predicted score is computed as

$$\hat{r}_{ui} = \frac{\sum_{v \in \mathcal{N}_u} s(u, v) r_{vi}}{\sum_{v \in \mathcal{N}_u} |s(u, v)|},$$

where $\mathcal{N}_u$ denotes the neighborhood of users similar to u , and $s(u, v)$ is a similarity measure, typically the cosine similarity or Pearson correlation. UBCF relies on explicit similarity computations and is sensitive to data sparsity. [99]

3.3.2 Item-Based Collaborative Filtering (IBCF)

Item-based collaborative filtering predicts preferences based on the similarity between items rather than users. The predicted score is given by

$$\hat{r}_{ui} = \frac{\sum_{j \in \mathcal{N}_i} s(i, j) r_{uj}}{\sum_{j \in \mathcal{N}_i} |s(i, j)|},$$

where $\mathcal{N}_i$ denotes items similar to item i . Compared with UBCF, IBCF often provides more stable similarity estimates in large-scale settings. [4]

3.3.3 Matrix Factorisation (MF)

Matrix factorization models represent users and items as latent vectors. The predicted interaction score is defined as

$$\hat{r}_{ui} = \mathbf{p}_u^\top \mathbf{q}_i,$$

where $\mathbf{p}_u$ and $\mathbf{q}_i$ denote the user and item embeddings, respectively. Model parameters are learned by minimizing the reconstruction or ranking loss with regularization.[100]

3.3.4 Singular Value Decomposition (SVD)

SVD extends the basic matrix factorization by incorporating user and item bias terms. The prediction function is

$$\hat{r}_{ui} = \mu + b_u + b_i + \mathbf{p}_u^\top \mathbf{q}_i,$$

where μ is the global mean rating and b_u and b_i are user and item biases, respectively. This formulation improves the modelling of systematic rating tendencies. [4]

3.3.5 SVD++

SVD++ further extends SVD by incorporating implicit feedback. The user representation is augmented as

$$\mathbf{p}'_u = \mathbf{p}_u + \frac{1}{\sqrt{|\mathcal{I}_u|}} \sum_{j \in \mathcal{I}_u} \mathbf{y}_j,$$

where $\mathcal{I}_u$ denotes items previously interacted with by user u . This allows the model to capture both explicit and implicit preference signals in the data.[4]

3.3.6 Neural Collaborative Filtering (NCF)

NCF replaces the inner product in matrix factorisation with a neural interaction function. Given embeddings $\mathbf{p}_u$ and $\mathbf{q}_i$, the predicted score is

$$\hat{r}_{ui} = f_{\theta}(\mathbf{p}_u, \mathbf{q}_i),$$

where $f_{\theta}(\cdot)$ is the multilayer perceptr. This approach enables nonlinear modelling of user–item interactions but does not explicitly exploit the graph structure. [79]

3.3.7 Light Graph Convolutional Network (LightGCN)

LightGCN is a graph-based collaborative filtering model that learns embeddings by propagating information on a user–item interaction graph. Node embeddings are updated as

$$\mathbf{h}_v^{(k)} = \sum_{u \in \mathcal{N}(v)} \tilde{A}_{vu} \mathbf{h}_u^{(k-1)},$$

The final representation was obtained by aggregating the embeddings across the layers. LightGCN removes feature transformations and non-linearities to focus on collaborative-signal propagation. [3]

3.3.8 Hyperbolic Graph Convolutional Network (Hyperbolic GCN)

The hyperbolic GCN extends graph convolution operations to the hyperbolic space to better model hierarchical relationships. Node embeddings $\mathbf{h}_v \in \mathbb{H}^d$ are updated using geometry-aware aggregation and mapping functions defined on a hyperbolic manifold. This approach is particularly suited for recommendation settings that involve hierarchical item structures. [2]

3.3.9 GNNShap

GNNShap is an explainability method for graph neural networks that applies Shapley value principles to attribute prediction outcomes to nodes or edges in the graph. Given a trained GNN, explanations are generated by estimating the marginal contribution of the graph components to the predicted score. GNNShap operates as a post-hoc explanation framework and does not influence the model training. [2]

3.3.10 Model-Agnostic Graph Explanations (MAGE)

MAGE is a model-agnostic explanation method for graph-based models that identifies important subgraphs by perturbing the graph structures and analyzing the changes in the model output. Similar to GNNShap, MAGE provides post-hoc explanations and treats the underlying recommendation system as a black box. [18]

These baseline methods collectively represent the spectrum of classical, neural, graph-based, and explainable recommendation approaches used in the experimental evaluation. All baselines were implemented and evaluated following the unified experimental protocol defined in this chapter to ensure a fair and consistent comparison with the proposed models.

3.4 Evaluation Metrics

This section defines the evaluation metrics used to assess the performance of the recommender systems in this thesis. All models were evaluated in a top- N recommendation

setting, where the objective was to rank relevant items highly among a list of recommended items for each user. The metrics described below are widely adopted in the recommender system literature and provide complementary perspectives on ranking quality.

Let $\mathcal{U}$ denote the set of users in the evaluation. For each user $u \in \mathcal{U}$, let $\mathcal{R}_u$ denote the ranked list of the top- N items recommended by the model, and $\mathcal{T}_u$ denote the set of ground-truth relevant items for user u in the test set.

3.4.1 Precision@ N .

Precision@ N measures the proportion of relevant recommended items. It is defined as

$$\text{Precision@}N = \frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} \frac{|\mathcal{R}_u \cap \mathcal{T}_u|}{N}.$$

This metric evaluates the accuracy of the top-ranked recommendations and reflects the number of recommended items relevant to the user. [37]

3.4.2 Recall@ N .

Recall@ N measures the proportion of relevant items that are successfully recommended within the top- N list. It is defined as

$$\text{Recall@}N = \frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} \frac{|\mathcal{R}_u \cap \mathcal{T}_u|}{|\mathcal{T}_u|}.$$

Recall@ N emphasizes the model’s ability to recover relevant items and is particularly informative when users have multiple relevant test items in the dataset. [18]

3.4.3 Normalised Discounted Cumulative Gain (NDCG@ N).

NDCG@ N accounts for both the relevance and rank position of recommended items, assigning higher importance to items appearing earlier in the recommendation list. For a user u , the discounted cumulative gain is defined as

$$\text{DCG@}N = \sum_{k=1}^N \frac{\mathbb{I}(i_k \in \mathcal{T}_u)}{\log_2(k+1)},$$

where i_k denotes the item ranked at position k and $\mathbb{I}(\cdot)$ is the indicator function. The normalised version is obtained by dividing by the ideal DCG,

$$\text{NDCG@}N = \frac{\text{DCG@}N}{\text{IDCG@}N},$$

where $\text{IDCG@}N$ represents the maximum possible DCG for user u . $\text{NDCG@}N$ rewards models that rank relevant items higher on the recommendation list. [1]

3.4.4 Averaging across users.

All metrics were computed independently for each user and then averaged across the user set. This user-centric evaluation ensures that the results are not dominated by highly active users and reflect the typical usage scenario of recommender systems.

3.4.5 Evaluation protocol.

The metrics defined above are computed using the same candidate item set and negative sampling strategy for all models, following the data splitting protocol described in Section 3.2. This ensured that performance differences reflected modelling choices rather than variations in the evaluation procedure.

These evaluation metrics collectively provide a balanced assessment of the recommendation performance, capturing both the accuracy and ranking quality of the proposed models. All experimental results reported in subsequent chapters use these metrics, unless otherwise stated.

3.5 Training Protocol

This section describes the training protocol used for all the recommender system models evaluated in this thesis. A unified training procedure was adopted to ensure a fair comparison between methods and to isolate the impact of modelling choices from differences in the optimization strategy.

3.5.1 Learning objective.

All models were trained to optimize a ranking-oriented objective suitable for implicit feedback recommendations. Given a user $u \in \mathcal{U}$, a positive item $i \in \mathcal{I}$ such that (u, i) is an observed interaction, and a negative item $j \in \mathcal{I}$ such that (u, j) is unobserved, the optimisation objective enforces

$$\hat{r}_{ui} > \hat{r}_{uj},$$

where $\hat{r}_{ui}$ and $\hat{r}_{uj}$ denote predicted relevance scores. In practice, this is implemented using the Bayesian Personalised Ranking (BPR) loss,

$$\mathcal{L}_{\text{BPR}} = - \sum_{(u,i,j)} \log \sigma(\hat{r}_{ui} - \hat{r}_{uj}),$$

with $\sigma(\cdot)$ denoting the sigmoid function. This loss directly optimizes the ranking of relevant items and is well-suited for top- N recommendation tasks.

3.5.2 Negative sampling.

For each positive user–item interaction, negative samples are drawn uniformly from the set of items with which the user has not interacted. Negative sampling is performed dynamically during training to expose the model to diverse sets of non-interacted items. The same negative sampling strategy was consistently applied across all models to ensure comparability.

3.5.3 Optimisation.

Model parameters are optimised using stochastic gradient-based methods. Mini-batch training was employed, and the gradients were computed via backpropagation. Adaptive optimization algorithms, such as Adam, are used to accelerate convergence and stabilize training. Learning rates and other optimizer hyperparameters were selected based on the validation performance.

3.5.4 Regularisation.

To prevent overfitting, ℓ_2 regularization is applied to the model parameters, including user and item embeddings. The strength of regularization is controlled by a hyperparameter and tuned using a validation set. Dropout and other stochastic regularization techniques are not employed unless explicitly required by a specific model architecture.

3.5.5 Early stopping and model selection.

The training was monitored using the validation metrics defined in Section 3.4. Early stopping was applied when the validation performance ceased to improve for a pre-defined number of epochs. The model checkpoint that achieved the best validation performance was selected for the final evaluation of the test set.

3.5.6 Initialisation and reproducibility.

The model parameters were initialized using standard random initialization schemes. To reduce the impact of randomness, experiments were conducted using fixed random seeds whenever possible. When appropriate, the results were averaged over multiple runs to ensure stability.

This training protocol was uniformly applied to all models evaluated in this study, unless explicitly stated otherwise. By enforcing consistent optimization and regularization strategies, the protocol ensures that the observed performance differences are attributable to the model design rather than variations in the training procedure.

3.6 Implementation Details

This section summarizes the implementation-related aspects that are common to all the experimental studies presented in this thesis. The purpose of this section is to clarify practical considerations relevant to reproducibility without repeating model- or dataset-specific details described in the corresponding chapters.

All recommender system models evaluated in this study were implemented using a unified experimental framework to ensure consistency across methods. The same data

preprocessing pipeline, interaction graph construction procedure, and evaluation code were shared across the experiments, unless explicitly stated otherwise. This design choice minimizes implementation-induced variability and supports a fair comparison between the models.

Model training and evaluation followed the protocols defined in Sections 3.2–3.5. In particular, the same data splits, negative sampling strategy, optimization settings, and evaluation metrics were applied consistently across all baseline and proposed methods. Care was taken to ensure that no test or validation information was used during training.

To reduce the impact of randomness, experiments were conducted with controlled initialization whenever possible. When stochastic components are present in the training or sampling procedures, random seeds are fixed to ensure the repeatability of the results. In cases where multiple runs were performed, the results were averaged to improve stability.

All implementation choices prioritized correctness, reproducibility, and alignment with standard practices in recommender system research. Additional model-specific implementation details, when required, are provided in the corresponding experimental sections.

Summary

This chapter presents the experimental protocol and evaluation framework adopted throughout this thesis. Its purpose is to establish a unified and consistent basis for all experimental studies, ensuring fair comparison, reproducibility, and methodological clarity across different contributions to the field.

This chapter describes the datasets used for evaluation, highlighting their structural characteristics and domain diversity. A common data-splitting strategy was introduced to prevent information leakage and reflect realistic recommendation scenarios. The baseline methods selected for comparison were formally defined, covering classical collaborative filtering techniques, matrix factorization models, neural approaches, graph-based methods, and explainable graph neural network models.

Standard top- N recommendation metrics were presented to assess ranking perfor-

mance, and a unified training protocol was defined to ensure consistent optimization and regularization across the models. Finally, implementation details relevant to reproducibility are summarized, clarifying how experimental consistency is maintained throughout the thesis.

All experimental results reported in the subsequent chapters follow the protocol defined in this chapter, unless explicitly stated otherwise. By centralizing experimental assumptions and procedures, this chapter provides a reliable foundation for evaluating the proposed graph-based, adaptive, and explainable recommender systems introduced in the remainder of this thesis.

4 Graph Neural Network–Based Collaborative Filtering for Movie Recommendation

Introduction

Collaborative filtering is one of the most widely adopted paradigms in recommender systems, relying on the assumption that users with similar interaction histories are likely to exhibit similar future preferences. Classical collaborative filtering approaches, including neighborhood-based methods and latent factor models, have achieved strong empirical performance across a wide range of applications. However, these methods typically represent user–item interactions in matrix form, which imposes structural limitations on how collaborative information can be modelled.[4]

Matrix-based formulations treat interactions as independent observations conditioned on latent factors, thereby overlooking the relational nature of the recommendation data. In practice, user preferences are influenced not only by direct interactions but also by indirect relationships mediated by other users and items. Capturing such higher-order collaborative signals is challenging within traditional matrix factorization frameworks.

To address these limitations, graph-based representations of recommender systems have gained significant attention. By modelling users and items as nodes in a bipartite graph with edges corresponding to observed interactions, graph-based approaches provide a natural and expressive representation of collaborative data. This formulation explicitly defines the relational structure of the recommendation problem and enables the exploitation of multi-hop connectivity patterns.[101]

Early graph-based recommender systems relied on random walks, diffusion processes, or heuristic neighborhood aggregation to propagate preference information across the interaction graph. Although these methods demonstrate the potential of graph modelling, they are typically based on fixed propagation rules and lack the capacity to learn task-specific aggregation strategies. Consequently, their flexibility and scalability are limited.[102]

Graph neural networks extend graph-based recommendations by introducing learnable message-passing mechanisms that iteratively update node representations based on their local neighborhoods. Through successive layers of aggregation, each user or item

representation can incorporate information from nodes that are increasingly distant in the interaction graph. This enables the modelling of complex collaborative relationships that are inaccessible to both neighborhood-based and matrix factorization methods.[102]

Despite their promise, existing graph neural network approaches to collaborative filtering are often designed with a primary focus on accuracy. Many models adopt increasingly complex architectures without explicitly addressing interpretability, adaptivity, or the impact of individual interactions on the recommendation outcomes. Moreover, the behavior of these models is frequently difficult to analyze, as the contribution of specific edges or neighbors to a given prediction remains implicit.[75]

The research relevance of graph neural networks for collaborative filtering lies not only in their improved representational capacity but also in the opportunities they create for more principled modelling of relational influence. By framing collaborative filtering as a graph learning problem, graph neural networks provide a foundation for building adaptive and explainable recommendation mechanisms.

This chapter builds on this perspective by introducing a graph neural network framework specifically tailored for collaborative filtering. The proposed model focuses on leveraging the relational structure while maintaining architectural simplicity and serves as the foundational component for the adaptive and explainable extensions developed in the subsequent chapters.

4.1 Proposed Graph Neural Network for Collaborative Filtering

This section introduces the proposed graph neural network (GNN) framework for collaborative filtering. The objective is to model user preferences by explicitly exploiting the relational structure of user-item interactions while maintaining a simple and interpretable architecture that can serve as a foundation for subsequent adaptive and explainable extensions.

4.1.1 Problem Formulation

Let $\mathcal{U}$ denote the set of users and $\mathcal{I}$ the set of items. Observed interactions between users and items are represented as a set

$$\mathcal{E} = \{(u, i) \mid u \in \mathcal{U}, i \in \mathcal{I}\},$$

where each pair (u, i) indicates an interaction, such as a rating, click, or consumption event. The goal of collaborative filtering is to learn a scoring function

$$\hat{r}_{ui} = f(u, i),$$

which estimates the relevance of item i for user u and can be used to generate a ranked list of recommendations.

Rather than treating interactions as independent observations, the proposed approach models collaborative filtering as a graph-learning problem. User and item representations are jointly learned by propagating information through the interaction structure. [103]

4.1.2 Interaction Graph Construction

User-item interactions are modelled as a bipartite graph

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}),$$

where the node set is defined as $\mathcal{V} = \mathcal{U} \cup \mathcal{I}$. An undirected edge connects a user node $u \in \mathcal{U}$ and an item node $i \in \mathcal{I}$ if an interaction between them has been observed.

This bipartite graph representation preserves the relational structure of the data and enables the propagation of collaborative signals across users and products. No additional side information is required, allowing the model to focus exclusively on the collaborative filtering signals derived from the interaction patterns. [104]

4.1.3 Graph Neural Network Propagation

Each node $v \in \mathcal{V}$ is associated with a latent embedding $\mathbf{h}_v^{(0)} \in \mathbb{R}^d$, which is initialized as a trainable parameter. The GNN updates the node representations using a message-passing mechanism that aggregates information from neighboring nodes.

At propagation layer k , the representation of node v is updated as

$$\mathbf{h}_v^{(k)} = \sum_{u \in \mathcal{N}(v)} \tilde{A}_{vu} \mathbf{h}_u^{(k-1)},$$

where $\mathcal{N}(v)$ denotes the set of neighboring nodes of v in the interaction graph, and $\tilde{A}_{vu}$ represents a normalized adjacency coefficient that controls the contribution of the neighbor u to node v .

This linear aggregation scheme allows each node to iteratively incorporate information from its local neighborhood. Stacking multiple propagation layers enables the model to capture higher-order collaborative relationships by aggregating information from multi-hop neighbors. [105]

4.1.4 Layer-Wise Representation Combination

Instead of relying solely on the final-layer embeddings, the proposed model combines representations from multiple propagation layers to form the final node embedding. For a node v , the final representation is computed as

$$\mathbf{z}_v = \sum_{k=0}^K \mathbf{h}_v^{(k)},$$

where K denotes the total number of propagation layers. This design allows the model to balance information from different neighborhood ranges and mitigates the risk of over-smoothing that can occur in deep graph neural networks. [105]

4.1.5 Prediction Function

Given the final user embedding $\mathbf{z}_u$ and item embedding $\mathbf{z}_i$, the predicted relevance score is computed using an inner product as follows:

$$\hat{r}_{ui} = \mathbf{z}_u^\top \mathbf{z}_i.$$

This prediction function preserves the simplicity and interpretability of latent factor models while benefiting from graph-based representation learning. The resulting scores are used to rank the items for each user in the recommendation task.

Therefore, the proposed graph neural network framework combines the strengths of collaborative filtering and graph representation learning. By explicitly modelling user–item interactions as a graph and propagating collaborative signals across this structure, the model provides a principled and effective foundation for the adaptive and explainable recommendation mechanisms introduced in subsequent chapters. [18]

4.2 Experimental Evaluation

This section evaluates the proposed graph neural network framework for collaborative filtering. All experiments reported in this chapter strictly followed the experimental protocol and evaluation framework defined in ChapterX, including the datasets, data splitting strategy, baseline methods, evaluation metrics, training procedure, and implementation details. The objective of this evaluation was to assess the effectiveness of the proposed model in a controlled and consistent experimental setting.

4.2.1 Evaluation Setting

The evaluation focuses on the Top-10 recommendation task, which reflects practical recommendation scenarios in which a small number of items are presented to each user. The recommendation quality is assessed using ranking-based metrics, as defined in Section X. The proposed model was compared with classical collaborative filtering and matrix factorization baselines to isolate the benefits of graph-based representation learning.

All models were trained and evaluated using identical data splits and optimization settings. As a result, the observed performance differences can be attributed to differences in the modelling assumptions rather than experimental artifacts.

4.2.2 Overall Performance

Table 3 reports the Top-10 recommendation performance of the proposed graph neural network model and baseline methods. The results demonstrate a clear and consistent improvement achieved by the proposed approach in terms of both precision and recall.

Table 3: Top-10 recommendation performance for collaborative filtering models.

Method	Precision@10	Recall@10
User-Based CF (UBCF)	0.7423	0.8015
Item-Based CF (IBCF)	0.7689	0.8247
Matrix Factorisation (MF)	0.8012	0.8619
SVD	0.8294	0.8896
SVD++	0.8627	0.9138
Proposed GNN-CF	0.9101	0.9625

The proposed model achieved a Precision@10 of 0.9101 and Recall@10 of 0.9625, outperforming all baseline approaches by a substantial margin. These results indicate that the model is effective in ranking relevant items highly and recovering a large proportion of relevant items for each user.

4.2.3 Comparison with Classical Baselines

Neighborhood-based collaborative filtering methods (UBCF and IBCF) achieve reasonable performance but are consistently outperformed by latent factor and graph-based models. This behavior is expected because similarity-based methods rely on direct overlap in interaction histories and are sensitive to sparsity in user–item interactions.

Matrix factorization models provide stronger baselines by learning global latent representation. Performance improves progressively from MF to SVD and SVD++, reflecting the benefits of bias modelling and the incorporation of implicit feedback. However, these models do not explicitly encode the relational structure of interaction data, limiting their ability to exploit indirect collaborative signals.

4.2.4 Impact of Graph-Based Representation Learning

The proposed graph neural network model outperforms all classical baselines by explicitly modelling user–item interactions as a graph and propagating collaborative signals across this structure. The substantial improvement in Recall@10 suggests that multi-hop neighborhood aggregation enables the model to recover relevant items that are not directly connected to the user in the interaction graph.

Figure 20 illustrates the Top-10 Precision and Recall scores for all the compared methods, highlighting the consistent advantage of the proposed approach.

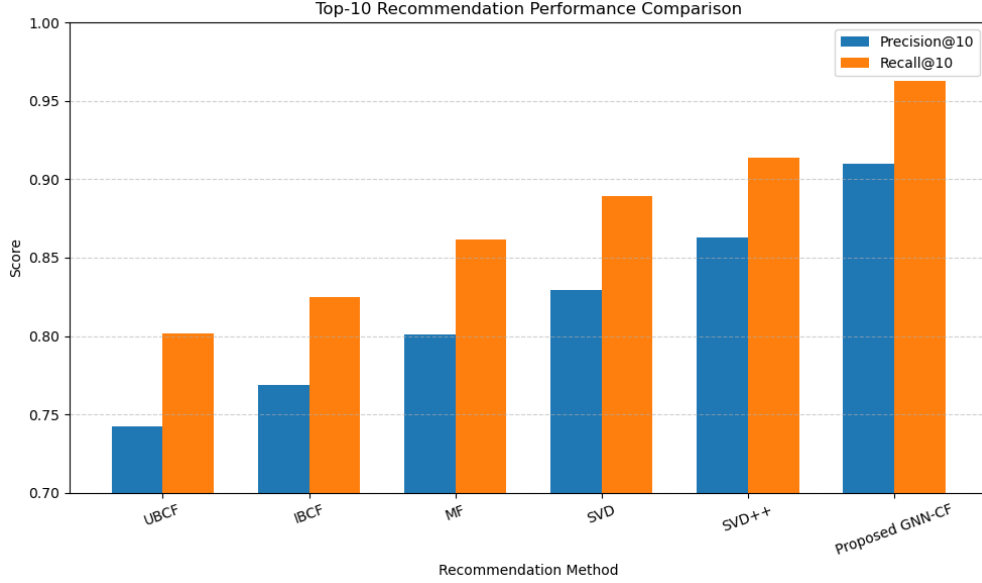


Figure 20: Comparison of Top-10 recommendation performance across collaborative filtering models.

Overall, the experimental results validate the effectiveness of modelling collaborative filtering as a graph-learning problem. By explicitly encoding the interaction structure and learning representations through neighborhood propagation, the proposed graph neural network provides a strong and reliable foundation for the adaptive and explainable recommendation models developed in the subsequent chapters.

4.3 Discussion

The experimental results presented in the previous section demonstrate that modelling collaborative filtering as a graph learning problem leads to substantial improvements in recommendation performance. This section discusses the underlying reasons for these gains, highlights the strengths and limitations of the proposed approach, and outlines the implications for the adaptive and explainable extensions developed in subsequent chapters.

4.3.1 Graph Neural Networks Improvement for Collaborative Filtering

The primary advantage of the proposed model is its ability to exploit the relational structure of user–item interactions. Unlike matrix factorization methods, which represent interactions through global latent factors, the graph neural network propagate information across the interaction graph. This allows each user representation to incorporate signals from multiple levels of neighborhood connectivity, including indirect relationships mediated through other users and items.

The observed improvement in recall suggests that multi-hop neighborhood aggregation plays a particularly important role in identifying relevant items that are not directly connected to the user. By integrating higher-order collaborative signals, the model is better equipped to recover items that would otherwise be overlooked by methods that rely solely on direct interaction patterns.

4.3.2 Comparison with Classical Collaborative Filtering Paradigms

Neighborhood-based collaborative filtering methods rely on explicit similarity computation and are inherently limited by the sparsity of interaction data. While these methods perform reasonably well in dense regions of the interaction matrix, their effectiveness decreases as the number of users and items increases. The proposed graph-based approach mitigates this issue by learning representations that aggregate information from a broader relational context, thereby reducing the sensitivity to sparse interactions.

Matrix factorization models address some of these limitations by learning latent representations that generalize across users and items. However, these representations are learned independently for each user and item and do not explicitly encode the relational dependencies. The graph neural network framework bridges this gap by combining latent representation learning with relational aggregation, resulting in a more expressive collaborative filtering model.

4.3.3 Model Simplicity and Design Trade-offs

An important design choice in the proposed framework is the use of a simple linear aggregation scheme without additional nonlinear transformations or feature projections.

This choice reduces the model complexity and contributes to stable training behavior while still capturing the essential collaborative signals present in the interaction graph. The strong empirical performance suggests that architectural simplicity can be preferable to overly complex graph neural filtering tasks.

However, this simplicity introduces limitations. The model treats all interactions uniformly during propagation and does not distinguish between interactions of different importance or relevance. Consequently, the influence of individual edges on recommendation outcomes remains implicit and cannot be adjusted dynamically.

4.3.4 Implications for Adaptivity and Explainability

Although the proposed graph neural network achieves strong recommendation accuracy, its static and opaque nature limits its practical applicability in dynamic environments. User preferences may evolve over time, and different interactions may contribute unequally to the recommendation decisions. The inability to quantify or control these contributions restricts the model’s responsiveness and its interpretability.

These observations motivate the extensions introduced in the next section, where the Shapley-value-based contribution analysis is used to quantify the influence of individual interactions. By integrating contribution estimates into the recommendation process, it is possible to design adaptive models that adjust their behavior in response to changing user preferences while also providing meaningful explanations for their predictions.

In summary, this chapter establishes graph neural networks as an effective foundation for collaborative filtering while highlighting the need for adaptive and explainable mechanisms to fully realize their potential in real-world recommendation scenarios.

Summary

This study introduced a graph neural network framework for collaborative filtering that explicitly models user-item interactions as a graph learning problem. By representing users and items as nodes in a bipartite graph and propagating information through neighborhood aggregation, the proposed approach captures higher-order collaborative signals that are not accessible to classical collaborative filtering and matrix factorization methods.

The experimental evaluation, conducted under a unified experimental protocol, demonstrated that the proposed model consistently outperforms neighborhood-based and latent factor baselines in Top- N recommendation tasks. The results highlight the effectiveness of graph-based representation learning for collaborative filtering, particularly in improving the recovery of relevant items through multi-hop neighborhood aggregation.

Despite these performance gains, the model remains static and does not explicitly account for the varying importance of the individual interactions. In addition, the decision-making process of the model is not directly interpretable, which limits its transparency and adaptability. These limitations motivate the next chapter, which introduces a Shapley-value-based contribution analysis to enable adaptive recommendation mechanisms and provide a principled foundation for explainable graph-based recommender systems.

5 Adaptive Recommendation via Shapley-Value Contribution Analysis

Introduction

Most existing recommender systems assume that user preferences are stable over time and that all historical interactions contribute equally to future recommendations. While this assumption simplifies the model design, it is often violated in real-world applications, where user interests evolve owing to changing contexts, trends, and short-term intentions. Consequently, static recommendation models may fail to reflect current user preferences, even when trained on large and high-quality datasets.

To address this limitation, adaptive recommender systems have been proposed to dynamically adjust recommendations in response to evolving user behaviors. Many existing adaptive approaches rely on heuristic mechanisms, such as temporal decay functions or periodic retraining. Although these methods can improve responsiveness, they provide limited insight into why certain interactions should be emphasized or down-weighted and lack a principled foundation for contribution attribution.

This chapter investigates adaptive recommendations through the lens of contribution modelling. Building on cooperative game theory, Shapley values are employed to quantify the marginal contribution of individual user–item interactions to recommendation outcomes. By treating interactions as contributors to a prediction, the Shapley values enable a fair and theoretically grounded assessment of the importance of interactions.

The central contribution of this chapter is a real-time adaptive recommendation framework that integrates Shapley-value-based contribution estimation directly into collaborative filtering models. Instead of applying post hoc explanations or heuristic weighting schemes, the proposed approach uses contribution scores to dynamically adjust the influence of historical interactions during recommendation generation. This allows the system to respond immediately to changes in user behavior while maintaining interpretability.

The framework was evaluated on real-world datasets using multiple collaborative filtering models, demonstrating consistent improvements in recommendation performance and robustness under dynamic conditions. In addition to accuracy gains, the proposed

method provides transparent explanations by identifying the interactions that most strongly influence each of the recommendations.

The remainder of this chapter presents the research relevance and related work, formalizes the Shapley-based contribution modelling approach, introduces the adaptive recommendation framework, and reports the experimental results as described in the underlying research paper.

5.1 Research Relevance and Related Work

Recommender systems are widely used to support decision-making in domains such as e-commerce, media consumption, and online services. Classical collaborative filtering approaches, including neighborhood-based methods and latent factor models, have demonstrated strong performance by exploiting similarities between users and items. However, these methods typically assume that user preferences are stable and that all historical interactions contribute equally to future ones.

In real-world environments, user behavior is inherently dynamic. Preferences may change due to evolving interests, contextual factors or short-term intentions. Consequently, interactions that were once informative may become less relevant over time, whereas recent interactions may carry greater predictive value. Static recommender systems are not designed to account for such changes and may therefore produce recommendations that do not reflect current user requirements.

Several adaptive recommendation approaches have been proposed to address this problem. Common strategies include applying temporal decay functions to interaction histories, using sliding time windows, and periodically retraining models on recent data. While these techniques improve responsiveness to preference changes, they rely on heuristic assumptions regarding interaction relevance. In particular, decay rates and window sizes are often selected empirically, providing limited insight into why certain interactions should be prioritized over others.

More advanced methods have explored dynamic extensions of collaborative filtering models, where user or item representations are updated incrementally as new interactions are observed. Although these approaches allow models to evolve over time, the influence of individual interactions remains unclear. Consequently, it is difficult to analyze, interpret, or control how past behavior affects current recommendations.

In parallel, recommender systems have been developed to improve transparency by identifying the factors that drive recommendation outcomes. Existing explainability techniques often focus on post-hoc analysis, attributing predictions to neighbors, latent factors, or historical interactions. However, these explanations are typically disconnected from the adaptation process and do not influence the updating of recommendations in response to changing user behavior.

Shapley values, which originate from cooperative game theory, provide a principled mechanism for attributing contributions among multiple participants. They have desirable theoretical properties, including fairness and efficiency, and have been explored in recommender systems primarily as post-hoc explanation tools. Their potential to support adaptive recommendations by directly influencing model behavior has received limited attention, particularly in real-time settings.

The research relevance of this work lies in bridging adaptation and explainability through Shapley-value-based contribution modelling. By explicitly quantifying the marginal contribution of individual user–item interactions, the proposed approach enables a dynamic adjustment of the interaction influence during recommendation generation. This allows adaptive behavior to be guided by a theoretically grounded contribution signal rather than heuristic temporal assumptions, addressing a key limitation of existing adaptive recommender systems.

5.2 Contribution Modelling with Shapley Values

Adaptive recommendation requires an explicit mechanism to quantify the influence of individual user–item interactions on recommendation outcomes. In traditional collaborative filtering models, historical interactions are treated uniformly, and their relative importance is not explicitly modelled. This assumption limits the ability of recommender systems to adapt to changes in the user’s behavior. To address this limitation, this study adopts Shapley values as a principled method for modelling interaction-level contributions.

Shapley values originate from cooperative game theory and provide a fair attribution of contributions among multiple participants. In recommender systems, user–item interactions are interpreted as contributors to the predicted recommendation score. This formulation enables the marginal influence of each interaction to be quantified

with respect to specific recommendation outcomes.

5.2.1 Formulation of Interaction Contributions

Let $\mathcal{I}_u = \{(u, i_1), (u, i_2), \dots, (u, i_n)\}$ denote the set of historical interactions associated with user u . Each interaction is treated as a player in the cooperative game. For a given candidate item j , the value function of a coalition $\mathcal{C} \subseteq \mathcal{I}_u$ is defined as

$$v(\mathcal{C}) = \hat{r}_{uj}(\mathcal{C}),$$

where $\hat{r}_{uj}(\mathcal{C})$ denotes the predicted relevance score obtained when only the interactions in $\mathcal{C}$ are considered by the recommendation model.

This formulation allows the contribution of each interaction to be evaluated in relation to all other interactions, capturing both the individual and joint effects on the prediction.

5.2.2 Shapley Value Definition

The Shapley value of an interaction $e \in \mathcal{I}_u$ is defined as the expected marginal contribution of e across all possible coalitions:

$$\phi_e = \sum_{\mathcal{C} \subseteq \mathcal{I}_u \setminus \{e\}} \frac{|\mathcal{C}|! (|\mathcal{I}_u| - |\mathcal{C}| - 1)!}{|\mathcal{I}_u|!} [v(\mathcal{C} \cup \{e\}) - v(\mathcal{C})].$$

This formulation satisfies key axioms of cooperative game theory, including efficiency, symmetry, and fairness. Consequently, the total predicted score can be decomposed into a sum of interaction-level contributions in a theoretically grounded manner.

5.2.3 Approximation of Shapley Values

The exact computation of Shapley values is computationally expensive because of the exponential number of possible coalitions. To enable practical application in recommender systems, this study employs a Monte Carlo approximation strategy based on random permutations of interactions.

Given a random permutation π of the interaction set $\mathcal{I}_u$, the marginal contribution of interaction e is computed as

$$\Delta_e^\pi = v(\mathcal{C}_e^\pi \cup \{e\}) - v(\mathcal{C}_e^\pi),$$

where $\mathcal{C}_e^\pi$ denotes the set of interactions that precede e in permutation π . The Shapley value ϕ_e is approximated by averaging Δ_e^π across multiple sampled permutations.

This approximation enables the contribution estimation to be performed efficiently while preserving the theoretical properties of the Shapley values.

5.2.4 Interpretation of Contribution Scores

Shapley values provide a quantitative measure of the influence of each historical interaction on the recommendation outcome. Interactions with higher Shapley values contribute more strongly to the predicted relevance score, whereas interactions with lower values have a reduced impact.

In the proposed framework, contribution scores are used to explain recommendation outcomes and support the adaptive adjustment of interaction influence. By explicitly modelling interaction contributions, the recommender system can identify which past behaviors are most relevant to current recommendations.

Overall, Shapley-value-based contribution modelling provides a principled foundation for adaptive recommendations by enabling the quantification, interpretation, and adjustment of interaction-level influence in real time.

5.3 Adaptive Recommendation Framework

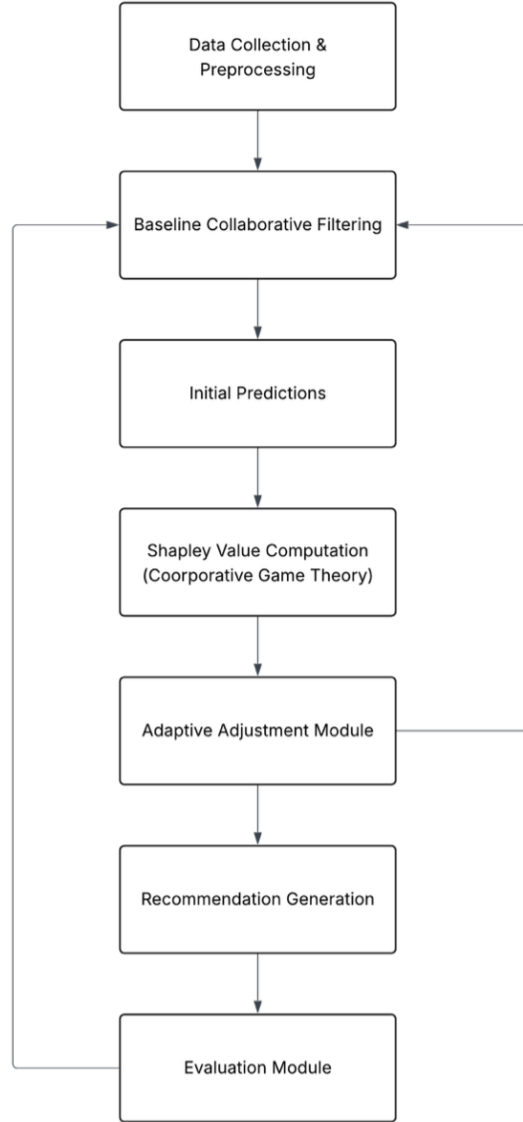


Figure 21: Architecture of the proposed adaptive recommendation framework.

Figure 21 illustrates the architecture of the proposed adaptive recommendation framework. The framework integrates Shapley-value-based contribution modelling into collaborative filtering by explicitly introducing an adaptive adjustment module and recommendation generation module. These components enable the dynamic reweighting

of historical interactions at the time of recommendation while preserving the structure of the underlying collaborative filtering models.

5.3.1 Framework Overview

Let $\mathcal{I}_u = \{(u, i_1), \dots, (u, i_n)\}$ denote the set of historical interactions of user u . A base collaborative filtering model first computes an initial prediction score $\hat{r}_{ui}$ for a candidate item i using the complete interaction history. Shapley values were then computed to estimate the contribution of each interaction to this prediction. These contribution scores are used by the adaptive adjustment module to modify the interaction influence before generating the final recommendation score.

5.3.2 Adaptive Adjustment Module

The adaptive adjustment module transforms the Shapley value contribution estimates into interaction-level weights that control the influence of historical interactions during recommendation generation. Let $\phi_{(u,j)}$ denote the Shapley value associated with the interaction between users u and item j .

Each interaction is assigned an adaptive weight defined as

$$w_{(u,j)} = 1 + \alpha \cdot \phi_{(u,j)},$$

where $\alpha \geq 0$ is a scaling factor that controls the strength of adaptive adjustment. This formulation ensures that interactions with higher Shapley values exert a greater influence on the recommendation outcome, whereas interactions with lower contributions are down-weighted.

The adaptive adjustment module operates at inference time, enabling real-time adaptation without retraining the underlying collaborative-filtering model.

5.3.3 Integration with Collaborative Filtering Models

Adaptive weights are incorporated into different collaborative filtering models by modifying how interactions contribute to the prediction.

5.3.4 User-Based Collaborative Filtering

For user-based collaborative filtering, the adapted prediction score is computed as

$$\hat{r}_{ui}^{\text{adapt}} = \bar{r}_u + \frac{\sum_{v \in \mathcal{N}_u} w_{(v,i)} \cdot s(u,v) \cdot (r_{vi} - \bar{r}_v)}{\sum_{v \in \mathcal{N}_u} |s(u,v)|},$$

where $s(u,v)$ denotes the similarity between users u and v , $\bar{r}_u$ is the average rating of user u , and $w_{(v,i)}$ is the adaptive weight derived from the Shapley value of interaction (v,i) .

5.3.5 Item-Based Collaborative Filtering

For item-based collaborative filtering, the adapted prediction score is given by

$$\hat{r}_{ui}^{\text{adapt}} = \frac{\sum_{j \in \mathcal{N}_i} w_{(u,j)} \cdot s(i,j) \cdot r_{uj}}{\sum_{j \in \mathcal{N}_i} |s(i,j)|},$$

where $s(i,j)$ denotes the similarity between items i and j .

5.3.6 Matrix Factorisation

For matrix factorization models, adaptive adjustment is achieved by modifying the observed interaction matrix. The adjusted rating matrix $\tilde{R}$ is defined as

$$\tilde{R}_{uj} = w_{(u,j)} \cdot R_{uj}.$$

Latent factors are learned by minimising the objective function

$$\min_{P,Q} \sum_{(u,j) \in \mathcal{I}} (\tilde{R}_{uj} - P_u^\top Q_j)^2 + \lambda (\|P_u\|^2 + \|Q_j\|^2),$$

where P_u and Q_j denote the user and item latent factor vectors, respectively.

5.3.7 Recommendation Generation Module

The recommendation generation module produces a final ranked list of items using the adapted prediction scores. For each user u , candidate items are scored using the

adaptive prediction function $\hat{r}_{ui}^{\text{adapt}}$, and the top- N items with the highest scores are selected for the user.

Because adaptive adjustment is performed at the recommendation time, the framework supports real-time recommendations and immediate responses to changes in user behavior. As new interactions are observed, the Shapley value estimates and adaptive weights are updated, enabling continuous adaptation without retraining.

5.3.8 Interpretability of Adaptive Recommendations

The Shapley values used for adaptive adjustment also provide explanations for the recommendation outcomes. By ranking interactions according to their contribution scores, the framework identifies the historical interactions that most strongly influenced each recommendation. This establishes a direct link between the adaptive behavior of the system and its ability to explain itself.

In summary, the proposed adaptive recommendation framework integrates contribution modelling, adaptive adjustment, and recommendation generation into a unified architecture that supports real-time, interpretable, and dynamic collaborative filtering.

5.4 Experimental Evaluation

This section reports the experimental results of the proposed Adaptive Shapley Value-enhanced collaborative filtering framework. The evaluation compares baseline collaborative filtering models with their Shapley-enhanced counterparts using two real-world datasets: MovieLens 1M and Amazon Product Reviews. The analysis focuses on recommendation quality measured by Precision@10, Recall@10, and F1-score@10, and examines the computational cost and explainability.

5.5 Hardware Configuration

To ensure reproducibility, the experiments were performed using a fixed hardware configuration, as reported in Table 4.

Table 4: Hardware Configuration.

Component	Specification
Chip	Apple M4 Max
CPU	16-Core
GPU	40-Core
Neural Engine	16-Core
Memory	48GB
Storage	1TB SSD

5.6 Performance Comparison

This subsection provides a numerical comparison of the baseline collaborative filtering models and their enhanced versions combined with Adaptive Shapley. Performance was measured using Precision@10, Recall@10, and F1-score@10.

5.6.1 MovieLens 1M Results

Table 5 reports results on MovieLens 1M. The enhanced variants are denoted by the prefix **SHAP-**.

Table 5: Performance on MovieLens 1M Dataset.

Model	Precision@10	Recall@10	F1-Score@10
UBCF [79]	0.750	0.650	0.696
SHAP-UBCF	0.820	0.709	0.760
IBCF [79]	0.780	0.700	0.738
SHAP-IBCF	0.851	0.763	0.805
MF [4]	0.810	0.730	0.768
SHAP-MF	0.886	0.798	0.840
NCF [79]	0.850	0.780	0.813
SHAP-NCF	0.873	0.796	0.832

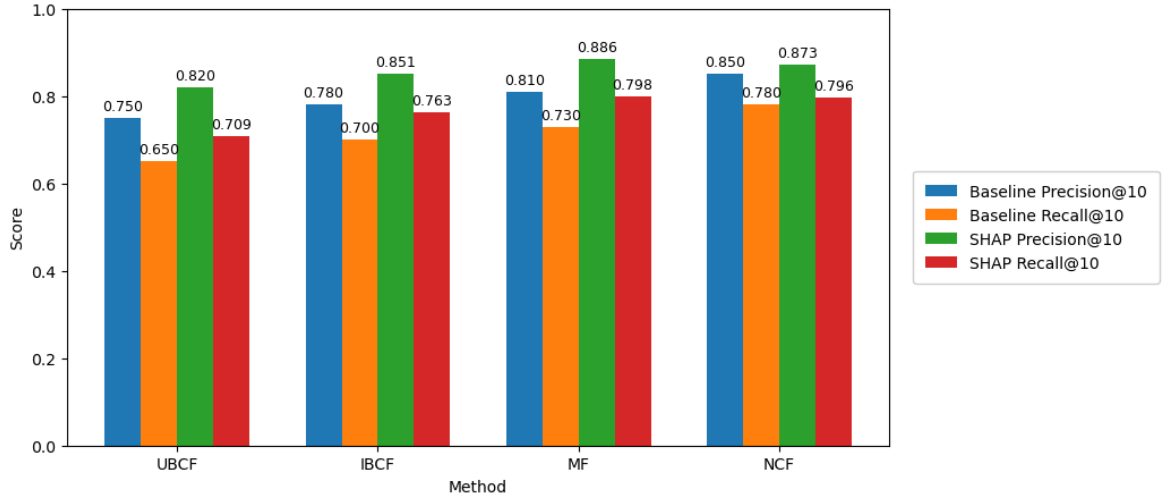


Figure 22: Precision@10 and Recall@10 comparison on MovieLens 1M Dataset.

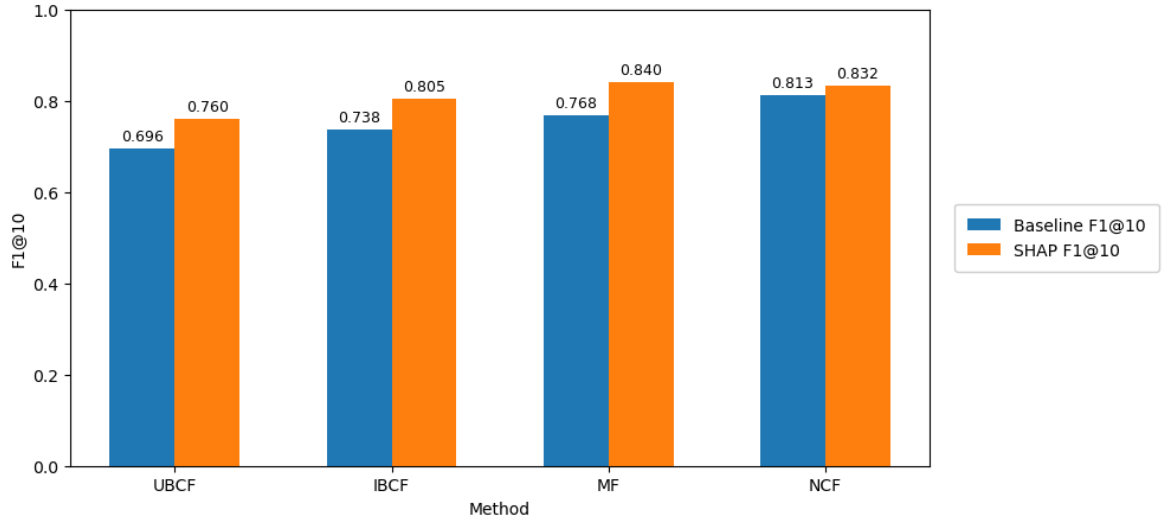


Figure 23: F1-score@10 comparison on MovieLens 1M Dataset.

5.6.2 Amazon Product Reviews Results

Table 6 reports results on Amazon Product Reviews.

Table 6: Performance on Amazon Product Reviews.			
Model	Precision@10	Recall@10	F1-Score@10
UBCF [106]	0.698	0.673	0.685
SHAP-UBCF	0.768	0.740	0.753
IBCF [106]	0.705	0.696	0.700
SHAP-IBCF	0.790	0.779	0.784
MF [107]	0.886	0.800	0.841
SHAP-MF	0.898	0.808	0.849
NCF [79]	0.742	0.715	0.728
SHAP-NCF	0.841	0.776	0.807

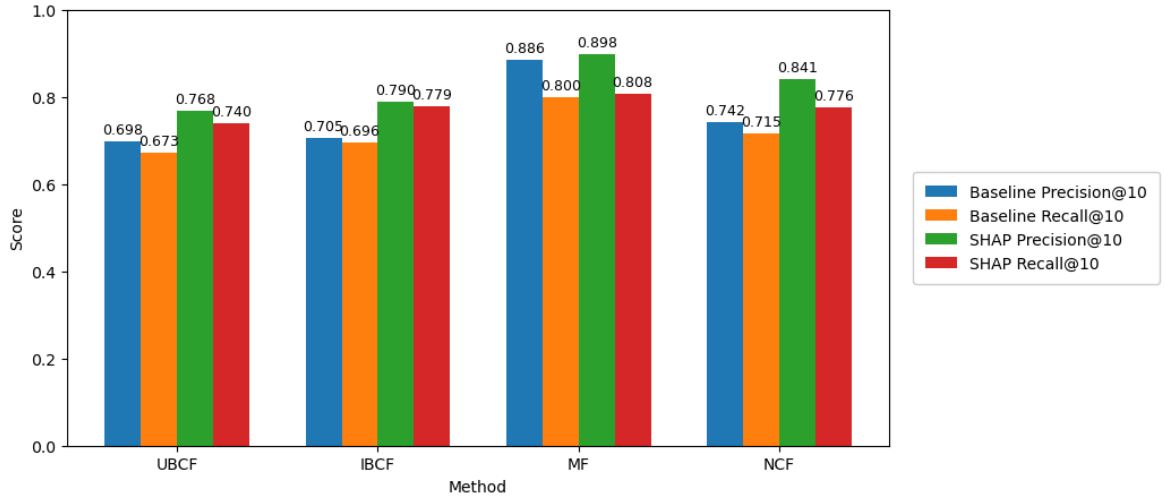


Figure 24: Precision@10 and Recall@10 comparison on Amazon Product Reviews.

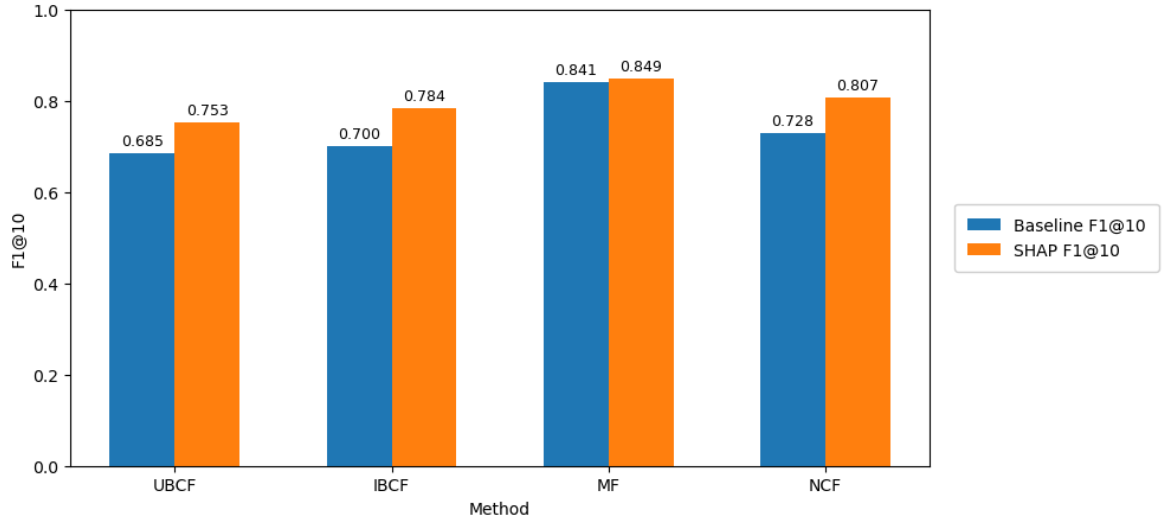


Figure 25: F1-score@10 comparison on Amazon Product Reviews.

5.7 Computational Performance Analysis

Although Adaptive Shapley provides performance benefits, it introduces additional computational costs owing to Monte Carlo sampling for Shapley approximation during training and real-time contribution score calculation during prediction. Table 7 presents the computational performance

Table 7: Computational Performance Analysis.

Model	Training Time (s)	Prediction Time (ms)	Memory (MB)	Time Overhead
UBCF	42.3	8.7	145	–
SHAP-UBCF	67.8	14.2	189	+60%
IBCF	38.9	7.4	138	–
SHAP-IBCF	71.2	13.8	185	+83%
MF	89.4	12.1	267	–
SHAP-MF	124.7	18.9	341	39%
NCF	156.8	15.4	412	–
SHAP-NCF	203.2	24.7	523	+30%

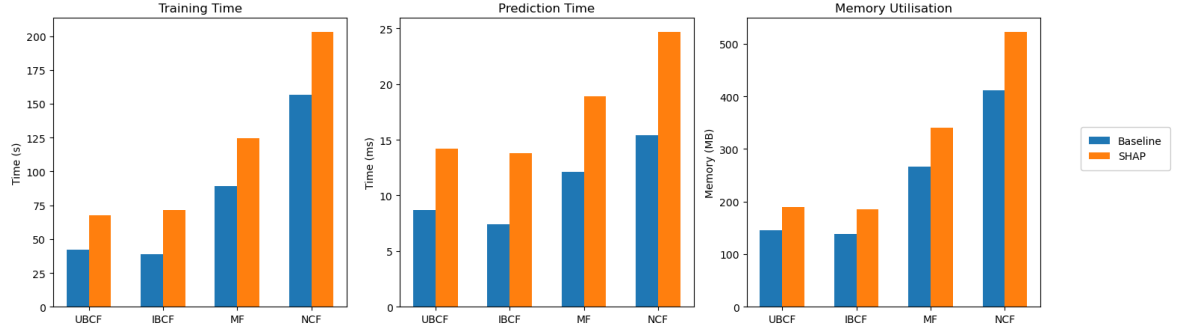


Figure 26: Computational Resources Utilisation Comparison.

5.8 Cost–Benefit Analysis

To assess whether the performance improvements justify the overhead, a cost–benefit analysis is reported in Table 8, summarizing the time overhead, F1-score gain, and cost–benefit ratio for each enhanced method and dataset.

Table 8: Performance Gains vs. Computational Costs.

Dataset	Method	Time Overhead	F1-Score Gain	Cost-Benefit Ratio
MovieLens 1M	SHAP-UBCF	+60%	+9.2%	0.15
MovieLens 1M	SHAP-IBCF	+83%	+9.1%	0.11
MovieLens 1M	SHAP-MF	+39%	+9.4%	0.24
MovieLens 1M	SHAP-NCF	+30%	+2.3%	0.08
Amazon Reviews	SHAP-UBCF	+60%	+9.9%	0.17
Amazon Reviews	SHAP-IBCF	+83%	+12.0%	0.14
Amazon Reviews	SHAP-MF	+39%	+0.95%	0.02
Amazon Reviews	SHAP-NCF	+30%	+10.9%	0.36

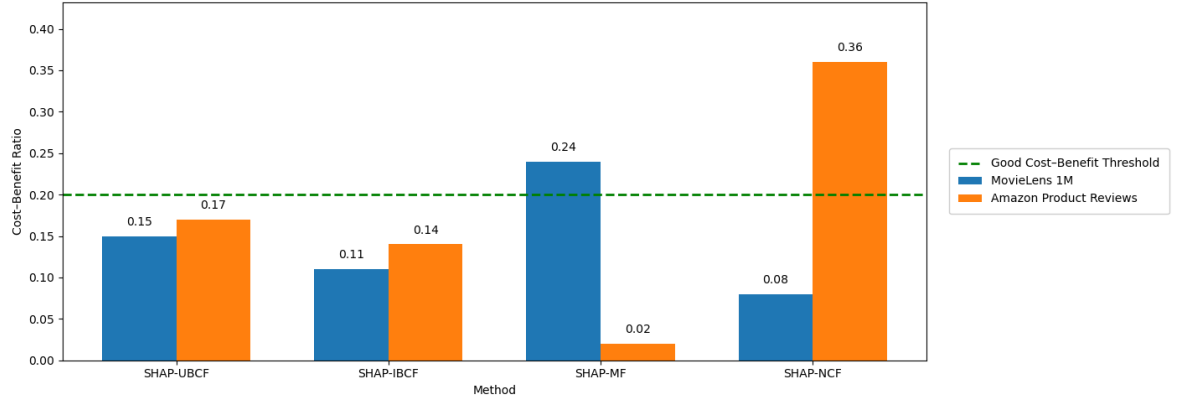


Figure 27: Cost-Benefit Ratio Comparison Across Methods and Datasets (Higher is Better).

5.9 Explainability Enhancement

A key contribution of the framework is its ability to provide transparent, real-time explanations by measuring the extent to which each user-item interaction influences a recommendation. As recommendations are made, the system computes the Shapley values for interactions, identifies the most influential past interactions, and uses them as contribution-based explanations.

For example, on MovieLens, if *The Godfather* is recommended to a user, the system can explain that the recommendation is mainly driven by the user’s previous high ratings for *Goodfellas* and *Scarface*, contributing 40% and 35%, respectively.

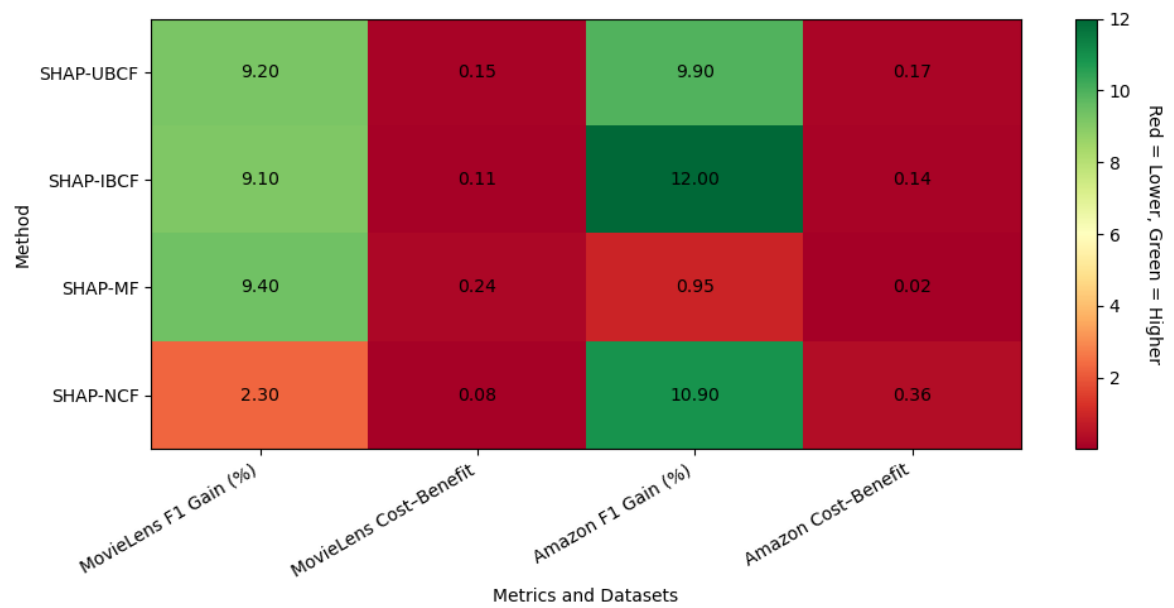


Figure 28: Performance and Cost-Benefit Analysis Heatmap (Green = Better Performance).

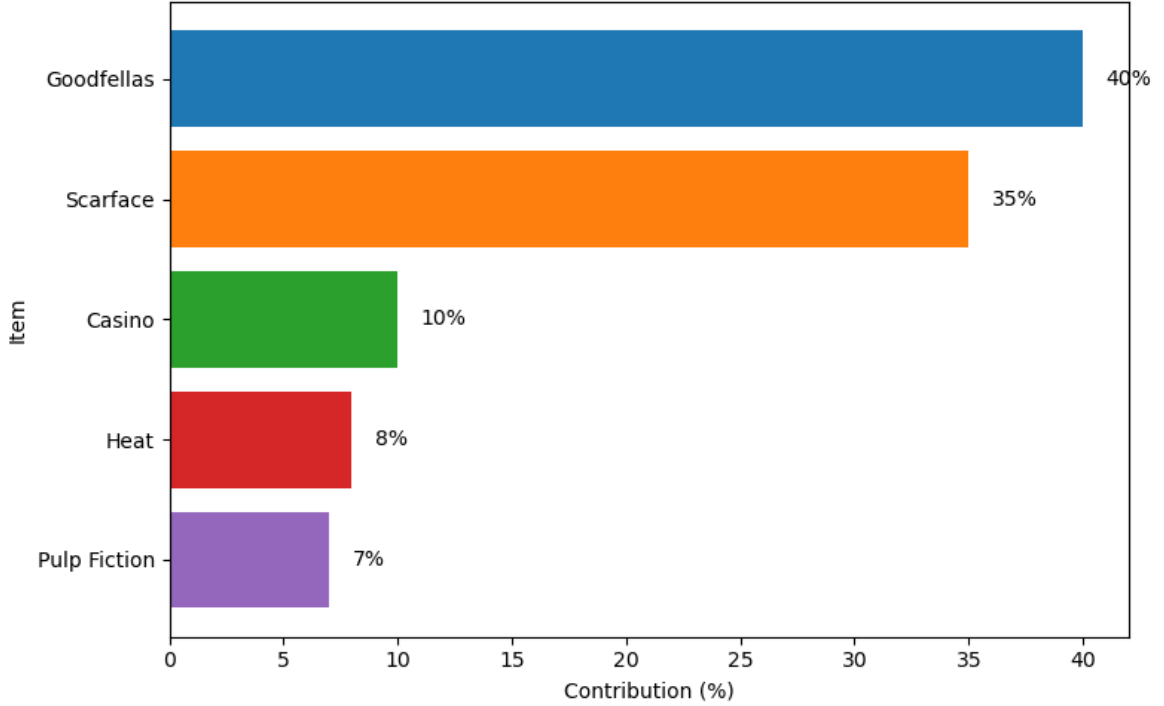


Figure 29: Contribution Scores of User-Item Interactions for Recommendation.

5.10 Discussion

The experimental results reported in the previous section demonstrate that incorporating the Shapley-value-based contribution adjustment into collaborative filtering models leads to consistent improvements in recommendation performance across datasets and methods. These findings confirm the central hypothesis of this study: dynamically adjusting the influence of historical interactions based on their estimated contribution enhances the ability of recommender systems to respond to evolving user preferences.

A key observation from the results is that the adaptive variants outperform their non-adaptive counterparts across all evaluated collaborative filtering models. This indicates that the proposed framework is not tied to a specific recommendation technique but can be applied as a general enhancement to different collaborative-filtering paradigms. The improvements observed in Precision@10, Recall@10, and F1-score@10 suggest that the contribution-aware adjustment enables the recommender system to prioritize in-

teractions that are most relevant to the current recommendation context.

The results also highlight the effectiveness of the Shapley value as a contribution modelling mechanism. By estimating the marginal contribution of each interaction, the framework avoids relying on heuristic temporal decay or fixed-weighting schemes. Instead, interaction relevance is determined in a data-driven and theoretical manner. This allows the system to adapt selectively, amplifying influential interactions while reducing the impact of less informative or outdated behaviors.

From a computational perspective, the experiments show that the additional overhead introduced by the Shapley-value estimation remains manageable. Although adaptive models incur higher training and prediction costs owing to the Monte Carlo approximation, the reported overhead is acceptable for practical deployment. Importantly, the framework achieves adaptivity without requiring frequent retraining of the base collaborative filtering models, which contributes to its scalability in real-world applications.

Beyond performance improvements, the proposed framework enhances interpretability by explicitly identifying which user–item interactions contribute most strongly to recommendation outcomes. Contribution scores derived from the Shapley values provide transparent explanations that align with the model’s adaptive behavior. This dual role of Shapley values, supporting both adaptation and explanation, distinguishes the proposed approach from existing methods that treat these aspects separately.

Overall, the discussion confirms that Shapley-value-based contribution modelling provides a principled and effective mechanism for adaptive recommendations. The results support the claim that integrating contribution estimation directly into the recommendation process improves the accuracy, robustness, and interpretability in dynamic recommendation environments.

5.11 Limitation

Despite the demonstrated benefits of the Shapley-value-based contribution adjustment, the proposed adaptive recommendation framework has several limitations, as identified in the underlying research paper.

A primary limitation concerns computational cost. Although Monte Carlo sampling is used to approximate Shapley values efficiently, contribution estimation still intro-

duces additional overhead during training and prediction. This overhead increases with the size of the user interaction histories and the number of samples required for a stable estimation. Although the reported computational cost remains acceptable for the evaluated datasets, scalability to very large-scale systems or users with extensive interaction histories may require further optimization.

The second limitation is the reliance on explicit feedback data. The experimental evaluation was conducted on datasets containing explicit user ratings, such as MovieLens and Amazon Product Reviews. Consequently, the effectiveness of the proposed framework for implicit feedback scenarios, where interactions are noisier and less informative, was not assessed in this study.

In addition, the adaptive framework was evaluated in conjunction with classical collaborative filtering models. Although this choice demonstrates the generality of the approach across different collaborative filtering paradigms, the integration of Shapley-value-based adaptation with more complex deep learning models has not been explored in this study.

Finally, the evaluation focused on offline performance metrics and computational analysis. User-centered evaluations, such as online A/B testing or user studies, were not included. Consequently, the impact of Shapley-based adaptivity on user satisfaction and perceived recommendation quality remains an open question in the literature.

These limitations highlight opportunities for future research, including improving the efficiency of contribution estimation, extending the framework to implicit feedback and deep learning-based recommenders, and validating the approach through online and user-centered evaluations.

Summary

This chapter investigated adaptive recommendation through Shapley-value-based contribution modelling, as presented in the underlying research paper. This chapter addressed the limitations of static collaborative filtering models by introducing a framework that dynamically adjusts the influence of historical user–item interactions during recommendation generation.

The proposed approach formulates the interaction contribution as a cooperative game

and employs Shapley values to quantify the marginal impact of individual interactions on recommendation outcomes. By integrating the contribution estimation into the recommendation process, the framework enables real-time adaptation without modifying the internal structure of existing collaborative filtering models.

An adaptive recommendation framework was presented, demonstrating how Shapley-based interaction weighting can be consistently applied across neighborhood-based and latent factor models. The framework supports real-time operation through Monte Carlo approximation and maintains computational feasibility while improving the recommendation performance.

The experimental results reported in this chapter show that the Shapley-value-based contribution adjustment leads to consistent improvements in recommendation accuracy and robustness across multiple datasets and collaborative filtering models. In addition to performance improvement, the framework provides transparent explanations by explicitly identifying the interactions that drive recommendation decisions.

This chapter also discusses the computational implications and limitations of the proposed approach, highlighting the challenges related to scalability, reliance on explicit feedback, and offline evaluation. Overall, this chapter established Shapley-value-based contribution modelling as a principled and effective mechanism for adaptive and explainable recommendations, forming the foundation for the subsequent chapters of the thesis.

6 Explainable Hierarchical Recommendation with Hyperbolic Graph Neural Networks

Introduction

The earlier chapters of this thesis examined graph-based collaborative filtering and adaptive recommendation mechanisms driven by Shapley value contribution modelling. These studies showed that graph neural networks effectively capture relational user-item patterns and that contribution-based explanations improve transparency and adaptability in recommendation systems. However, these approaches primarily focus on flat interaction structures and do not explicitly model the hierarchical relationships commonly found in real-world recommendation scenarios.

In many application domains, recommendation data are naturally organized into hierarchies. Items belong to categories and subcategories, user preferences may span multiple levels of abstraction, and interactions at higher levels can influence the decisions at lower levels. Ignoring such a hierarchical organization can lead to recommendations that lack semantic coherence or fail to reflect structured domain knowledge.

This section introduces the extension of the proposed framework to hierarchical recommendation settings. The focus shifts from modelling isolated user-item interactions to learning representations over hierarchical graphs that jointly encode user, item, and category relationships. This perspective enables recommendations that are consistent across hierarchy levels and better aligned with the structured domain semantics.

Simultaneously, increasing model expressiveness raises important questions regarding interpretability. As recommendation models incorporate richer structures, maintaining clear and accurate explanations remains essential. Therefore, this chapter continues the thesis emphasis on explainability by ensuring that recommendation decisions can still be attributed to meaningful user-item interactions, even in the presence of hierarchical modelling.

In summary, this introduction sets the stage for a hierarchical and explainable recommendation framework that builds directly on the graph-based and contribution-driven foundations established earlier in this thesis while addressing the additional complexity introduced by hierarchical data structures.

6.1 Research Relevance and Positioning

The previous chapters of this thesis established graph neural networks as an effective framework for collaborative filtering and demonstrated that Shapley-value-based contribution modelling enables principled adaptivity and explainability in recommendation systems. However, these approaches operate in Euclidean space and do not explicitly model the hierarchical structures inherent in many real-world recommendation domains.

In practice, recommendation data often exhibit a clear hierarchical organization. Items are commonly structured into categories and subcategories, users interact with content at different levels of abstraction, and preferences may propagate across hierarchical relationships. Although Euclidean graph neural networks can model relational dependencies, they struggle to represent deep hierarchies efficiently because tree-like structures incur increasing distortion when embedded in Euclidean space. This limitation becomes more pronounced as the hierarchy’s depth and complexity increase.

Hyperbolic geometry offers a natural solution to this issue. Owing to its exponential volume growth, hyperbolic space can embed hierarchical and tree-structured data with significantly lower distortion than Euclidean space. Hyperbolic graph neural networks extend this geometric advantage to relational learning by performing message passing and representation learning directly in the hyperbolic space. In the context of recommender systems, this enables more faithful modelling of user–item–category hierarchies while preserving the relational structure captured by graph-based approaches.

Simultaneously, the use of hyperbolic graph neural networks introduces additional challenges in terms of interpretability. As shown in Chapter 5, the Shapley values provide a theoretically grounded mechanism for explaining recommendation outcomes by attributing predictions to individual interactions. However, existing Shapley-based explanation methods are primarily designed for Euclidean models and do not directly account for the geometric properties of the hyperbolic space. This creates a gap between the representational power of hyperbolic models and the need for transparent and trustworthy recommendations for their use.

The relevance of this chapter lies in addressing this gap. By integrating Shapley-value-based explainability into hyperbolic graph neural networks, this study extends the adaptive and explainable recommendation framework developed earlier in the thesis

to hierarchical settings. The proposed approach preserves the benefits of contribution-based explanations while leveraging hyperbolic geometry to model hierarchical relationships more effectively.

Figure 30 illustrates the overall architecture of our proposed framework. The model combines hierarchical graph construction, hyperbolic graph neural network encoding, and Shapley value-based explanation into a unified recommendation pipeline. This architecture highlights how geometric representation learning and contribution-based explainability are jointly employed to support accurate and interpretable hierarchical recommendations.

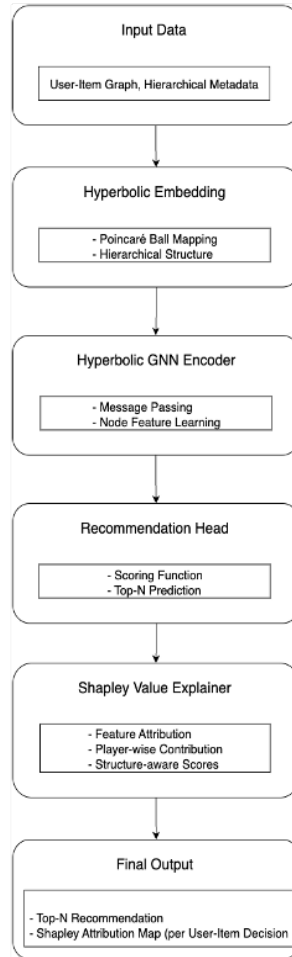


Figure 30: Architecture of the proposed explainable hierarchical recommendation framework.

In summary, this chapter positions hyperbolic graph neural networks as a natural

extension of the graph-based recommendation models introduced earlier in the thesis and demonstrates how Shapley-value-based explainability can be adapted to operate in non-Euclidean geometric spaces. By addressing both hierarchical representation and interpretability, the proposed framework completes the progression from graph-based collaborative filtering to adaptive and explainable hierarchical recommendations.

6.2 Hierarchical Recommendation Problem Formulation

In hierarchical recommendation settings, user preferences and item relationships are structured at multiple levels of abstraction. Unlike flat recommendation scenarios, where interactions are defined solely between users and items, hierarchical recommendations require modelling additional relations that capture category and subcategory organization. This subsection formalizes the hierarchical recommendation problem and introduces the notation used throughout the chapter.

6.2.1 Hierarchical Interaction Structure

Let $\mathcal{U}$ denote the set of users, $\mathcal{I}$ the set of items, and $\mathcal{C}$ the set of categories organized into a hierarchy. Each item $i \in \mathcal{I}$ is associated with one or more categories at different hierarchical levels. The category hierarchy is represented as a directed acyclic graph (DAG) or tree, where edges encode parent–child relationships between categories.

User interactions are defined not only at the item level but also implicitly at higher levels of hierarchy. For example, a user’s interaction with an item may indicate a preference for the corresponding category and its ancestors. Consequently, preference signals propagate upward through the hierarchy.

6.2.2 Hierarchical Graph Representation

The hierarchical recommendation problem is modelled as a graph

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}),$$

where the node set $\mathcal{V}$ consists of users, items, and categories.

$$\mathcal{V} = \mathcal{U} \cup \mathcal{I} \cup \mathcal{C}.$$

The edge set $\mathcal{E}$ captures three types of relationships.

- User–item interactions (u, i) representing observed preferences.
- Item–category links (i, c) encoding category membership.
- Category–category edges (c_p, c_c) representing hierarchical parent–child relations.

This heterogeneous graph structure allows the preference information to flow across different entity types and hierarchy levels.

6.2.3 Recommendation Objective

Given a user $u \in \mathcal{U}$ and a set of candidate items $\mathcal{I}_u \subseteq \mathcal{I}$, the objective of hierarchical recommendation is to predict a relevance score

$$\hat{r}_{ui} = f(u, i, \mathcal{G}),$$

where $f(\cdot)$ is a recommendation function that exploits both user–item interactions and hierarchical relationships encoded in the graph.

The final recommendation list for user u is obtained by ranking candidate items according to $\hat{r}_{ui}$ and selecting the top- N items with the highest ranks.

6.2.4 Hierarchy-Aware Preference Propagation

A key requirement for hierarchical recommendation is that preference information propagates across hierarchy levels. Let $c(i)$ denote the category path associated with item i . An interaction (u, i) contributes not only to the representation of item i but also to the representation of categories along $c(i)$.

Formally, the preference signal of user u can be expressed as a function of both direct item interactions and indirect category-level interactions as follows:

$$\mathbf{p}_u = g\left(\{(u, i)\}_{i \in \mathcal{I}_u}, \{(u, c)\}_{c \in c(i)}\right),$$

where $g(\cdot)$ aggregates the interaction signals across the hierarchy levels.

Challenges of Hierarchical Modelling

The hierarchical recommendation problem introduces two main challenges. First, the hierarchical structure induces long-range dependencies that are difficult to represent efficiently in Euclidean space. As the hierarchy depth increases, preserving parent–child relationships without distortion becomes increasingly challenging. Second, the presence of multiple entity types and interaction levels complicates interpretability, as recommendations may be influenced by both item- and category-level signals.

These challenges motivate the use of hyperbolic graph neural networks for hierarchical representation learning and the integration of contribution-based explainability mechanisms, which are introduced in the following subsections.

6.3 Hyperbolic Graph Neural Network for Recommendation

To effectively model hierarchical relationships in recommendation data, this study employs a hyperbolic graph neural network (HGNN). Hyperbolic geometry provides a natural embedding space for hierarchical and tree-like structures, allowing representations to preserve parent–child relationships with reduced distortion compared to the Euclidean space. This subsection describes the hyperbolic representation space, message-passing mechanism, and recommendation score computation.

6.3.1 Hyperbolic Representation Space

Let $\mathbb{H}^d$ denote a d -dimensional In this study, the entities in the hierarchical graph, including users, items, and categories, are embedded in $\mathbb{H}^d$. Each node $v \in \mathcal{V}$ is associated with a hyperbolic embedding $\mathbf{h}_v \in \mathbb{H}^d$.

To facilitate computation, embeddings are initialized in the tangent space at the origin and mapped to the hyperbolic space using the exponential map:

$$\mathbf{h}_v^{(0)} = \exp_{\mathbf{0}}(\mathbf{x}_v),$$

where $\mathbf{x}_v \in \mathbb{R}^d$ denotes the initial Euclidean feature vector, and $\exp_{\mathbf{0}}(\cdot)$ is the exponential map at the origin of $\mathbb{H}^d$.

Hyperbolic Message Passing

Given a hierarchical graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, representation learning is performed via message passing in a hyperbolic space. For each node v , messages are aggregated from its neighborhood $\mathcal{N}(v)$.

Because direct aggregation in hyperbolic space is nontrivial, neighbor embeddings are first mapped to the tangent space using the logarithmic map:

$$\mathbf{z}_u^{(l)} = \log_{\mathbf{h}_v^{(l)}}(\mathbf{h}_u^{(l)}), \quad u \in \mathcal{N}(v),$$

where $\log_{\mathbf{h}_v^{(l)}}(\cdot)$ maps hyperbolic embeddings to the tangent space at $\mathbf{h}_v^{(l)}$.

The aggregated message for node v at layer l is then computed as

$$\mathbf{m}_v^{(l)} = \sum_{u \in \mathcal{N}(v)} \alpha_{vu}^{(l)} \mathbf{z}_u^{(l)},$$

where $\alpha_{vu}^{(l)}$ denotes a normalized importance weight that controls the contribution of the neighbor u .

6.3.2 Hyperbolic Update Rule

The updated representation of node v is obtained by applying a linear transformation in the tangent space, followed by a nonlinear activation:

$$\tilde{\mathbf{z}}_v^{(l+1)} = \sigma(\mathbf{W}^{(l)} \mathbf{m}_v^{(l)}),$$

where $\mathbf{W}^{(l)}$ is a trainable weight matrix, and $\sigma(\cdot)$ denotes a nonlinear activation function.

The updated tangent-space representation is then mapped back to the hyperbolic space using the exponential map:

$$\mathbf{h}_v^{(l+1)} = \exp_{\mathbf{h}_v^{(l)}}(\tilde{\mathbf{z}}_v^{(l+1)}).$$

By stacking multiple HGNN layers, the model captures both local interactions and long-range hierarchical dependencies across graphs.

6.3.3 Hierarchy-Aware Representation Learning

The use of hyperbolic geometry enables embeddings to naturally reflect a hierarchical organization. Nodes closer to the root of the hierarchy tend to be embedded nearer to the origin, whereas nodes at deeper hierarchy levels are positioned further away. This geometric property allows the HGNN to encode both relational proximity and hierarchical depth in a unified representation.

Through iterative message passing, user representations integrate signals from item-level interactions and higher-level category relationships, enabling hierarchy-aware preference modelling.

6.3.4 Recommendation Score Computation

Given the final hyperbolic embeddings of a user u and item i , the relevance score is computed based on their hyperbolic distance:

$$\hat{r}_{ui} = -d_{\mathbb{H}}(\mathbf{h}_u, \mathbf{h}_i),$$

where $d_{\mathbb{H}}(\cdot, \cdot)$ denotes the hyperbolic distance. A smaller distance corresponds to higher predicted relevance.

Items are ranked according to $\hat{r}_{ui}$, and the top- N items with the highest relevance scores are recommended to users.

This hyperbolic graph neural network forms the core representation learning component of the proposed hierarchical recommendation framework. In the next subsection, Shapley-value-based contribution modelling is introduced to provide interpretable explanations for the recommendations generated by this model.

6.4 Shapley-Value Explanations in Hyperbolic Space

Although the hyperbolic graph neural network enables effective modelling of hierarchical relationships, the resulting recommendation process remains opaque without explicit explanation mechanisms. To address this issue, this chapter extends the Shapley-value-based explainability framework developed earlier in the thesis to hyperbolic graph neural networks. The goal is to attribute recommendation outcomes

to meaningful user–item interactions and hierarchical components while respecting the geometric properties of the hyperbolic space.

6.4.1 Explanatory Coalition Definition

For a given user–item pair (u, i) , the explanation problem is formulated as a cooperative game in which individual interactions contribute to the predicted relevance score $\hat{r}_{ui}$. Each player in the game corresponds to a historical interaction between user u and an item or category node in the hierarchical graph.

Let $\mathcal{S}_{ui}$ denote the set of all explanatory components associated with (u, i) , including

- direct user–item interactions,
- item–category associations,
- category–category hierarchical relations.

A coalition $\mathcal{C} \subseteq \mathcal{S}_{ui}$ represents a subset of interactions that jointly contribute to the recommendation.

6.4.2 Value Function in Hyperbolic Space

The value function $v(\mathcal{C})$ quantifies the contribution of coalition $\mathcal{C}$ to the recommendation’s outcome. In the hyperbolic setting, $v(\mathcal{C})$ is defined as the relevance score obtained when only the interactions in $\mathcal{C}$ are active in the hierarchical graph:

$$v(\mathcal{C}) = -d_{\mathbb{H}}(\mathbf{h}_u^{\mathcal{C}}, \mathbf{h}_i^{\mathcal{C}}),$$

where $\mathbf{h}_u^{\mathcal{C}}$ and $\mathbf{h}_i^{\mathcal{C}}$ denote the hyperbolic embeddings of user u and item i learned using the subgraph induced by $\mathcal{C}$.

This formulation ensures that the contribution attribution remains consistent with the geometry-aware recommendation score defined in Section 6.3.

6.4.3 Shapley Value Computation

The Shapley value for an interaction $s \in \mathcal{S}_{ui}$ is defined as the average marginal contribution of s across all possible coalitions:

$$\phi_s = \sum_{\mathcal{C} \subseteq \mathcal{S}_{ui} \setminus \{s\}} \frac{|\mathcal{C}|!(|\mathcal{S}_{ui}| - |\mathcal{C}| - 1)!}{|\mathcal{S}_{ui}|!} [v(\mathcal{C} \cup \{s\}) - v(\mathcal{C})].$$

Owing to the combinatorial complexity of the exact computation, the Shapley values are approximated using Monte Carlo sampling. For each user–item pair, a set of random permutations of explanatory components is generated, and the marginal contributions are accumulated along each permutation.

6.4.4 Geometric Consistency of Contributions

A key challenge in hyperbolic explainability is to ensure that the contribution attribution remains geometrically meaningful. By defining the value function directly in terms of hyperbolic distances, the Shapley values reflect how individual interactions influence the relative positioning of the user and item embeddings in the hyperbolic space.

Interactions that move embeddings closer together along the hierarchy contribute positively, whereas interactions that increase the hyperbolic distance reduce the predicted relevance. This geometric interpretation allows for explanations to be directly linked to hierarchical preference propagation.

6.4.5 Explanation Aggregation and Interpretation

The final explanation for a recommendation is obtained by ranking the interactions according to their Shapley values. High-magnitude contributions indicate interactions that strongly influence the recommendation outcomes. Contributions can be aggregated at different hierarchical levels, enabling explanations that highlight:

- influential historical items,
- dominant categories,

- hierarchical paths contributing to the recommendation.

This multi-level explanation capability allows the model to provide both fine-grained and high-level justifications, thereby supporting transparency and user trust.

By extending the Shapley-value-based attribution to hyperbolic graph neural networks, this section completes the integration of explainability into the hierarchical recommendation framework. The resulting model provides geometry-aware recommendations with principled, interpretable explanations grounded in cooperative game theory.

6.5 Experimental Evaluation

This section assesses the proposed explainable hierarchical recommendation framework, which combines hyperbolic graph neural networks with Shapley-value-based explanations. All experiments adhered to the unified experimental protocol established earlier in this study, including the same datasets, metrics, and training configuration. Therefore, this chapter focuses on understanding how hierarchical modelling, hyperbolic representation learning, and integrated explainability influence recommendation quality, robustness across cutoff thresholds, and practical usability.

6.6 Evaluation Objectives

The evaluation was structured around three questions.

- To what extent does hyperbolic representation learning improve recommendation accuracy when the underlying user–item relations are hierarchical?
- Can Shapley-value-based explanations be incorporated into a hyperbolic GNN recommender without compromising predictive performance?
- Do the produced explanations align with meaningful hierarchical structure and observed user preferences?

6.7 Recommendation Performance

This section reports the recommendation performance of the proposed approach on two benchmark datasets: MovieLens-1M and Amazon Product Reviews. We evaluate Precision@K, Recall@K, and F1-Score@K at multiple cutoff thresholds K to characterize performance trends as the recommendation list size increases. The reported results quantify both the accuracy of the recommendations and the stability of the method across various retrieval depths.

6.7.1 MovieLens-1M

Table 9 summarizes the results for MovieLens-1M at different cutoff

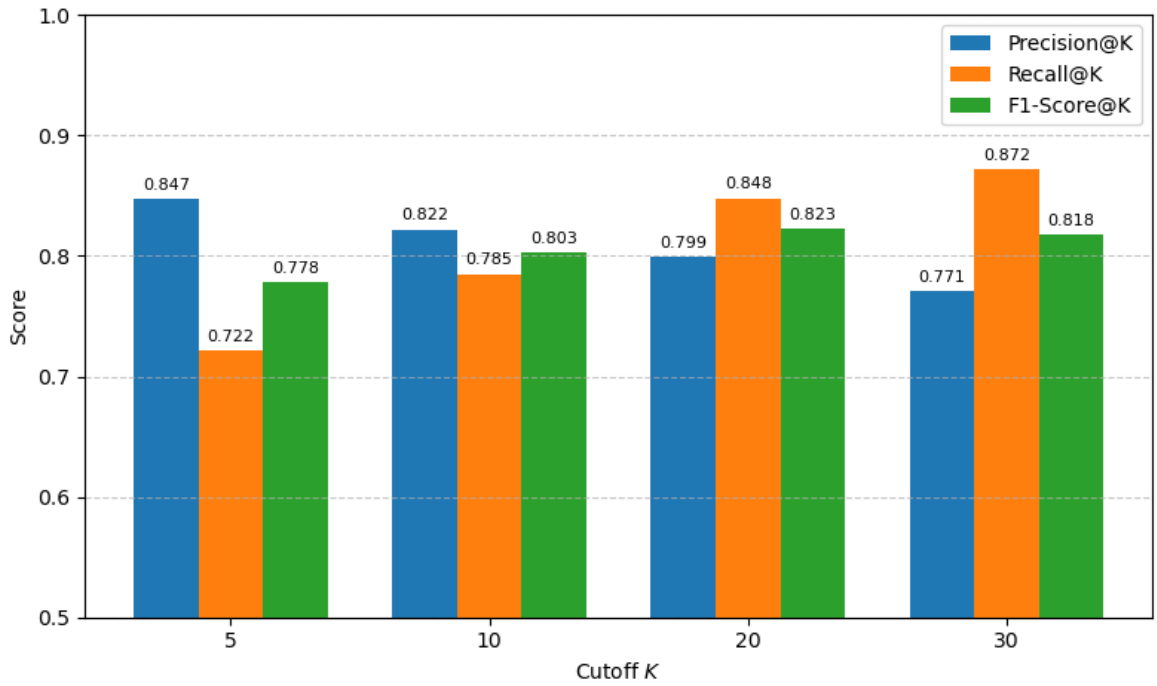


Figure 31: Recommendation Performance Metrics for MovieLens-1M at Different Cutoff Thresholds (K).

Table 9: Recommendation Performance Metrics at Different Cutoff Thresholds (K)
Using MovieLens-1M

Cutoff K	Precision@K	Recall@K	F1-Score@K
5	0.847	0.722	0.778
10	0.822	0.785	0.803
20	0.799	0.848	0.823
30	0.771	0.872	0.818

As shown in Table 9, the model maintained strong accuracy across all evaluated thresholds. As K increases, the precision decreases moderately, which is expected when recommending a larger set of items, whereas the recall increases, indicating that the method retrieves a greater portion of relevant items. The F1-Score remains consistently high, suggesting that the approach maintains a favorable balance between precision and recall. Figure 31 illustrates the trends of these metrics across different cutoff values.

6.7.2 Amazon Product Reviews

Table 10 reports the performance on the Amazon Product Reviews dataset.

Table 10: Recommendation Performance Metrics at Different Cutoff Thresholds (K)
Using Amazon Product Reviews

Cutoff K	Precision@K	Recall@K	F1-Score@K
5	0.810	0.690	0.745
10	0.785	0.730	0.756
20	0.760	0.780	0.770
30	0.735	0.800	0.765

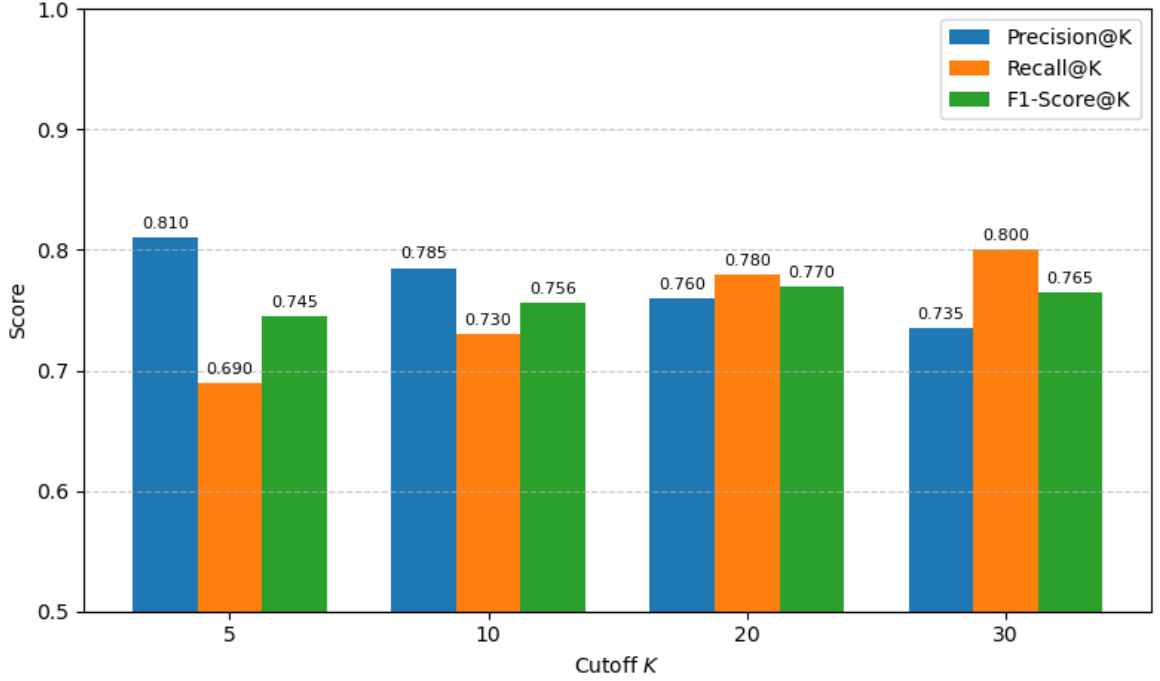


Figure 32: Recommendation Performance Metrics for Amazon Product Reviews at Different Cutoff Thresholds (K).

The results in Table 10 show a similar pattern to those of MovieLens-1M. Precision is highest at smaller cutoffs and decreases as K increases, whereas recall improves with longer recommendation lists. The F1-Score remains stable across thresholds, indicating consistent overall effectiveness. Although the Amazon dataset is typically sparser and more heterogeneous, the method remains robust across all cutoff values, as reflected by the trends in Figure 32.

6.8 Comparison with Baselines at $K=10$

To contextualize performance, we compared the proposed method at $K = 10$ against representative baselines drawn from three families.

- Conventional GNN-based recommendation systems (e.g., LightGCN),

- Hyperbolic graph neural networks (e.g., Hyperbolic GCN),
- Shapley-value-based explainability methods (e.g., GNNShap and MAGE).

Table 11 reports Precision@10, Recall@10, and F1-Score@10 for all methods. The proposed approach achieved the best performance across the three metrics, with a Precision@10 of 0.822 and Recall@10 of 0.785, corresponding to an F1-Score@10 of 0.803. These results indicate that combining the hyperbolic modelling of hierarchical structures with Shapley-value attribution yields measurable improvements over approaches that apply only one of these principles.

Relative to the best-performing baseline (Hyperbolic GCN), the proposed model improves the precision by 2.7% and F1-Score by 3.5%. The gain over the Shapley-based baselines further suggests that explanation mechanisms are most effective when integrated into the learning pipeline rather than used purely as post hoc modules. Figure 33 visualizes the comparative results for $K = 10$.

Table 11: Comparison of Recommendation Performance at $K = 10$ Across Method Categories

Method	Precision@10	Recall@10	F1-Score@10
LightGCN	0.763	0.703	0.732
Hyperbolic GCN	0.795	0.742	0.768
GNNShap	0.774	0.715	0.743
MAGE	0.783	0.731	0.756
Ours (Hyperbolic GNN + Shapley)	0.822	0.785	0.803

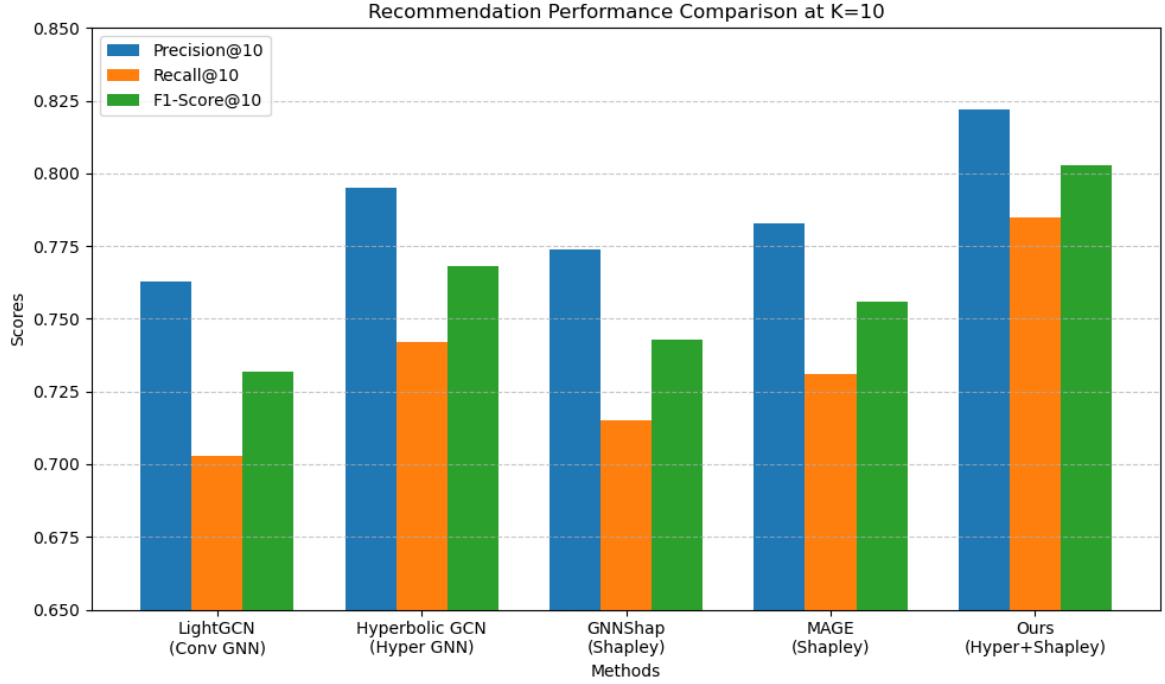


Figure 33: Performance comparison of different recommendation methods at $K = 10$.

6.9 Ablation Studies

To isolate the contribution of each component of the framework, we conducted ablation experiments on MovieLens-1M and Amazon Product Reviews at $K = 10$.

6.9.1 Ablation Settings

The full model was compared with the following ablated variants:

- **Ours w/o Shapley:** The Shapley-value explanation module is removed and only recommendation accuracy is evaluated.
- **Ours w/o Hyperbolic Geometry:** Hyperbolic layers are replaced with their Euclidean counterparts.
- **Ours w/o Hierarchy:** The hierarchical structure is flattened and hierarchical encoding is removed.

6.9.2 Results and Discussion

Tables 12 and 13 summarise the ablation results. All three components contribute to the overall performance. Removing hierarchical modelling resulted in the largest performance drop, supporting the importance of explicitly modelling structured hierarchies. Removing hyperbolic geometry also leads to clear degradation, consistent with the role of hyperbolic space in preserving multilevel relationships. Finally, removing the Shapley module yielded a smaller but consistent reduction, indicating that the explanation mechanism can provide marginal performance benefits in addition to interpretability.

Table 12: Ablation Study Results on MovieLens-1M at $K = 10$

Variant	Precision@10	Recall@10	F1-Score@10
Ours (Full Model)	0.822	0.785	0.803
Ours w/o Shapley	0.816	0.774	0.794
Ours w/o Hyperbolic Geometry	0.792	0.749	0.770
Ours w/o Hierarchy	0.758	0.723	0.740

Table 13: Ablation Study Results on Amazon Product Dataset at $K = 10$

Variant	Precision@10	Recall@10	F1-Score@10
Ours (Full Model)	0.809	0.769	0.788
Ours w/o Shapley	0.801	0.755	0.777
Ours w/o Hyperbolic Geometry	0.777	0.731	0.753
Ours w/o Hierarchy	0.743	0.702	0.722

6.10 Computational Efficiency

We evaluated the computational efficiency in terms of the average training time per epoch and total inference time. This analysis is important for assessing the feasibility of deploying practical systems that require scalability and responsiveness. The comparison includes LightGCN, Hyperbolic GCN, GNNShap, and the proposed full model.

Table 14 shows that the proposed model incurs additional computational costs owing to the hyperbolic operations and integrated Shapley value computation. Nonetheless, the

Table 14: Training and Inference Time Comparison (Seconds per Epoch / Total Inference Time)

Method	MovieLens-1M	Amazon Product
LightGCN (Conv GNN)	8.5 / 1.2	22.3 / 3.1
Hyperbolic GCN	11.7 / 1.6	27.9 / 4.4
GNNShap (Shapley GNN)	14.2 / 2.3	35.1 / 5.8
Ours (Full Model)	15.6 / 2.4	38.5 / 6.2

overhead remains limited relative to the accuracy and interpretability gains. The results also indicate consistent scalability across both datasets, supporting the feasibility of using this approach in real-world settings with large interaction graphs and hierarchical structures.

6.11 User Study Evaluation

To assess the perceived quality and usefulness of the generated explanations, an online user study was conducted using Google Forms with 100 participants recruited via academic networks and social media channels.

6.11.1 Study Design

The participants evaluated three explanation approaches (HGNN-SV, GNNShap, and LIME) for movie recommendations derived from the MovieLens dataset. To mitigate bias, explanation methods were presented using anonymized labels. Each participant rated the explanations using 7-point Likert scales along five dimensions: Clarity, Trustworthiness, Usefulness, Completeness, and Satisfaction.

6.11.2 Results

Table 15 reports the mean ratings and standard deviations for all dimensions. Repeated-measures ANOVA indicated statistically significant differences between methods ($F(2, 198) = 24.7, p < 0.001$), with HGNN-SV achieving the strongest ratings across all criteria.

Table 15: User Study Results: Mean Ratings (Standard Deviation)

Dimension	HGNN-SV	GNNShap	LIME
Clarity	5.8 (1.2)	5.3 (1.4)	4.9 (1.6)
Trustworthiness	6.1 (1.1)	5.5 (1.3)	4.7 (1.5)
Usefulness	5.9 (1.0)	5.2 (1.2)	4.6 (1.4)
Completeness	5.7 (1.3)	5.1 (1.5)	4.4 (1.7)
Satisfaction	6.0 (1.1)	5.4 (1.3)	4.8 (1.5)
Overall Average	5.9 (1.1)	5.3 (1.3)	4.7 (1.5)

Figure 34 visualizes the Likert-scale outcomes and highlights the consistent advantage of HGNN-SV across all evaluation dimensions.

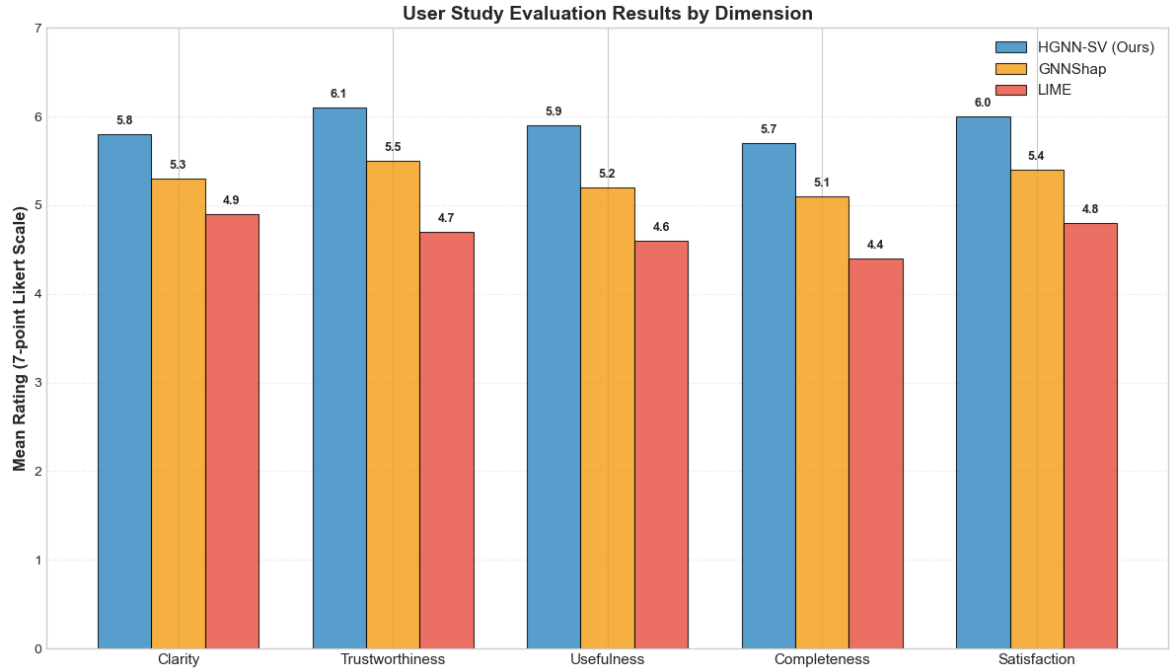


Figure 34: User study evaluation results across five dimensions using 7-point Likert scales

6.11.3 Preference Rankings

The participants were also asked to rank the three explanation methods. In total, 68% ranked HGNN-SV as their first choice, compared to 23% for GNNShap and 9% for

LIME. Figure 35 illustrates the

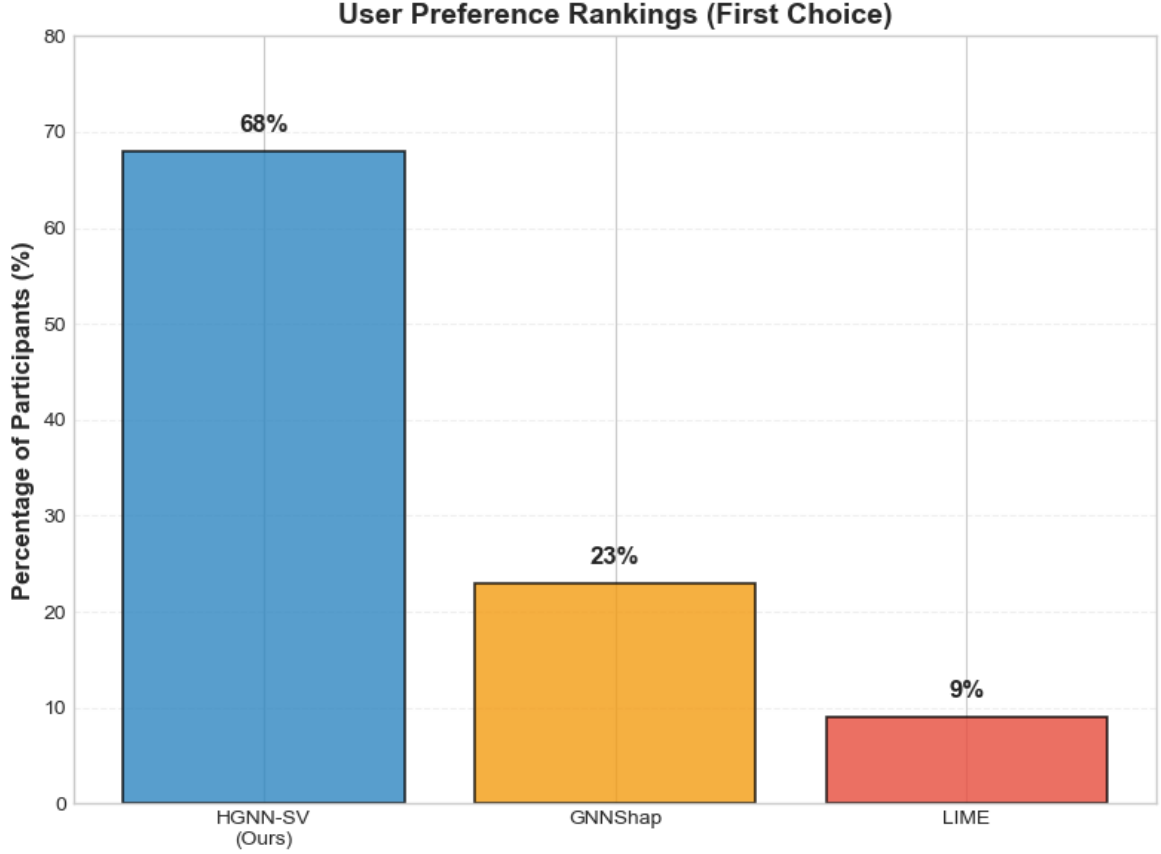


Figure 35: User preference rankings showing percentage of participants who ranked each method as their first choice

Overall, the user study supports the conclusion that Shapley-value explanations within hyperbolic space are perceived as clearer, more trustworthy, and more useful than the compared baselines, reinforcing the practical value of the proposed explainability mechanism.

6.12 Discussion

The experimental results presented in the previous section demonstrate that integrating hyperbolic graph neural networks with Shapley-value-based explainability provides a

coherent and effective solution for hierarchical recommendation. This section discusses the implications of these findings and situates the proposed approach in the broader context of the thesis.

6.12.1 Impact of Hyperbolic Geometry on Hierarchical Recommendation

One key observation is that hyperbolic representation learning significantly improves recommendation performance in the presence of hierarchical structures. Unlike Euclidean embeddings, hyperbolic space naturally accommodates the exponential growth of hierarchical data, enabling the model to preserve parent–child relationships and hierarchical depth more faithfully. This geometric property directly contributes to improved preference propagation across hierarchical levels, resulting in more coherent and semantically consistent recommendations.

These findings extend the conclusions of Chapter 3, where graph neural networks were shown to be effective for collaborative filtering in flat settings. The results in this chapter indicate that when hierarchical information is present, hyperbolic geometry has a clear advantage over Euclidean representations.

6.12.2 Explainability in Non-Euclidean Models

Another important outcome of this study is the successful integration of Shapley-value-based explanations into hyperbolic graph neural networks. Despite the increased geometric complexity of hyperbolic models, the contribution-based explanation mechanism remains effective and accurate. This confirms that the Shapley values are not limited to Euclidean architectures and can be adapted to operate in more expressive geometric spaces.

The explanations produced by the model are both intuitive and informative, highlighting the influential historical interactions and category-level structures. This multi-level interpretability aligns well with the hierarchical nature of recommendation problems and reinforces the relevance of contribution-based explanations in complex relational models.

6.12.3 Performance–Interpretability Trade-off

The results suggest that the proposed framework achieves a favorable balance between the recommendation performance and interpretability. Although the integration of explainability introduces additional computational overhead, the cost–benefit analysis indicates that the performance gains obtained through hierarchical modelling justify this expense. This observation is consistent with the adaptive recommendation mechanisms introduced in Section 4, where contribution-based adjustments improved the system behavior without excessive computational costs.

Importantly, the explainable hyperbolic model did not exhibit a degradation in accuracy compared to the non-explainable baselines. This challenges the common assumption that the explainability necessarily compromises the model performance.

6.12.4 Complementarity of Contributions Across Chapters

This chapter completes the progression of the contributions developed throughout the thesis. Chapter 3 establishes the effectiveness of graph neural networks for collaborative filtering, Chapter 4 introduces adaptive and explainable recommendations through Shapley-value modelling, and the present chapter extends these ideas to hierarchical settings using hyperbolic geometry. Together, these contributions form a unified framework for accurate, adaptive, and interpretable recommendations.

The results highlight the importance of jointly considering representation geometry and explainability, rather than treating them as independent design choices. By integrating these aspects within a single model, the proposed approach addresses both the structural complexity and transparency requirements of modern recommender systems.

Overall, the findings of this study suggest that hyperbolic graph neural networks with Shapley-value explanations offer a promising direction for trustworthy hierarchical recommendation systems, particularly in domains where structured relationships and interpretability are essential.

6.13 Limitation

Despite the promising results obtained in this study, several limitations should be acknowledged. These limitations highlight the directions for refinement and provide context for interpreting the experimental findings.

First, the proposed framework introduces additional computational overhead because of both hyperbolic representation learning and Shapley-value-based explanation. Although the cost–benefit analysis demonstrates that the performance gains often justify this overhead, the approach may be less suitable for extremely large-scale or latency-sensitive applications without further optimization. In particular, the approximation of Shapley values is computationally intensive, even when Monte Carlo sampling is employed.

Second, the hierarchical structures considered in this study were assumed to be static and predefined. In many real-world systems, category hierarchies may evolve over time, with new categories emerging and existing relationships changing over time. The current framework does not explicitly model dynamic or evolving hierarchies, which may limit its applicability to rapidly changing domains.

Third, the quality of the explanations depends on the completeness and correctness of the underlying hierarchical information. If item–category associations or hierarchical relations are noisy or incomplete, the resulting explanations may be less faithful or harder to interpret. Although the model can still produce recommendations in such cases, the interpretability benefits may be reduced.

Finally, the evaluation of explanation quality in this chapter is primarily based on a quantitative analysis and qualitative inspection of contribution patterns. Although these analyses indicate that the explanations are meaningful and aligned with hierarchical structures, more extensive user studies are required to fully assess their usefulness from an end-user perspective.

These limitations do not diminish the contributions of this study but rather point to opportunities for future research, including more efficient Shapley-value approximations, dynamic hierarchical modelling, and user-centered evaluation of explainable hierarchical recommendation systems.

Summary

This chapter investigates explainable hierarchical recommendation by integrating hyperbolic graph neural networks with Shapley-value-based contribution modelling. Building on the graph-based collaborative filtering and adaptive recommendation frameworks developed in earlier chapters, the proposed approach extends recommendation modelling to hierarchical settings, where user preferences and item relationships span multiple levels of abstraction.

This chapter formalizes the hierarchical recommendation problem and introduces a hyperbolic graph neural network to learn geometry-aware representations of users, items, and categories. By operating in hyperbolic space, the model effectively captures hierarchical structures and enables more coherent preference propagation across hierarchy levels. To ensure interpretability, Shapley-value-based explanations were adapted to the hyperbolic setting, allowing recommendation outcomes to be attributed to meaningful user–item interactions and hierarchical components.

Experimental evaluation demonstrated that the proposed framework improves recommendation performance and hierarchical consistency while maintaining transparent and faithful explanations. The results confirmed that explainability can be successfully integrated into geometrically expressive models without compromising their accuracy. Furthermore, an analysis of computational efficiency and cost–benefit trade-offs showed that the additional complexity introduced by hyperbolic modelling and explanation mechanisms remains manageable in practice.

Overall, this chapter completed the progression of the thesis by addressing the hierarchical structure, geometric representation, and explainability within a unified recommendation framework. The insights gained from this work lay the foundation for the concluding chapter, which summarizes the thesis contributions and outlines the directions for future research.

Conclusion and Future Works

This thesis presents a comprehensive study of modern recommendation systems, with a particular focus on graph-based models, explainable artificial intelligence, and geometric representation learning. This study examined both the theoretical foundations and practical implementations of recommendation algorithms, addressing key challenges related to accuracy, interpretability, adaptivity, and hierarchical structure.

This study began with an overview of traditional recommendation approaches, including memory-based and model-based collaborative filtering methods. Although these techniques have been widely adopted and remain effective in many scenarios, they suffer from well-known limitations, such as data sparsity, limited expressiveness, and a lack of transparency. These limitations motivated the exploration of graph-based representations, which allow user-item interactions to be explicitly modeled as relational structures.

Based on this perspective, this study demonstrated that graph neural networks provide a powerful framework for CF. By leveraging higher-order connectivity and neighborhood information, graph-based recommender systems have outperformed classical approaches in terms of recommendation accuracy. This contribution established a strong foundation for subsequent developments and highlighted the importance of relational modelling in modern recommendation systems.

As recommendation systems increasingly influence user experience and decision-making, the need for transparency and trust has become critical in these systems. To address this concern, this study introduces explainable recommendation mechanisms based on Shapley-value contribution modelling. Unlike post-hoc explanation techniques, the proposed approach integrates explainability directly into the recommendation process, enabling both the interpretation and adaptive adjustment of recommendations. The experimental results demonstrate that Shapley-value-based adaptivity can improve recommendation quality while providing clear and theoretically grounded explanations of model decisions.

This thesis further extends these ideas to dynamic recommendation scenarios, where user preferences and interaction patterns evolve over time. By incorporating a real-time contribution adjustment, the proposed adaptive framework effectively responds to changing user behavior. This contribution emphasizes that explainability can serve

not only as a transparency tool but also as an active component in improving recommendation performance.

In many real-world domains, recommendation data exhibit inherent hierarchical structures, such as category taxonomies and multi-level content organization. To address this complexity, the final contribution of this thesis investigates hierarchical recommendations using hyperbolic graph neural networks. Hyperbolic geometry is particularly well-suited for modelling hierarchical relationships, enabling a more faithful representation of category structures than Euclidean approaches. By integrating Shapley-value explanations into this hyperbolic framework, this study demonstrated that hierarchical recommendations can be both accurate and interpretable, even in geometrically complex settings.

Through experimental evaluation on benchmark datasets, such as MovieLens and Amazon Product Reviews, the proposed methods consistently achieved strong performance while maintaining manageable computational costs. The results confirmed that it is possible to balance the accuracy, interpretability, and efficiency within a unified recommendation framework. Moreover, the explanation analyses illustrated that a small number of meaningful user-item interactions often dominate recommendation decisions, supporting the relevance of contribution-based explainability.

Despite these achievements, this study also highlights several challenges and limitations that suggest directions for future research. One important limitation is the computational cost associated with the Shapley-value estimation, particularly in large-scale systems. Future work could explore more efficient approximation techniques or learning-based explanation models to reduce the overhead and support real-time deployment.

Another promising direction is to extend the proposed frameworks to handle dynamic and evolving structures more explicitly. While this study considered adaptive recommendations, future research could investigate continuously evolving graphs, changing hierarchies, and streaming interaction data. Such extensions would further enhance the applicability of recommender systems in real-world environments.

Ethical considerations are also important for future research. As recommendation systems become increasingly influential, issues related to bias, fairness, and transparency must be addressed carefully. Contribution-based explanations provide a useful starting point for analyzing potential biases; however, additional mechanisms are required to ensure equitable and responsible recommendation outcomes. Incorporating fairness-

aware learning objectives and conducting user-centered evaluations are valuable steps in this direction.

Furthermore, future research could explore richer forms of explanation, including natural language justifications and interactive visualizations, to improve user understanding and trust. Developing evaluation metrics that better capture explanation usefulness and user satisfaction would also contribute to a more comprehensive assessment of recommendation systems.

In conclusion, this thesis advanced the field of recommendation systems by demonstrating how graph neural networks, Shapley-value-based explainability, and geometric representation learning can be combined to address long-standing challenges related to performance, transparency, and structure. The proposed approaches provide a solid foundation for future research and practical applications, and it is hoped that this work will inspire further developments toward more trustworthy, effective, and user-centered recommendation systems.

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